005A: MSID,9060 08/24/2026 08:49

```
 IIIIIIII      DDDDDDDDDDD                      0000000         0000000      5555555555555     AAAAAAAAAAA
 IIIIIIII      DDDDDDDDDDDD                    000000000       000000000     5555555555555    AAAAAAAAAAAAA
    II         DD        DD                   00      00      00      00     55               AA        AA
    II         DD        DD    =============  00        00    00        00   55               AA        AA
    II         DD        DD    =============  00        00    00        00   55               AA        AA
    II         DD        DD                   00        00    00        00   555555555555     AAAAAAAAAAAAA
    II         DD        DD                   00        00    00        00   5555555555555    AAAAAAAAAAAAA
    II         DD        DD    =============  00        00    00        00              55    AA        AA
    II         DD        DD    =============  00        00    00        00              55    AA        AA
    II         DD        DD                   00      00      00      00     55        55     AA        AA
 IIIIIIII      DDDDDDDDDDDD                    000000000       000000000     5555555555555    AA        AA
 IIIIIIII      DDDDDDDDDDD                      0000000         0000000       55555555555     AA        AA
```

```
 99999999999        0000000        666666666666       0000000
9999999999999      000000000      6666666666666      000000000
99         99     00       00     66               00       00
99         99    00         00    66              00         00
99         99    00         00    66              00         00
9999999999999    00         00    666666666666    00         00
 999999999999    00         00    6666666666666   00         00
           99    00         00    66         66   00         00
           99    00         00    66         66   00         00
           99     00       00     66         66    00       00
9999999999999      000000000      6666666666666      000000000
999999999999        0000000        66666666666        0000000
```

```
  IIIIIIII       DDDDDDDDDDD                        0000000          0000000       5555555555555     AAAAAAAAAAA
  IIIIIIII       DDDDDDDDDDDD                      000000000        000000000      5555555555555    AAAAAAAAAAAAA
     II          DD         DD                    00       00      00       00     55               AA         AA
     II          DD         DD    =============   00         00    00         00   55               AA         AA
     II          DD         DD    =============   00         00    00         00   55               AA         AA
     II          DD         DD                    00         00    00         00   555555555555     AAAAAAAAAAAAA
     II          DD         DD                    00         00    00         00   5555555555555    AAAAAAAAAAAAA
     II          DD         DD    =============   00         00    00         00               55   AA         AA
     II          DD         DD    =============   00         00    00         00               55   AA         AA
     II          DD         DD                     00       00      00       00     55         55   AA         AA
  IIIIIIII       DDDDDDDDDDDD                      000000000        000000000      5555555555555    AA         AA
  IIIIIIII       DDDDDDDDDDD                        0000000          0000000        55555555555     AA         AA
```

```
 99999999999        0000000        666666666666       0000000
9999999999999      000000000      6666666666666      000000000
99         99     00       00     66                00       00
99         99    00         00    66               00         00
99         99    00         00    66               00         00
9999999999999    00         00    666666666666     00         00
 999999999999    00         00    6666666666666    00         00
           99    00         00    66         66    00         00
           99    00         00    66         66    00         00
           99     00       00     66         66     00       00
9999999999999      000000000      6666666666666      000000000
999999999999        0000000        66666666666        0000000
```

```
 08:49 AUG 24,'26 ID=005A
JOB 9060,MSID,7. FILE=J:LA-ALL, ID=9,  LINE=1
LIMIT (CO,64),(TIME,999)
SET M:SI/S:PRINT
SET M:BO/R:PRINT
AP SI,BO,SD,LO
```

```
                                TITLE   'S:PRINT - FORMAT AND PRINT'
                                PCC     0
                                SYSTEM  SIG7FDP
                                SYSTEM  BPM
                            * PRINT ROUTINES
                            * ALL ASSUME WA(SPD) IN R0, LINK WITH R1
                            * REGISTERS 2-7 ARE GENERALLY VOLATILE
                            * NUMERIC FORMAT WORD:
                            ** BITS 8-16 OPTIONS: SEE BELOW
                            ** BITS 17-19 FORMAT CODE: 0,CHAR,DATE,DECIMAL,HEX,FLOAT,TIME
                            ** BITS 20-24 DECIMALS (0-31)
                            ** BITS 25-31 FIELD WIDTH (0-127)
                            *
                            * OPTIONS FOR NUMERICS
        00000001            OP_SHORT        EQU     X'01'
        00000002            OP_LONG         EQU     X'02'
        00000004            OP_LEADSIGN     EQU     X'04'
        00000008            OP_LEADZEROS    EQU     X'08'
        00000010            OP_UNSIGNED     EQU     X'10'
        00000020            OP_PRINTABLE    EQU     X'20'
                            *
                            * OPTIONS FOR DATES AND TIMES
                            * SHORT AND LONG AS ABOVE, THEN:
        00000004            OT_DATEONLY     EQU     X'04'
        00000008            OT_TIMEONLY     EQU     X'08'
        00000010            OT_MM           EQU     X'10'   NUMERIC MONTH, SLASH SEPARATORS
        00000010            OT_NOSECS       EQU     X'10'
        00000020            OT_MILLISECS    EQU     X'20'
        00000040            OT_YMD          EQU     X'40'
                            *
                            ** SYSTEM REGISTERS:
                            *
        00000008            SR1             EQU     8
        00000009            SR2             EQU     9
        0000000A            SR3             EQU     10
        0000000B            SR4             EQU     11
                            *
                            *
                                REF     M:LO
```

```
                                REF     M:UC
                                REF     FP:INTERPRET
                                REF     INT:FROMTEXT
                                REF     UT:MONTH3
                        *
                                DEF     PD:DCB
                                DEF     PD:LINEWIDTH
                        *
                                DEF     PI:TEXTC        IMMEDIATE, R2=WA(TEXTC TO PRINT)
                                DEF     PI:TEXT         IMMEDIATE, R2=BA(TEXT), R3=LENGTH
                                DEF     PI:PAGE         IMMEDIATE, PAGE EJECT
                                DEF     PI:SPACE        IMMEDIATE, BLANK LINE
                        *
                                DEF     PB:INIT         RETURNS R4=BUFFER WA, R5=BUFFER BYTE LENGTH
                                DEF     PB:DONE
                                DEF     PB:CLEAR        CLEAR BUFFER
                                DEF     PB:SETPOS       SET POSITION WITHIN LINEBUFFER
                                DEF     PB:TABPOS       TAB TO SPECIFIC POSITION
                                DEF     PB:TEXT8        R8=(LEN,BA)
                                DEF     PB:TEXTC        R2=WA(TEXTC)
                                DEF     PB:TEXT         R2=BA(TEXT), R3=LENGTH
                                DEF     PB:CHAR         R2=CHAR
                                DEF     PB:WORD         R2=VALUE, R3=FORMAT
                                DEF     PB:DWORD        R4/5=VALUE, R3=FORMAT
                                DEF     PB:FLUSH        PRINT BUFFER
                                DEF     PB:WRAP         PRINT BUFFER WORDWRAPPED
                        *                               R6: 24-31: CONTINUATION LINE INDENT
                                DEF     PB:GETBUF       GET ADDRESS AND LENGTH OF BUFFER
                        *
                                BOUND   8
  01 00000    00000000 A   PTC     DATA    0,0             PREVIOUS TRAP STATE
  01 00001    00000000 A
                                BOUND   4
  01 00002    00000000 A   BUFFER  DATA    0               BUFFER WORD ADDRESS
  01 00003    00000000 A   BUFMAX  DATA    0               TOTAL BYTES ALLOCATED
  01 00004    00000000 A   BUFLEN  DATA    0               BUFFER USED
  01 00005    00000000 A   BUFX    DATA    0               CURRENT BUFFER POSITION
  01 00006    00000000 A   INDENT  DATA    0               NUMBER OF CHARS TO INDENT
                        *
```

```
         00000080          FORMATW EQU     128             MAX WIDTH OF A FORMAT ITEM
01 00007    00000000 A   FORMAT  DATA    0               WA FORMATTING BUFFER
                         **
01 00008    00000000 N   PD:DCB  DATA    M:LO            CURRENT OUTPUT DCB
01 00009    00000078 A   PD:LINEWIDTH DATA 120
01 0000A    0000004E A   PD:TERMWIDTH DATA 78
                         **
04 00000                         CSECT   1
04 00000    40404040 A   BLANKS  DATA    X'40404040'
04 00001    0000007F A   M7F     DATA    X'0000007F'
04 00002    000000F0 A   MF0     DATA    X'000000F0'
04 00003    000000FF A   MFF     DATA    X'000000FF'
04 00004    00FFFFFF A   MFFFFFF DATA    X'00FFFFFF'
04 00005    FFFFFFFF A   MALL    DATA    X'FFFFFFFF'
                         **
                                 BOUND   4
04 00006    0000000A A   I10     DATA    10
04 00007    00000014 A   I20     DATA    20
                                 BOUND   8
04 00008    00000000 A   D0      DATA,8  0
            00000000
04 0000A    00000000 A   D1      DATA,8  1
            00000001
```

```
                                **
                                ** THESE FUNCTIONS ARE IMMEDIATE AND INDEPENDENT OF THE BUFFER.
                                **
        04 0000C                PI:TEXTC EQU    $
04 0000C    02200040 A                  LCI     4
04 0000D    8B100000 A                  PSM,1   *0
04 0000E    F2300002 A                  LB,3    *2
04 0000F    22400001 A                  LI,4    1
04 00010    68000016                    B       PI:WRITE
        04 00011                PI:TEXT EQU     $
04 00011    02200040 A                  LCI     4
04 00012    8B100000 A                  PSM,1   *0
04 00013    22400003 A                  LI,4    X'3'
04 00014    4B400002 A                  AND,4   2
04 00015    2520007E A                  SLS,2   -2
        04 00016                PI:WRITE EQU    $
04 00016    04100000 02                 M:WRITE *PD:DCB,(BUF,*2),(BTD,*4),(SIZE,*3),(WAIT)
02 00000    91000008 N
            34000010
02 00002    80000002 A
02 00003    80000003 A
02 00004    80000004 A
04 00017    02200040 A                  LCI     4
04 00018    8A100000 A                  PLM,1   *0
04 00019    68020000 A                  B       0,1
04 0001A    04100005 02         PI:PAGE M:DEVICE *PD:DCB,(PAGE)
02 00005    84000008 01
04 0001B    68020000 A                  B       0,1
04 0001C    02200040 A          PI:SPACE LCI    4
04 0001D    8B100000 A                  PSM,1   *0
04 0001E    22200007 A                  LI,2    X'7'
04 0001F    4B300002 A                  AND,3   2
04 00020    04100006 02                 M:WRITE *PD:DCB,(BUF,BLANKS),(SIZE,1),(WAIT)
02 00006    91000008 N
            30000010
02 00008    00000000 04
02 00009    00000001 A
04 00021    64300020                    BDR,3   $-1
```

```
                          **
                          ** CALL PR:INIT TO ALLOCATE AND INITIALIZE A PRINT LINE AND FORMAT BUFFER.
                          **
                          *
        04 00022          PB:INIT EQU     $
04 00022    32400002 01          LW,4    BUFFER          ANYTHING ALLOCATED?
04 00023    6930003A             BNEZ    INITD           YES/ ALREADY INITIALIZED
04 00024    89800000 A           PSW,SR1 *0
04 00025    89900000 A           PSW,SR2 *0
04 00026    0480000A 02          M:GP    1               GET A PAGE FOR THE BUFFER
02 0000A    08000001 A
04 00027    69800038             BCS,8   INITF           FAILED
04 00028    2580000B A           SLS,SR1 11
04 00029    208FFF80 A           AI,SR1  -FORMATW
04 0002A    35800003 01          STW,SR1 BUFMAX          SAVE BYTE LENGTH
04 0002B    32400009 A           LW,4    SR2
04 0002C    35400002 01          STW,4   BUFFER          SAVE WORD ADDRESS
04 0002D    2580007E A           SLS,SR1 -2
04 0002E    30900008 A           AW,SR2  SR1             FORMAT BUFFER WORD ADDRESS
04 0002F    35900007 01          STW,SR2 FORMAT
                          * SET UP WIDTH BASED ON CURRENT OUTPUT DCB
04 00030    22500006 A           LI,5    6
04 00031    D25A0008 01          LH,5    *PD:DCB,5
04 00032    2550007F A           SLS,5   -1              DEFAULT RECORD SIZE
04 00033    0410000B 02          M:DEVICE *PD:DCB,(CORRES,M:UC)
02 0000B    AB000008 01
02 0000C    00000000 X
04 00034    20800000 A           AI,SR1  0
04 00035    68300037             BEZ     $+2             NOT M:UC
04 00036    2250000A 01          LI,5    PD:TERMWIDTH    USE THIS INSTEAD
04 00037    35500009 01          STW,5   PD:LINEWIDTH
                          *
04 00038    88900000 A   INITF   PLW,SR2 *0
04 00039    88800000 A           PLW,SR1 *0
04 0003A    22500000 A   INITD   LI,5    0
04 0003B    35500004 01          STW,5   BUFLEN          CURRENT LENGTH
04 0003C    35500005 01          STW,5   BUFX            CURRENT POSITION
04 0003D    32500003 01          LW,5    BUFMAX
04 0003E    68020000 A           B       0,1
```

```
                          *
        04 0003F          PB:DONE EQU     $
04 0003F    32400002 01         LW,4    BUFFER          ANY ALLOCATED?
04 00040    68320000 A          BEZ     0,1             NO/ RETURN
04 00041    0480000D 02         M:FVP   *4              FREE IT
02 0000D    85000004 A
04 00042    22500000 A          LI,5    0
04 00043    35500002 01         STW,5   BUFFER
04 00044    35500003 01         STW,5   BUFMAX
04 00045    68020000 A          B       0,1
                          *
        04 00046          PB:WRAP EQU     $
04 00046    022000B0 A          LCI     11
04 00047    8B100000 A          PSM,1   *0              SAVE R1-R11
04 00048    2240007F A          LI,4    X'7F'
04 00049    4B400006 A          AND,4   6               INDENT
04 0004A    31400009 01         CW,4    PD:LINEWIDTH    CAN'T BE BIGGER THAN LINE WIDTH
04 0004B    6910004D            BL      $+2             OK
04 0004C    22400000 A          LI,4    0
04 0004D    35400006 01         STW,4   INDENT          SAVE FOR LATER
04 0004E    32700004 01         LW,7    BUFLEN
                          *
04 0004F    31700002 A   WRAPLN CW,7    2               DOES THIS FIT?
04 00050    68200074            BLE     WRAPFLUSH       YES
04 00051    32600002 A   WRAPNXT LW,6   2
04 00052    22500040 A          LI,5    ' '
04 00053    F15C0002 01  WRAPP  CB,5    *BUFFER,6
04 00054    68300057            BE      WRAPB
04 00055    64600053            BDR,6   WRAPP
04 00056    32600002 A          LW,6    2               NO BLANK FOUND
04 00057    0410000E 02  WRAPB  M:WRITE *PD:DCB,(BUF,*BUFFER),(BTD,0),(SIZE,*6),(WAIT)
02 0000E    91000008 N
            34000010
02 00010    80000002 01
02 00011    80000006 A
02 00012    00000000 A
                          * NOW SHIFT EVERYTHING IN THE BUFFER, AFTER INSERTING INDENT SPACES
04 00058    32400006 01         LW,4    INDENT          NUMBER OF CHARS TO INDENT
04 00059    6820005E            BLEZ    WRAPNI          NONE
```

```
04 0005A    22500040 A          LI,5    ' '
04 0005B    F5580002 01  WRAPFI  STB,5   *BUFFER,4
04 0005C    6440005B            BDR,4   WRAPFI
04 0005D    F5500002 01         STB,5   *BUFFER
                         *
04 0005E    38700006 A   WRAPNI  SW,7    6               NUMBER OF CHARACTERS LEFT
04 0005F    32400002 01         LW,4    BUFFER          WA
04 00060    25400002 A          SLS,4   2               BA
04 00061    32500004 A          LW,5    4
04 00062    30400006 A          AW,4    6               SOURCE
04 00063    30500006 01         AW,5    INDENT          DESTINATION
                         * SKIP LEADING WHITESPACE ON CONTINUATION
04 00064    22300040 A          LI,3    ' '
04 00065    71380000 A   WRAPSKP CB,3    0,4
04 00066    6930006B            BNE     WRAPMV
04 00067    20400001 A          AI,4    +1
04 00068    64700065            BDR,7   WRAPSKP
04 00069    35700004 01         STW,7   BUFLEN
04 0006A    68000074            B       WRAPFLUSH       POP STACK, DONT PRINT THIS LINE
                         * MOVE 1 CHAR AT A TIME, THERE MAY BE MORE THAN 256.
04 0006B    32600007 A   WRAPMV  LW,6    7
04 0006C    72380000 A   WRAPMVB LB,3    0,4
04 0006D    753A0000 A          STB,3   0,5
04 0006E    20400001 A          AI,4    +1
04 0006F    20500001 A          AI,5    +1
04 00070    6460006C            BDR,6   WRAPMVB
04 00071    30700006 01         AW,7    INDENT          NEW BUFLEN
04 00072    35700004 01         STW,7   BUFLEN          SET IT
04 00073    6800004F            B       WRAPLN          DO NEXT LINE
                         *
       04 00074          WRAPFLUSH EQU   $
04 00074    022000B0 A          LCI     11
04 00075    8A100000 A          PLM,1   *0
       04 00076          PB:FLUSH EQU    $
04 00076    32500004 01         LW,5    BUFLEN          CHECK LENGTH
04 00077    68200079            BLEZ    PB:CLEAR
04 00078    04100013 02         M:WRITE *PD:DCB,(BUF,*BUFFER),(BTD,0),(SIZE,*BUFLEN),(WAIT)
02 00013    91000008 N
            34000010
```

```
         02 00015    80000002 01
         02 00016    80000004 01
         02 00017    00000000 A
                                 *
         04 00079                PB:CLEAR EQU    $
   04 00079    22500000 A                LI,5    0
   04 0007A    35500005 01               STW,5   BUFX
   04 0007B    35500004 01               STW,5   BUFLEN
   04 0007C    68020000 A                B       0,1
                                 *
         04 0007D                PB:GETBUF EQU   $               RETURN R2=WA(BUFFER), R3=LENGTH
   04 0007D    B2200002 01               LW,2    *BUFFER
   04 0007E    32300004 01               LW,3    BUFLEN          CURRENT E=LENGTH
   04 0007F    68020000 A                B       0,1
                                 *
         04 00080                PB:SETPOS EQU   $               R3=POSITION
   04 00080    4B300001                  AND,3   M7F
   04 00081    31300004 01               CW,3    BUFLEN
   04 00082    69200085                  BG      PB:TABPOS       ADD BLANKS
   04 00083    35300005 01               STW,3   BUFX            JUST SET POSTITION
   04 00084    68020000 A                B       0,1             DONE
                                 *
         04 00085                PB:TABPOS EQU   $               R3=POSITION
   04 00085    022000B0 A                LCI     11
   04 00086    8B100000 A                PSM,1   *0              SAVE R1-11
                                 *
   04 00087    4B300001                  AND,3   M7F
   04 00088    31300003 01               CW,3    BUFMAX          TOO FAR?
   04 00089    6910008B                  BL      TABTO           NO, OK
   04 0008A    32300003 01               LW,3    BUFMAX
         04 0008B                TABTO   EQU     $
   04 0008B    38300005 01               SW,3    BUFX
   04 0008C    6810008F                  BGE     PAD             ADD BLANKS
   04 0008D    30300005 01               AW,3    BUFX
   04 0008E    6A100076                  BAL,1   PB:FLUSH
   04 0008F    22800040 A      PAD       LI,8    X'40'
   04 00090    32700005 01               LW,7    BUFX
   04 00091    F58E0002 01     ADDPAD    STB,8   *BUFFER,7
   04 00092    20700001 A                AI,7    1
```

```
04 00093    64300091            BDR,3   ADDPAD
04 00094    35700005 01         STW,7   BUFX            SAVE CURRENT POSITION
04 00095    31700004 01         CW,7    BUFLEN          NEW
04 00096    68200229            BLE     PB:RETURN
04 00097    35700004 01         STW,7   BUFLEN
04 00098    68000229            B       PB:RETURN
                          *
                          *
     04 00099           PB:TEXT8 EQU $                  R8=LENGTH(8 BITS),BA(24 BITS)
04 00099    022000B0 A          LCI     11
04 0009A    8B100000 A          PSM,1   *0              SAVE R1-11
04 0009B    32200004            LW,2    MFFFFFF
04 0009C    4B200008 A          AND,2   8               BA
04 0009D    72300008 A          LB,3    8               LEN
04 0009E    680000A7            B       P$:TEXT
     04 0009F           PB:TEXTC EQU $                  R2=WA(TEXTC)
04 0009F    022000B0 A          LCI     11
04 000A0    8B100000 A          PSM,1   *0              SAVE R1-11
04 000A1    F2300002 A   P$:TEXTC LB,3  *2              TAIL ENTRY WITH REGS PUSHED
04 000A2    25200002 A          SLS,2   2
04 000A3    20200001 A          AI,2    1
04 000A4    680000A7            B       P$:TEXT
     04 000A5           PB:TEXT EQU $                   R2=BA(TEXT), R3=LENGTH
04 000A5    022000B0 A          LCI     11
04 000A6    8B100000 A          PSM,1   *0              SAVE R1-11
     04 000A7           P$:TEXT EQU     $               TAIL ENTRY WITH REGS PUSHED
04 000A7    32500003 A          LW,5    3
04 000A8    68200229            BLEZ    PB:RETURN       STUPID LENGTH
                          *
04 000A9    30500005 01         AW,5    BUFX            TOTAL LENGTH
04 000AA    38500009 01         SW,5    PD:LINEWIDTH    OVERFLOW
04 000AB    682000AD            BLEZ    ADDSEG
04 000AC    38300005 A          SW,3    5               BYTES TO MOVE
04 000AD    32700005 01  ADDSEG LW,7    BUFX
04 000AE    72640000 A   ADDBYTE LB,6   0,2             GET SOURCE BYTE
04 000AF    20200001 A          AI,2    1
04 000B0    F56E0002 01         STB,6   *BUFFER,7
04 000B1    20700001 A          AI,7    1
04 000B2    643000AE            BDR,3   ADDBYTE
```

```
04 000B3    35700005 01          STW,7   BUFX            SAVE CURRENT POSITION
                            ** AT THIS POINT ALL BYTES HAVE BEEN MOVED, OR THE BUFFER IS FULL.
04 000B4    31500000 A           CW,5    0
04 000B5    692000BA             BGZ     ADDFULL         FULL
04 000B6    31700004 01          CW,7    BUFLEN          UPDATE?
04 000B7    68200229             BLE     PB:RETURN       NO
04 000B8    35700004 01          STW,7   BUFLEN          SET NEW LENGTH
04 000B9    68000229             B       PB:RETURN
                            ** FULL, PRINT IT AND CONTINUE ADDING.
04 000BA    32300005 A   ADDFULL LW,3    5               COPY REMAINING LENGTH
04 000BB    89100000 A           PSW,1   *0              SAVE CALLER
04 000BC    6A100076             BAL,1   PB:FLUSH
04 000BD    88100000 A           PLW,1   *0
04 000BE    680000A7             B       P$:TEXT
                            *
04 000BF    022000B0 A   PB:CHAR LCI     11
04 000C0    8B100000 A           PSM,1   *0
04 000C1    25200018 A           SLS,2   24              MOVE TO HIGH BYTE
04 000C2    20200008 A           AI,2    8               POINT AT IT
04 000C3    22300001 A           LI,3    1               MOVE 1 BYTE
04 000C4    680000A7             B       P$:TEXT
                            *
        04 000C5            PB:WORD EQU $
04 000C5    022000B0 A           LCI     11
04 000C6    8B100000 A           PSM,1   *0              SAVE R1-11
04 000C7    22400000 A           LI,4    0
04 000C8    32500002 A           LW,5    2
04 000C9    681000CB             BGEZ    $+2
04 000CA    224FFFFF A           LI,4    -1              EXTEND SIGN
04 000CB    680000CE             B       DWORD1
                            *
        04 000CC            PB:DWORD EQU    $
04 000CC    022000B0 A           LCI     11
04 000CD    8B100000 A           PSM,1   *0              SAVE R1-11
04 000CE    22A00FFF A   DWORD1  LI,10   X'FFF'
04 000CF    4BA00003 A           AND,10  3               LENGTHS
04 000D0    25300074 A           SLS,3   -12
04 000D1    22B00007 A           LI,11   X'7'
04 000D2    4BB00003 A           AND,11  3               FORMAT CODE
```

```
04 000D3    2530007D A          SLS,3   -3              OPTIONS
04 000D4    4630000B A          XW,3    11
04 000D5    21300007 A          CI,3    DWORDD-DWORDV
04 000D6    682600D7            BLE     $+1,3           DISPATCH
                         *
04 000D7    680001C9    DWORDV  B       DDTRUNC         BAD FORMAT CODE
04 000D8    680000DE            B       DCHAR
04 000D9    680000F7            B       DDATE
04 000DA    68000158            B       DDECIMAL
04 000DB    680001D5            B       DHEX
04 000DC    680001EF            B       DFLOAT
04 000DD    68000228            B       DTIME
       04 000DE         DWORDD  EQU     $
                         **
                         ** FORMAT THE LOW BYTE OF R5 AS A CHARACTER
                         **
04 000DE    21B00020 A  DCHAR   CI,11   OP_PRINTABLE    PRINTABLE ONLY?
04 000DF    684000E6            BAZ     DCHAR1          NO/ PRINT IT
04 000E0    4B500003            AND,5   MFF
04 000E1    21500040 A          CI,5    X'40'           LOWEST PRINTABLE
04 000E2    691000E5            BL      DCHAR2
04 000E3    215000F9 A          CI,5    X'F9'           HIGHEST PRINTABLE
04 000E4    682000E6            BLE     DCHAR1
04 000E5    2250004B A  DCHAR2  LI,5    '.'             SUBSTITUTE
04 000E6    32200007 01 DCHAR1  LW,2    FORMAT
04 000E7    25200002 A          SLS,2   2               GET BA
04 000E8    75540000 A          STB,5   0,2             SAVE IT
04 000E9    22300001 A          LI,3    1               LENGTH
04 000EA    680000A7            B       P$:TEXT
```

```
                          **
                          ** FORMAT A FILE-SYSTEM CP-V DATE (EBCDIC MMDDYYHH IN R4/5)
                          ** IF OP_LONG IS SET, R2 CONTAINS EBCIDIC HHMM
                          **
                          ** FORMAT BUFFER: HHMMDDMMYY
                          ******************************** CHANGE KWD TO FILEDATE
                          ******************************** REPLACE WITH CALL TO JD ROUTINE
         00000001         DTMIN   EQU     1
         00000000         DTHOUR  EQU     0
         00000002         DTDAY   EQU     2
         00000003         DTMONTH EQU     3
         00000004         DTYEAR  EQU     4
                          *
   04 000EB    F0F0F0F0 A   DTMIDNGT DATA   X'F0F0F0F0'
   04 000EC    05D5C5E5 A   DTNEVER5 TEXTC  'NEVER'
   04 000ED    C5D94040 A
   04 000EE    075CD5C5 A   DTNEVER7 TEXTC  '*NEVER*'
   04 000EF    E5C5D95C A
   04 000F0    095C40D5 A   DTNEVER TEXTC   '* NEVER *'
   04 000F1    C5E5C5D9 A
   04 000F2    405C4040 A
   04 000F3    017A4040 A   DTCOLON TEXTC   ':'
   04 000F4    01604040 A   DTDASH  TEXTC   '-'
   04 000F5    016F4040 A   DTQ     TEXTC   '?'
   04 000F6    01404040 A   DTSPACE TEXTC   ' '
         EXT                DTMTAB  EQU     UT:MONTH3
                            *
   04 000F7    21B00002 A   DDATE   CI,11   OP_LONG         HHMM INCLUDED?
   04 000F8    694000FA             BANZ    DDATEL          YES
   04 000F9    322000EB             LW,2    DTMIDNGT        NO, INCLUDE SOMETHING
   04 000FA    22700001 A   DDATEL  LI,7    DTMIN
   04 000FB    492000EB             OR,2    DTMIDNGT        IF BLANKS, MAKE ZEROS
   04 000FC    D52E0007 01          STH,2   *FORMAT,7       SAVE MINUTE IF PRESENT
   04 000FD    22700004 A           LI,7    DTYEAR
   04 000FE    D55E0007 01          STH,5   *FORMAT,7       SAVE YEAR
   04 000FF    22700002 A           LI,7    DTDAY
   04 00100    D54E0007 01          STH,4   *FORMAT,7       SAVE DAY
   04 00101    25500070 A           SLS,5   -16             ISOLATE HOUR
   04 00102    22700000 A           LI,7    DTHOUR
```

```
04 00103    D55E0007 01          STH,5   *FORMAT,7
04 00104    25400070 A           SLS,4   -16             ISOLATE MONTH
04 00105    2140D5C5 A           CI,4    'NE'            NEVER?
04 00106    6830013F             BE      DDATEV          YES, PRINT THAT
04 00107    22700003 A           LI,7    DTMONTH
04 00108    D54E0007 01          STH,4   *FORMAT,7       SAVE TEXT, RAW FORMAT COMPLETE
04 00109    4B40022C             AND,4   =X'0F0F'        BARE DIGITS
04 0010A    68200154             BLEZ    DDATEX          BAD MONTH
04 0010B    21400100 A           CI,4    X'100'          10,11, OR 12?
04 0010C    6910010E             BL      DDATEM          NO, R4 HAS MONTH BINARY
04 0010D    204FFF0A A           AI,4    X'A'-X'100'
04 0010E    2140000C A   DDATEM  CI,4    12              CHECK AGAIN
04 0010F    69200154             BG      DDATEX          BAD MONTH
                         *
04 00110    204FFFFF N           AI,4    DTMTAB-1        WA(TEXTC(MONTH))
04 00111    22300002 A           LI,3    2               MOST WIDTHS
                         *
04 00112    4BA00001             AND,10  M7F
04 00113    68300154             BEZ     DDATEX          PRINT RAW FORMAT BUFFER
04 00114    21A00005 A           CI,10   5               MINIMUM SIZE
04 00115    6910014F             BL      DDATEQ          TOO SMALL
04 00116    21A00007 A           CI,10   7
04 00117    6910013A             BL      DDATE5
04 00118    21A00009 A           CI,10   9
04 00119    69100137             BL      DDATE7
04 0011A    21A0000E A           CI,10   14
04 0011B    6910012B             BL      DDATE9
                         *
                         * PRINT TIME AND DATE
                         *
04 0011C    21B00002 A           CI,11   OP_LONG         HHMM?
04 0011D    6840012B             BAZ     DDATE9          NO, JUST PRINT DATE
                         **TODO: ADD BLANKS IF WIDTH > 15.
04 0011E    22700000 A           LI,7    DTHOUR*2
04 0011F    4420019E             ANLZ,2  ANLZBX7         2=BA(HH)
04 00120    6A1000A5             BAL,1   PB:TEXT
04 00121    222000F3             LI,2    DTCOLON
04 00122    6A10009F             BAL,1   PB:TEXTC
04 00123    22700002 A           LI,7    DTMIN*2
```

```
04 00124    4420019E            ANLZ,2  ANLZBX7         2=BA(MM)
04 00125    6A1000A5            BAL,1   PB:TEXT
04 00126    222000F6            LI,2    DTSPACE
04 00127    6A10009F            BAL,1   PB:TEXTC
04 00128    21A0000E A          CI,10   14              CAN WE AFFORD ANOTHER SPACE?
04 00129    6820012B            BLE     $+2             NO
04 0012A    6A10009F            BAL,1   PB:TEXTC        YES, PRINT ANOTHER
                          *
                          * PRINT DD-MMM-YY
04 0012B    22700004 A   DDATE9  LI,7    DTDAY*2
04 0012C    4420019E            ANLZ,2  ANLZBX7         2=BA(DD)
04 0012D    6A1000A5            BAL,1   PB:TEXT
04 0012E    222000F4            LI,2    DTDASH
04 0012F    6A10009F            BAL,1   PB:TEXTC
04 00130    32200004 A          LW,2    4               MONTH
04 00131    6A10009F            BAL,1   PB:TEXTC
04 00132    222000F4            LI,2    DTDASH
04 00133    6A10009F            BAL,1   PB:TEXTC
04 00134    22700008 A          LI,7    DTYEAR*2
04 00135    4420019E            ANLZ,2  ANLZBX7         2=BA(YY)
04 00136    680000A7            B       P$:TEXT         PRINT AND RETURN
                          *
                          * PRINT DDMMMYY
04 00137    22700004 A   DDATE7  LI,7    DTDAY*2
04 00138    4420019E            ANLZ,2  ANLZBX7         2=BA(DD)
04 00139    6A1000A5            BAL,1   PB:TEXT
                          * PRINT MMMYY
04 0013A    32200004 A   DDATE5  LW,2    4               MONTH
04 0013B    6A10009F            BAL,1   PB:TEXTC
04 0013C    22700008 A          LI,7    DTYEAR*2
04 0013D    4420019E            ANLZ,2  ANLZBX7         2=BA(YY)
04 0013E    680000A7            B       P$:TEXT         PRINT AND RETURN
                          *
                          *
04 0013F    21A00009 A   DDATEV  CI,10   9
04 00140    68100148            BGE     DDATEV9         PRINT LONGER ONE
04 00141    21A00007 A          CI,10   7
04 00142    68100145            BGE     DDATEV7         MIDDLE ONE
04 00143    222000EC            LI,2    DTNEVER5
```

```
04 00144    680000A1            B       P$:TEXTC
04 00145    20AFFFF9 A   DDATEV7 AI,10   -7              WHAT IS LEFT
04 00146    222000EE            LI,2    DTNEVER7
04 00147    680000A1            B       P$:TEXTC
04 00148    20AFFFF7 A   DDATEV9 AI,10   -9              WHAT IS LEFT
04 00149    6820014D            BLE     DDATENF         NO BLANK FILL
04 0014A    222000F6            LI,2    DTSPACE
04 0014B    6A10009F            BAL,1   PB:TEXTC        PRINT THAT
04 0014C    64A0014B            BDR,10  $-1             PRINT MORE
04 0014D    222000F0     DDATENF LI,2   DTNEVER
04 0014E    680000A1            B       P$:TEXTC        PRINT AND RETURN
                         *
04 0014F    4BA00001     DDATEQ  AND,10  M7F
04 00150    222000F5            LI,2    DTQ
04 00151    6A10009F     DDFILL  BAL,1   PB:TEXTC
04 00152    64A00151            BDR,10  $-1
04 00153    68000229            B       PB:RETURN
                         *
04 00154    32200007 01  DDATEX  LW,2    FORMAT          WA(RAW DATE)
04 00155    25200002 A          SLS,2   2
04 00156    2230000A A          LI,3    10
04 00157    680000A7            B       P$:TEXT         PRINT THAT
```

```
                          ** FORMAT A DOUBLEWORD VALUE IN R4/5 AS DECIMAL
                          ** START BY GENERATING RAW DECIMAL VALUE
                          ******************************** REPLCE WITH CALL TO INT:TOTEXT
04 00158    2270007F A    DDECIMAL LI,7   FORMATW-1
04 00159    04800018 02          M:TRAP  ,(IGNORE,BOTH)
02 00018    14000000 A
02 00019    00000003 A
04 0015A    15800000 01          STD,SR1 PTC
04 0015B    12800004 A           LD,8    4               COPY VALUE UNNECESSARILY
04 0015C    68100163             BGEZ    DDIGIT
04 0015D    21B00010 A           CI,11   OP_UNSIGNED
04 0015E    69400163             BANZ    DDIGIT
04 0015F    7310000B A           MTB,+1  11              REMEMBER NEGATIVE
04 00160    48800005             EOR,8   MALL
04 00161    48900005             EOR,9   MALL
04 00162    1080000A             AD,8    D1              COMPLETE COMPLEMENT
04 00163    21800000 A    DDIGIT CI,8    0               HIGH WORD STILL RELEVANT?
04 00164    68300174             BEZ     DDIGITL         NO, REST IS EASIER
04 00165    89900000 A           PSW,9   *0
04 00166    25800160 A           SLD,8   -32             HIGH ORDER TO 9
04 00167    36800006             DW,8    I10
04 00168    B2600000 A           LW,6    *0              GET TOP OF STACK
04 00169    C6900006 A           XW,9    *6              R8-9 = (BORROW, LOW ORDER)
                          **
                          ** Division by 10 here can overflow, because the quotient is signed,
                          ** So we'll divide by 20 and then multiply by 2, fixing up the
                          ** remainder after
                          **
04 0016A    36800007             DW,8    I20             REMAINDER(0-19) IN R8
04 0016B    25900001 A           SLS,9   1               MULTIPLY QUOTIENT BY 2
04 0016C    2180000A A           CI,8    10              CHECK REMAINDER
04 0016D    69100170             BL      GODIGIT         OK, USE IT
04 0016E    20900001 A           AI,9    1               ONE MORE FOR THE QUOTIENT
04 0016F    208FFFF6 A           AI,8    -10             CORRECT REMAINER
04 00170    208000F0 A    GODIGIT AI,8   '0'             GET A CHARACTER VALUE
04 00171    F58E0007 01          STB,8   *FORMAT,7
04 00172    88800000 A           PLW,8   *0              RETRIEVE HIGH QUOTIENT
04 00173    64700163             BDR,7   DDIGIT
                          *
```

```
04 00174    21900000 A   DDIGITL CI,9    0               DONE YET?
04 00175    6830017B             BEZ     DDECLX          YES/ GO FINISH
04 00176    22800000 A           LI,8    0
04 00177    36800006             DW,8    I10             DIGIT TO R8, REST TO R9
04 00178    208000F0 A           AI,8    '0'             GET CHAR VALUE
04 00179    F58E0007 01          STB,8   *FORMAT,7
04 0017A    64700174             BDR,7   DDIGITL
      04 0017B           DDECLX  EQU     $
04 0017B    228000F0 A           LI,8    '0'
04 0017C    2260007F A           LI,6    FORMATW-1
04 0017D    38600007 A           SW,6    7               NUMBER OF DIGITS
04 0017E    69200182             BGZ     DDECNZ          MORE THAN ZERO
04 0017F    F58E0007 01          STB,8   *FORMAT,7       PUT IN A ZERO
04 00180    207FFFFF A           AI,7    -1
04 00181    20600001 A           AI,6    +1
      04 00182           DDECNZ  EQU     $
04 00182    21A00F80 A           CI,10   X'F80'          ANY FRACTION?
04 00183    684001A4             BAZ     DDECALGN        NO
04 00184    3290000A A           LW,9    10
04 00185    25900079 A           SLS,9   -7              GET FRACTION DIGITS
04 00186    31600009 A   DDECCF  CW,6    9               ARE THERE ENOUGH ?
04 00187    69200199             BG      DDECIF          MORE THAN ENOUGH, DECIMAL MUST BE INSERTED
04 00188    6830018C             BE      DDECGF
                         ** ADD LEADING ZEROS SO THERE ARE ENOUGH FOR THE FRACTION.
04 00189    F58E0007 01          STB,8   *FORMAT,7
04 0018A    20600001 A           AI,6    +1
04 0018B    64700186             BDR,7   DDECCF
      04 0018C           DDECGF  EQU     $
04 0018C    2280004B A           LI,8    '.'
04 0018D    F58E0007 01          STB,8   *FORMAT,7
04 0018E    20600001 A           AI,6    +1
04 0018F    207FFFFF A           AI,7    -1
04 00190    4BA00001             AND,10  M7F             DESTINATION LENGTH
04 00191    68300194             BEZ     DDECAZ          VARIABLE
04 00192    3160000A A           CW,6    10              CAN WE TOLERATE A LEADING ZERO
04 00193    681001A4             BGE     DDECALGN        B/NO, WILL START WITH DECIMAL
04 00194    228000F0 A   DDECAZ  LI,8    '0'
04 00195    F58E0007 01          STB,8   *FORMAT,7
04 00196    20600001 A           AI,6    +1
```

```
04 00197    207FFFFF A          AI,7    -1
04 00198    680001A4            B       DDECALGN        READY FOR ALIGNMENT
                          ** INSERT A DECIMAL
04 00199    32200006 A   DDECIF  LW,2    6               TOTAL DIGITS
04 0019A    38200009 A          SW,2    9               NUMBER TO MOVE
04 0019B    30700002 A          AW,7    2
04 0019C    20200001 A          AI,2    +1              INCLUDE DECIMAL
04 0019D    2280004B A          LI,8    '.'
      04 0019E          ANLZBX7 EQU     $               NB/ REFERENCED BY ANLZ INST.
04 0019E    F23E0007 01  DDECIN  LB,3    *FORMAT,7
04 0019F    F58E0007 01         STB,8   *FORMAT,7
04 001A0    32800003 A          LW,8    3
04 001A1    207FFFFF A          AI,7    -1
04 001A2    6420019E            BDR,2   DDECIN
04 001A3    20600001 A          AI,6    +1
                          ** ALIGN A DECIMAL STRING
      04 001A4          DDECALGN EQU    $
04 001A4    22800060 A          LI,8    '-'
                          ** IF NEGATIVE INSERT SIGN NOW.
04 001A5    7300000B A          MTB,0   11              IS IT?
04 001A6    693001AA            BNEZ    DDECNEG         YES
04 001A7    21B00004 A          CI,11   OP_LEADSIGN     PRINT SIGN ANYWAY?
04 001A8    684001AD            BAZ     DDECLEN         NO
04 001A9    2280004E A          LI,8    '+'             ADJUST SIGN
04 001AA    F58E0007 01  DDECNEG STB,8   *FORMAT,7
04 001AB    20600001 A          AI,6    +1
04 001AC    207FFFFF A          AI,7    -1
      04 001AD          DDECLEN EQU     $
04 001AD    4BA00001            AND,10  M7F
04 001AE    683001D0            BEZ     DDECMOV         DONE, MOVE TO LINE BUFFER
04 001AF    31A00006 A          CW,10   6
04 001B0    691001C9            BL      DDTRUNC         OOPS, TOO BIG
04 001B1    683001D0            BE      DDECMOV         JUST RIGHT
                          ** PREPEND BLANKS OR ZEROS
04 001B2    38A00006 A          SW,10   6               NUMBER TO FILL
04 001B3    21B00008 A          CI,11   OP_LEADZEROS
04 001B4    694001BB            BANZ    DDECZ
04 001B5    22800040 A          LI,8    ' '
04 001B6    F58E0007 01  DDECFIB STB,8   *FORMAT,7
```

```
04 001B7    20600001 A          AI,6    +1
04 001B8    207FFFFF A          AI,7    -1
04 001B9    64A001B6            BDR,10  DDECFIB
04 001BA    680001D0            B       DDECMOV         MOVE IT
                          ** PREPEND LEADING ZEROS
04 001BB    7300000B A  DDECZ   MTB,0   11              NEGACTIVE?
04 001BC    683001BE            BEZ     DDECZP          NO
04 001BD    20700001 A          AI,7    1               START FILL OVER SIGN
04 001BE    228000F0 A  DDECZP  LI,8    '0'
04 001BF    F58E0007 01 DDECFI0 STB,8   *FORMAT,7
04 001C0    20600001 A          AI,6    +1
04 001C1    207FFFFF A          AI,7    -1
04 001C2    64A001BF            BDR,10  DDECFI0
04 001C3    7300000B A          MTB,0   11              NEGATIVE
04 001C4    683001D0            BEZ     DDECMOV         NO DONE
04 001C5    22800060 A          LI,8    '-'
04 001C6    F58E0007 01         STB,8   *FORMAT,7       STICK IN FRONT
04 001C7    207FFFFF A          AI,7    -1
04 001C8    680001D0            B       DDECMOV         GO MOVE IT
                          **
                          ** COME HERE WHEN FORMATTED NUMBER IS TOO BIG
        04 001C9          DDTRUNC EQU     $
04 001C9    2270007F A          LI,7    FORMATW-1
04 001CA    4BA00001            AND,10  M7F
04 001CB    3260000A A          LW,6    10
04 001CC    2280005C A          LI,8    '*'             MY GOD IT'S FULL OF STARS
04 001CD    F58E0007 01 DDECFIS STB,8   *FORMAT,7
04 001CE    207FFFFF A          AI,7    -1
04 001CF    64A001CD            BDR,10  DDECFIS
                          **
                          ** FINALLY, WE HAVE A STRING
04 001D0    32300006 A  DDECMOV LW,3    6               LENGTH
04 001D1    20700001 A  MVFMT   AI,7    +1              INDEX FIRST BYTE
04 001D2    4420019E            ANLZ,2  ANLZBX7         GET ADDRESS OF STRING
04 001D3    0480001A 02         M:TRAP  (RESTORE,PTC)
02 0001A    14000000 01
02 0001B    80000000 A
04 001D4    680000A7            B       P$:TEXT
```

```
                          **
                          ** FORMAT A DOUBLEWORD VALUE IN R4/5 AS HEX
                          **
04 001D5    2270007F A    DHEX    LI,7    FORMATW-1
04 001D6    4BA00001              AND,10  M7F             GET LENGTH
04 001D7    2280000F A    DHEXH   LI,8    X'F'
04 001D8    4B800005 A            AND,8   5               GET LOW ORDER HEXIT
04 001D9    2180000A A            CI,8    X'A'
04 001DA    681001DC              BGE     $+2
04 001DB    20800039 A            AI,8    '0'-('A'-X'A')
04 001DC    208000B7 A            AI,8    'A'-X'A'
04 001DD    F58E0007 01           STB,8   *FORMAT,7
04 001DE    207FFFFF A            AI,7    -1
04 001DF    2540017C A            SLD,4   -4              SLIDE TO NEXT HEXIT
04 001E0    11400008              CD,4    D0              IS IT ZERO NOW?
04 001E1    683001E6              BEZ     DHEX0           DONE
04 001E2    20A00000 A            AI,10   0               WAS FORMAT WIDTH SPECIFIED?
04 001E3    683001D7              BEZ     DHEXH           NO GO DO NEXT
04 001E4    64A001D7              BDR,10  DHEXH
04 001E5    680001EC              B       DHEXMOV         DONE, MOVE TO BUFFER
04 001E6    20AFFFFF A    DHEX0   AI,10   -1
04 001E7    682001EC              BLEZ    DHEXMOV
04 001E8    228000F0 A            LI,8    '0'
04 001E9    F58E0007 01   DHEXF   STB,8   *FORMAT,7
04 001EA    207FFFFF A            AI,7    -1
04 001EB    64A001E9              BDR,10  DHEXF
04 001EC    2230007F A    DHEXMOV LI,3    FORMATW-1
04 001ED    38300007 A            SW,3    7               LENGTH
04 001EE    680001D1              B       MVFMT
```

```
                          **
                          ** FORMAT A DOUBLEWORD VALUE IN R4/5 AS FLOAT
                          ** R10: FORMAT LENGTHS, R11: OPTIONS (SHIFTED)
04 001EF    89E00000 A    DFLOAT  PSW,14  *0
04 001F0    89F00000 A            PSW,15  *0
                          **
04 001F1    12200004 A            LD,2    4
04 001F2    6A100000 X            BAL,1   FP:INTERPRET
                          ** R4: SIGN, R5: EXP (POWER OF 10), R14/15: PACKED DECIMAL
                          ** BUILD EXPONENT FIRST
04 001F3    2270007C A            LI,7    FORMATW-4
04 001F4    228000C5 A            LI,8    'E'
04 001F5    F58E0007 01           STB,8   *FORMAT,7
04 001F6    20700001 A            AI,7    +1
04 001F7    2280004E A            LI,8    '+'             EXPONENT SIGN
04 001F8    32300005 A            LW,3    5               COPY EXPONENT
04 001F9    681001FC              BGEZ    DFPPE           B/IT IS POSITIVE
04 001FA    3A300005 A            LCW,3   5               NEGATE IT
04 001FB    22800060 A            LI,8    '-'             SIGN
04 001FC    F58E0007 01   DFPPE   STB,8   *FORMAT,7
04 001FD    20700001 A            AI,7    +1
04 001FE    22200000 A            LI,2    0
04 001FF    36200006              DW,2    I10             GET EXPONENT DIGITS
04 00200    49300002              OR,3    MF0             MAKE EBCIDIC DIGIT
04 00201    F53E0007 01           STB,3   *FORMAT,7
04 00202    20700001 A            AI,7    +1
04 00203    49200002              OR,2    MF0             MAKE EBCIDIC DIGIT
04 00204    F52E0007 01           STB,2   *FORMAT,7
                          ** NOW DO FRACTION PART
                          ** REMOVE TRAILING ZEROES FROM FRACTION
                          ** TODO
04 00205    2270007B A            LI,7    FORMATW-5
04 00206    2260000F A            LI,6    15              PROCESS 15 DIGITS OF FRACTION
04 00207    2280000F A    DFPDIG  LI,8    X'F'
04 00208    4B80000F A            AND,8   15              GET LOW DIGIT
04 00209    25E0037C A            SCD,14  -4              PREP FOR NEXT TIME
04 0020A    49800002              OR,8    MF0             MAKE EBCDIC DIGIT
04 0020B    F58E0007 01           STB,8   *FORMAT,7       SET DIGIT
04 0020C    207FFFFF A            AI,7    -1              POSITION NEXT
```

```
04 0020D    68200224            BLEZ    DFPSTAR         **BUG
04 0020E    64600207            BDR,6   DFPDIG          DO NEXT DIGIT
                          **
04 0020F    2280004B A          LI,8    '.'
04 00210    F58E0007 01         STB,8   *FORMAT,7
04 00211    207FFFFF A          AI,7    -1              POSITION NEXT
04 00212    68200224            BLEZ    DFPSTAR         **BUG
04 00213    2280000F A          LI,8    X'F'
04 00214    4B80000F A          AND,8   15              GET NON FRACTION
04 00215    49800002            OR,8    MF0             EBCDIC
04 00216    F58E0007 01         STB,8   *FORMAT,7       SET
04 00217    207FFFFF A          AI,7    -1
04 00218    69100224            BLZ     DFPSTAR         ** BUG
                          **
04 00219    22800060 A          LI,8    '-'
04 0021A    20400000 A          AI,4    0               SIGNED?
04 0021B    6910021F            BL      DFPSIGN         -
04 0021C    21B00004 A          CI,11   OP_LEADSIGN     PRINT ANYWAY?
04 0021D    68400221            BAZ     DFPNS           B/NO
04 0021E    208FFFEE A          AI,8    '+'-'-'         '+'
04 0021F    F58E0007 01  DFPSIGN STB,8  *FORMAT,7       SET SIGN
04 00220    207FFFFF A          AI,7    -1
04 00221    88F00000 A   DFPNS  PLW,15  *0
04 00222    88E00000 A          PLW,14  *0
04 00223    680001EC            B       DHEXMOV
                          **
                          ** PRINT WHOLE FIELD OF STARS WHEN ERROR
                          **
04 00224    88F00000 A   DFPSTAR PLW,15 *0
04 00225    88E00000 A          PLW,14  *0
04 00226    88300000 A          PLW,3   *0
04 00227    680001C9            B       DDTRUNC         GO PRINT *'S
                          **
                          ** LATER
                          **
04 00228    680001D5     DTIME  B       DHEX
                          *
                          *
        04 00229          PB:RETURN EQU  $
```

```
    775   04 00229    022000B0 A          LCI     11
    776   04 0022A    8A100000 A          PLM,1   *0
    777   04 0022B    68020000 A          B       0,1
    778                           *
    779                           *
    780                                   END
          04 0022C    00000F0F A
```

CONTROL SECTION SUMMARY: 01 0000B PT 0 02 0001C PT 0 03 00000 PT 1 04 0022D PT 1

```
*    SYMBOL VALUES
        ADDBYTE*04 000AE         ADDFULL*04 000BA         ADDPAD*04 00091          ADDSEG*04 000AD
        ANLZBX7*04 0019E         BLANKS*04 00000          BUFFER*01 00002          BUFLEN*01 00004
        BUFMAX*01 00003          BUFX*01 00005            DCHAR*04 000DE           DCHAR1*04 000E6
        DCHAR2*04 000E5          DDATE*04 000F7           DDATEL*04 000FA          DDATEM*04 0010E
        DDATENF*04 0014D         DDATEQ*04 0014F          DDATEV*04 0013F          DDATEV7*04 00145
        DDATEV9*04 00148         DDATEX*04 00154          DDATE5*04 0013A          DDATE7*04 00137
        DDATE9*04 0012B          DDECALGN*04 001A4        DDECAZ*04 00194          DDECCF*04 00186
        DDECFIB*04 001B6         DDECFIS*04 001CD         DDECFI0*04 001BF         DDECGF*04 0018C
        DDECIF*04 00199          DDECIMAL*04 00158        DDECIN*04 0019E          DDECLEN*04 001AD
        DDECLX*04 0017B          DDECMOV*04 001D0         DDECNEG*04 001AA         DDECNZ*04 00182
        DDECZ*04 001BB           DDECZP*04 001BE          DDFILL*04 00151          DDIGIT*04 00163
        DDIGITL*04 00174         DDTRUNC*04 001C9         DFLOAT*04 001EF          DFPDIG*04 00207
        DFPNS*04 00221           DFPPE*04 001FC           DFPSIGN*04 0021F         DFPSTAR*04 00224
        DHEX*04 001D5            DHEXF*04 001E9           DHEXH*04 001D7           DHEXMOV*04 001EC
        DHEX0*04 001E6           DTCOLON*04 000F3         DTDASH*04 000F4          DTDAY*00000002
        DTHOUR*00000000          DTIME*04 00228           DTMIDNGT*04 000EB        DTMIN*00000001
        DTMONTH*00000003         DTMTAB*EXT               DTNEVER*04 000F0         DTNEVER5*04 000EC
        DTNEVER7*04 000EE        DTQ*04 000F5             DTSPACE*04 000F6         DTYEAR*00000004
        DWORDD*04 000DE          DWORDV*04 000D7          DWORD1*04 000CE          D0*04 00008
        D1*04 0000A              FORMAT*01 00007          FORMATW*00000080         GODIGIT*04 00170
        INDENT*01 00006          INITD*04 0003A           INITF*04 00038           I10*04 00006
        I20*04 00007             MALL*04 00005            MFF*04 00003             MFFFFFF*04 00004
        MF0*04 00002             MVFMT*04 001D1           M7F*04 00001             OP_LEADSIGN*00000004
        OP_LEADZEROS*00000008    OP_LONG*00000002         OP_PRINTABLE*00000020    OP_SHORT*00000001
        OP_UNSIGNED*00000010     OT_DATEONLY*00000004     OT_MILLISECS*00000020    OT_MM*00000010
        OT_NOSECS*00000010       OT_TIMEONLY*00000008     OT_YMD*00000040          P$:TEXT*04 000A7
        P$:TEXTC*04 000A1        PAD*04 0008F             PB:CHAR-04 000BF         PB:CLEAR-04 00079
        PB:DONE-04 0003F         PB:DWORD-04 000CC        PB:FLUSH-04 00076        PB:GETBUF-04 0007D
        PB:INIT-04 00022         PB:RETURN*04 00229       PB:SETPOS-04 00080       PB:TABPOS-04 00085
        PB:TEXT-04 000A5         PB:TEXTC-04 0009F        PB:TEXT8-04 00099        PB:WORD-04 000C5
        PB:WRAP-04 00046         PD:DCB-01 00008          PD:LINEWIDTH-01 00009
        PD:TERMWIDTH*01 0000A    PI:PAGE-04 0001A         PI:SPACE-04 0001C        PI:TEXT-04 00011
        PI:TEXTC-04 0000C        PI:WRITE*04 00016        PTC*01 00000             SR1*00000008
        SR2*00000009             SR3*0000000A             SR4*0000000B             TABTO*04 0008B
        WRAPB*04 00057           WRAPFI*04 0005B          WRAPFLUSH*04 00074       WRAPLN*04 0004F
        WRAPMV*04 0006B          WRAPMVB*04 0006C         WRAPNI*04 0005E          WRAPNXT*04 00051
        WRAPP*04 00053           WRAPSKP*04 00065
*    EXTERNAL SYMBOLS
```

```
        FP:INTERPRET-REF              INT:FROMTEXT-REF              M:LO-REF       M:UC-REF
        PB:CHAR-DEF    PB:CLEAR-DEF   PB:DONE-DEF    PB:DWORD-DEF   PB:FLUSH-DEF   PB:GETBUF-DEF
        PB:INIT-DEF    PB:SETPOS-DEF  PB:TABPOS-DEF  PB:TEXT-DEF    PB:TEXTC-DEF   PB:TEXT8-DEF
        PB:WORD-DEF    PB:WRAP-DEF    PD:DCB*DEF     PD:LINEWIDTH-DEF              PI:PAGE-DEF
        PI:SPACE-DEF   PI:TEXT-DEF    PI:TEXTC-DEF   UT:MONTH3-REF
* NO UNDEFINED SYMBOLS
*    ERROR SEVERITY LEVEL: 0
* NO ERROR LINES
```

```
SET M:SI/S:MATCH
SET M:BO/R:MATCH
AP SI,SD,BO,LO
```

```
                                *
                                * PATTERN MATCHER
                                       DEF     MATCH
                                *
                                * R0: WA(STACK POINTER DW)
                                * R1: LINK REGISTER
                                * R2: LENGTH AND INPUT BYTE ADDRESS (STRING TO BE TESTED)
                                * R3: OPTIONS AND WORD ADDRESS OF PATTERN(TEXTC)
                                *     OPTIONS:
         00000001               MO:CASE EQU      X'01'       CASE SENSITIVE.
                                *
                                *
                                * RETURNS CC3 - 1: INPUT CHARACTERS REMAIN
                                *         CC4 - 1: PATTERN CHARACTERS REMAIN
                                *         R2: UPDATED LENGTH AND BYTE ADDRESS OF INPUT.
                                *         R0: HIGH BYTE SET TO PATTERN BYTE INDEX IF NOT MATCHED
                                **
                                * THE PATTERN CONSISTS OF A TEXTC STRING CONTAINING CHARACTERS TO MATCH
                                * AND SPECIAL CHARACTERS WHICH ARE:
                                *         \     ESCAPE, MATCH NEXT CHARACTER VERBATIM
                                *         *     MATCH ANY NUMBER OF CHARACTERS
                                *         ?     MATCH A SINGLE CHARACTER
                                *         +     MATCH A NUMERIC CHARACTER
                                *         !     MUST NOT MATCH FOLLOWING PATTERN
                                *
                                *
   02 00000                            CSECT   1
         00000000               SPD     EQU     0
         00000001               NOT     EQU     1               INVERT RESULT
         00000002               RI      EQU     2               INPUT
         00000003               RP      EQU     3               PATTERN
         00000005               PX      EQU     5               PATTERN INDEX
         0000000B               NR      EQU     11
                                *
   02 00000    FFFFFFFF A       MALL    DATA    X'FFFFFFFF'
   02 00001    01000000 A       MCASE   GEN,8,24 MO:CASE,0
                                *
         02 00002               MATCH   EQU     $
   02 00002    022000B0 A               LCI     NR              SAVE R1-NR
```

```
02 00003    8B100000 A          PSM,1   *SPD
                            *
02 00004    22500001 A          LI,PX   1               INITIALIZE PATTERN INDEX
02 00005    22800000 A          LI,8    0               INITIALIZE RESTART COUNT
02 00006    22100000 A          LI,NOT  0               NOT A NOT PATTERN
                            *
02 00007    F1500003 A  MCHAR   CB,PX   *RP             ANY PATTERN LEFT?
02 00008    6920003A            BG      NOPAT           NO
02 00009    22900000 A          LI,9    0               NOT A PATTERN LITERAL
02 0000A    F26A0003 A          LB,6    *RP,PX          GET CURRENT PATTERN CHAR
02 0000B    20500001 A          AI,PX   1
02 0000C    2160005C A          CI,6    '*'
02 0000D    68300034            BE      MSTAR           GO SET UP * MATCH
                            *
02 0000E    2160005A A          CI,6    '!'             NOT?
02 0000F    69300012            BNE     MCHARL          NO, PROCEED
02 00010    48100000            EOR,NOT MALL
02 00011    68000007            B       MCHAR           CONTINUE
                            *
02 00012    216000B1 A  MCHARL  CI,6    '\'             ESCAPE?
02 00013    69300017            BNE     MCHAR1          NO
02 00014    F26A0003 A          LB,6    *RP,PX          GET ESCAPED CHAR
02 00015    20500001 A          AI,PX   1
02 00016    22900001 A          LI,9    1               SET LITERAL FLAG
                            *
02 00017    73000002 A  MCHAR1  MTB,0   RI              ANY TEST STRING LEFT?
02 00018    6830003C            BEZ     FAIL            NO
02 00019    72740000 A          LB,7    0,RI            GET CURRENT TEST CHAR
02 0001A    73F00002 A          MTB,-1  RI              DECREMENT LENGTH
02 0001B    20200001 A          AI,RI   1               BUMP POINTER
                            *
02 0001C    20900000 A          AI,9    0               WAS IT A LITERAL?
02 0001D    69300027            BNE     MLITERAL        YES
                            *
02 0001E    2160006F A          CI,6    '?'             ANY CHAR?
02 0001F    68300007            BE      MCHAR           OK, NEXT
                            *
02 00020    2160004E A          CI,6    '+'             NUMERIC?
02 00021    69300027            BNE     MCHAR2          NO
```

```
02 00022    217000F0 A          CI,7    '0'
02 00023    6910003C            BL      FAIL
02 00024    217000F9 A          CI,7    '9'
02 00025    6920003C            BG      FAIL
02 00026    68000007            B       MCHAR           OK, NEXT
                            *
        02 00027            MCHAR2  EQU     $
        02 00027            MLITERAL EQU    $
02 00027    31600007 A          CW,6    7
02 00028    68300007            BE      MCHAR           OK, NEXT
02 00029    32400001            LW,4    MCASE           MASK FOR CASE SENSITIVE FLAG
02 0002A    4B400003 A          AND,4   3
02 0002B    6930003C            BNEZ    FAIL            IS CASE SENSITIVE, NO MATCH
02 0002C    21700081 A          CI,7    'a'             CAN THIS BE UPSHIFTED?
02 0002D    6910003C            BL      FAIL            NO
02 0002E    217000A9 A          CI,7    'z'
02 0002F    6920003C            BG      FAIL            NO
02 00030    20700040 A          AI,7    X'40'           UPSHIFT
02 00031    31600007 A          CW,6    7               CHECK AGAIN
02 00032    68300007            BE      MCHAR           OK MATCH
02 00033    6800003C            B       FAIL
                            *
02 00034    F1500003 A  MSTAR   CB,PX   *RP              ANY MORE PATTERN?
02 00035    6922005B            BG      MATCHED,NOT     NO, * MATCHES REST OF INPUT.
                            *
                            * TRY MATCHING ZERO CHARACTERS
                            *
02 00036    89200000 A  RSTAR   PSW,RI  *SPD
02 00037    89500000 A          PSW,PX  *SPD
02 00038    20800001 A          AI,8    1
02 00039    68000007            B       MCHAR
                            *
02 0003A    73000002 A  NOPAT   MTB,0   RI               NO PATTERN LEFT, ANY INPUT?
02 0003B    6832005B            BEZ     MATCHED,NOT     NO, IT MATCHED
                            *
02 0003C    21800000 A  FAIL    CI,8    0
02 0003D    68320047            BE      NOMATCH,NOT     OUT OF OPTIONS
                            *
02 0003E    88500000 A          PLW,PX  *SPD
```

```
02 0003F    88200000 A          PLW,RI  *SPD
02 00040    208FFFFF A          AI,8    -1
                         *
02 00041    73000002 A          MTB,0   RI              CAN * MATCH MORE?
02 00042    6830003C            BEZ     FAIL            NO, SEE IF PREVIOUS *
                         *
                         * MATCH AN ADDITIONAL CHARACTER
02 00043    20200001 A          AI,RI   1
02 00044    73F00002 A          MTB,-1  RI
02 00045    68000036            B       RSTAR
                         *
02 00046    6800005B            B       MATCHED
      02 00047           NOMATCH EQU  $
02 00047    B2700000 A          LW,7    *SPD
02 00048    352FFFF7 A          STW,RI  RI-NR,7
02 00049    75500000 A          STB,PX  0
                         *
02 0004A    022000A0 A          LCI     NR-1
02 0004B    8A200000 A          PLM,2   *SPD
                         *
02 0004C    73000002 A          MTB,0   RI
02 0004D    68300057            BEZ     NMNOINPUT
02 0004E    72100000 A          LB,1    0
02 0004F    F1100003 A          CB,1    *RP
02 00050    69200054            BG      NMNOPAT
                         *
02 00051    88100000 A          PLW,1   *SPD
02 00052    02200030 A          LCI     3               BOTH INPUT AND PATTERN LEFT
02 00053    68020000 A          B       0,1
                         *
      02 00054           NMNOPAT EQU     $
02 00054    88100000 A          PLW,1   *SPD
02 00055    02200020 A          LCI     2               MUST BE INPUT LEFT
02 00056    68020000 A          B       0,1
                         *
      02 00057           NMNOINPUT EQU   $
02 00057    88100000 A          PLW,1   *SPD
02 00058    02200010 A          LCI     1               MUST BE PATTERN LEFT
02 00059    68020000 A          B       0,1
```

```
                                *
02 0005A    68000047            B       NOMATCH
        02 0005B           MATCHED EQU     $
02 0005B    21800000 A          CI,8    0
02 0005C    68200061            BLE     MATCHR
02 0005D    88500000 A          PLW,PX  *SPD
02 0005E    88500000 A          PLW,PX  *SPD
02 0005F    208FFFFF A          AI,8    -1
02 00060    6800005B            B       MATCHED
                                *
02 00061    B2700000 A  MATCHR  LW,7    *SPD
02 00062    352FFFF7 A          STW,RI  RI-NR,7
                                *
02 00063    022000B0 A          LCI     NR
02 00064    8A100000 A          PLM,1   *SPD
                                *
02 00065    02200000 A          LCI     0
02 00066    68020000 A          B       0,1
                                *
                                        END
```

CONTROL SECTION SUMMARY: 01 00000 PT 0 02 00067 PT 1

```
*    SYMBOL VALUES
        FAIL*02 0003C           MALL*02 00000           MATCH-02 00002          MATCHED*02 0005B
        MATCHR*02 00061         MCASE*02 00001          MCHAR*02 00007          MCHARL*02 00012
        MCHAR1*02 00017         MCHAR2*02 00027         MLITERAL*02 00027       MO:CASE*00000001
        MSTAR*02 00034          NMNOINPUT*02 00057      NMNOPAT*02 00054        NOMATCH*02 00047
        NOPAT*02 0003A          NOT*00000001            NR*0000000B             PX*00000005
        RI*00000002             RP*00000003             RSTAR*02 00036          SPD*00000000
*    EXTERNAL SYMBOLS
        MATCH*DEF
* NO UNDEFINED SYMBOLS
*    ERROR SEVERITY LEVEL: 0
* NO ERROR LINES
```

```
SET M:SI/S:UTILS
SET M:BO/R:UTILS
AP SI,SD,BO,LO
```

```
                                TITLE   'S:UTILS - UTILITY ROUTINES FOR NUMBERS, DATES, TIMES'
                                PCC     0
                        **
                        ** UTILITY ROUTINES
                        **
                                SYSTEM  BPM
                                SYSTEM  SIG7FDP
                                SYSTEM  SYSUTILS
                        **
                                DEF     UT:MONTH3       NAMES OF MONTHS (3 CHARACTERS)
                                DEF     UT:MONTHNAME    FULL NAMES OF MONTHS
                                DEF     UT:DAYNAME      FULL NAMES OF DAYS
                        **
                                DEF     FP:POW10        RETURNS 10^N
                                DEF     FP:INTERPRET    DECOMPOSE FLOAT INTO COMPONENTS
                                DEF     FP:COMPUTE      COMPUTE FLOAT FROM COMPONENTS
                                DEF     FP:FROMTEXT     CONVERT TEXT TO FLOAT DOUBLE
                        **
                                DEF     INT:FROMTEXT    CONVERT TEXT TO INTEGER
                                DEF     INT:TOTEXT      CONVERT 64-BIT INTEGER TO TEXT
                                DEF     INT32:TOTEXT    CONVERT 32-BIT INTEGER TO TEXT
                        **
                                DEF     JD:DOW          GET DAY OF WEEK
                                DEF     JD:INTERPRET    COMPUTE DATE COMPONENTS FROM JD
                                DEF     JD:COMPUTE      COMPUTE JULIAN DAY
                                DEF     JD:FROMTEXT     CONVERT TEXT TO JULIAN DAY
                                DEF     JD:FROMCPV      CONVERT CPV DATE TO JULIAN DAY
                                DEF     JD:TOTEXT       GENERATE DATE PART OF ISO STRING YYYY-MM-DD
                        **
                                DEF     DT:NOW          GET CURRENT DATE AND TIME
                                DEF     TS:NOW          GET TS FOR CURRENT DATE/TIME
                                DEF     TS:INTERPRET    COMPUTE DATETIME COMPONENTS FROM TS
                                DEF     TS:COMPUTE      COMPUTE TIMESTAMP FROM COMPONENTS
                                DEF     TS:COMPOSE      GET TIMESTAMP FROM JD AND MS OF DAY
                                DEF     TS:FROMTEXT     CONVERT TEXT TO TIMESTAMP
                                DEF     TS:TOTEXT       CONVERT TIMESTAMP TO VERBOSE TEXT
                        **      DEF     TS:ISO8601      GENERATE ISO STRING FOR TS
                        **
                                DEF     MSPERDAY
```

```
                                  DEF     MSPERHOUR
                          **
                          **      DEF     UID:GET         GET A UNIQUE ID
```

```
04 00000                UT:DATA DSECT   0
                        ** SHOULD BE DYNAMICALLY ALLOCATED? OR PRE-USE DESTINATION BUFFER?
      00000010          BUFSIZE EQU     16
04 00000                BUFFER  RES     BUFSIZE
      04 00000          TIMEBUF EQU     BUFFER
                        **
      00000040          FORMATW EQU     BUFSIZE*4       MAX BYTES FOR FORMATTED NUMBER
      04 00000          FORMAT  EQU     BUFFER          BUFFER FOR INT:TOTEXT
04 00010    0000004B A  FCHARD  DATA    '.'             DECIMAL CHARACTER
04 00011    0000006B A  FCHART  DATA    ','             THOUSANDS SEPARATOR
                        ** END OF VARIABLE DATA
                        **



                        **
                        ** CONSTANTS
                        **
05 00000                S:UTILS CSECT   1
      05 00000          UT:MONTH3 EQU   $
05 00000    03D1C1D5 A          TEXTC   'JAN'
05 00001    03C6C5C2 A          TEXTC   'FEB'
05 00002    03D4C1D9 A          TEXTC   'MAR'
05 00003    03C1D7D9 A          TEXTC   'APR'
05 00004    03D4C1E8 A          TEXTC   'MAY'
05 00005    03D1E4D5 A          TEXTC   'JUN'
05 00006    03D1E4D3 A          TEXTC   'JUL'
05 00007    03C1E4C7 A          TEXTC   'AUG'
05 00008    03E2C5D7 A          TEXTC   'SEP'
05 00009    03D6C3E3 A          TEXTC   'OCT'
05 0000A    03D5D6E5 A          TEXTC   'NOV'
05 0000B    03C4C5C3 A          TEXTC   'DEC'
                        **
05 0000C    07D18195 A  MNJAN   TEXTC   'January'
05 0000D    A48199A8 A
05 0000E    08C68582 A  MNFEB   TEXTC   'February'
05 0000F    99A48199 A
05 00010    A8404040 A
05 00011    05D48199 A  MNMAR   TEXTC   'March'
```

```
          05 00012    83884040 A
     78   05 00013    05C19799 A   MNAPR   TEXTC   'April'
          05 00014    89934040 A
     79   05 00015    03D481A8 A   MNMAY   TEXTC   'May'
     80   05 00016    04D1A495 A   MNJUN   TEXTC   'June'
          05 00017    85404040 A
     81   05 00018    04D1A493 A   MNJUL   TEXTC   'July'
          05 00019    A8404040 A
     82   05 0001A    06C1A487 A   MNAUG   TEXTC   'August'
          05 0001B    A4A2A340 A
     83   05 0001C    09E28597 A   MNSEP   TEXTC   'September'
          05 0001D    A3859482 A
          05 0001E    85994040 A
     84   05 0001F    07D683A3 A   MNOCT   TEXTC   'October'
          05 00020    96828599 A
     85   05 00021    08D596A5 A   MNNOV   TEXTC   'November'
          05 00022    85948285 A
          05 00023    99404040 A
     86   05 00024    08C48583 A   MNDEC   TEXTC   'December'
          05 00025    85948285 A
          05 00026    99404040 A
     87                           **
     88   05 00027    0000000C     UT:MONTHNAME DATA MNJAN,MNFEB,MNMAR,MNAPR,MNMAY,MNJUN
          05 00028    0000000E
          05 00029    00000011
          05 0002A    00000013
          05 0002B    00000015
          05 0002C    00000016
     89   05 0002D    00000018             DATA    MNJUL,MNAUG,MNSEP,MNOCT,MNNOV,MNDEC
          05 0002E    0000001A
          05 0002F    0000001C
          05 00030    0000001F
          05 00031    00000021
          05 00032    00000024
     90                           **
     91   05 00033    06E2A495 A   DNSUN   TEXTC   'Sunday'
          05 00034    8481A840 A
     92   05 00035    06D49695 A   DNMON   TEXTC   'Monday'
          05 00036    8481A840 A
```

```
     93   05 00037    07E3A485 A   DNTUE   TEXTC   'Tuesday'
          05 00038    A28481A8 A
     94   05 00039    09E68584 A   DNWED   TEXTC   'Wednesday'
          05 0003A    9585A284 A
          05 0003B    81A84040 A
     95   05 0003C    08E388A4 A   DNTHU   TEXTC   'Thursday'
          05 0003D    99A28481 A
          05 0003E    A8404040 A
     96   05 0003F    06C69989 A   DNFRI   TEXTC   'Friday'
          05 00040    8481A840 A
     97   05 00041    08E281A3 A   DNSAT   TEXTC   'Saturday'
          05 00042    A4998481 A
          05 00043    A8404040 A
     98   05 00044    11C9D5E5 A   DNXXX   TEXTC   'INVALID TIMESTAMP'
          05 00045    C1D3C9C4 A
          05 00046    40E3C9D4 A
          05 00047    C5E2E3C1 A
          05 00048    D4D74040 A
     99                           **
    100                           UT:DAYNAME DATA DNMON,DNTUE,DNWED,DNTHU,;
    101   05 00049    00000035                    DNFRI,DNSAT,DNSUN,DNXXX
          05 0004A    00000037
          05 0004B    00000039
          05 0004C    0000003C
          05 0004D    0000003F
          05 0004E    00000041
          05 0004F    00000033
          05 00050    00000044
    102                           **
    103   05 00051    6B404040 A   COMMASPACE TEXT ', '
    104                           **
    105                                   BOUND   8
    106   05 00052    00000000 A   D0      DATA,8  0
                      00000000
    107   05 00054    0000000A A   I10     DATA    10
    108   05 00055    00000014 A   I20     DATA    20
    109   05 00056    0000003C A   I60     DATA    60
    110   05 00057    000003E8 A   I1000   DATA    1000
    111   05 00058    000F4240 A   I1000000 DATA   1000000
```

```
05 00059    00989680 A   I10000000 DATA  10000000
                         **
                         **
05 0005A    00000040 A   MCASE   DATA    X'40'
                         **
05 0005B    0000000F A   MF      DATA    X'0000000F'
05 0005C    000000FF A   MFF     DATA    X'000000FF'
05 0005D    0000FFFF A   MFFFF   DATA    X'0000FFFF'
05 0005E    00FFFFFF A   MFFFFFF DATA    X'00FFFFFF'
```

```
                                **
                                ** SIGMA DOUBLE PRECISION POWERS OF 10. (10^-77 ... 10^75)
                                **
                                        BOUND   8
         05 00060               FPP10   EQU     $
         05 00060                QDATA,8  X'011286D80EC190DC',X'01B94470938FA898'
         05 00064                QDATA,8  X'0273CAC65C39C960',X'03485EBBF9A41DDC'
         05 00068                QDATA,8  X'042D3B357C0692AA',X'051C45016D841BAA'
         05 0006C                QDATA,8  X'0611AB20E472914A',X'06B0AF48EC79ACE8'
         05 00070                QDATA,8  X'076E6D8D93CC0C10',X'084504787C5F878C'
         05 00074                QDATA,8  X'092B22CB4DBBB4B6',X'0A1AF5BF109550F2'
         05 00078                QDATA,8  X'0B10D9976A5D5297',X'0BA87FEA27A539E8'
         05 0007C                QDATA,8  X'0C694FF258C74434',X'0D41D1F7777C8AA0'
         05 00080                QDATA,8  X'0E29233AAAADD6A4',X'0F19B604AAACA626'
         05 00084                QDATA,8  X'101011C2EAABE7D8',X'10A0B19D2AB70E70'
         05 00088                QDATA,8  X'11646F023AB26904',X'123EC56164AF81A4'
         05 0008C                QDATA,8  X'13273B5CDEEDB106',X'1418851A0B548EA4'
         05 00090                QDATA,8  X'14F53304714D9268',X'15993FE2C6D07B80'
         05 00094                QDATA,8  X'165FC7EDBC424D30',X'173BDCF495A9703E'
         05 00098                QDATA,8  X'18256A18DD89E626',X'1917624F8A762FD8'
         05 0009C                QDATA,8  X'19E9D71B689DDE70',X'1A9226712162AB08'
         05 000A0                QDATA,8  X'1B5B5806B4DDAAE4',X'1C391704310A8ACE'
         05 000A4                QDATA,8  X'1D23AE629EA696C2',X'1E164CFDA3281E39'
         05 000A8                QDATA,8  X'1EDF01E85F912E38',X'1F8B61313BBABCE0'
         05 000AC                QDATA,8  X'20571CBEC554B60C',X'213671F73B54F1C8'
         05 000B0                QDATA,8  X'2222073A8515171E',X'23154484932D2E72'
         05 000B4                QDATA,8  X'23D4AD2DBFC3D078',X'2484EC3C97DA6248'
         05 000B8                QDATA,8  X'255313A5DEE87D70',X'2633EC47AB514E66'
         05 000BC                QDATA,8  X'272073ACCB12D100',X'2814484BFEEBC2A0'
         05 000C0                QDATA,8  X'28CAD2F7F5359A38',X'297EC3DAF9418064'
         05 000C4                QDATA,8  X'2A4F3A68DBC8F040',X'2B318481895D9628'
         05 000C8                QDATA,8  X'2C1EF2D0F5DA7DD9',X'2D1357C299A88EA7'
         05 000CC                QDATA,8  X'2DC16D9A00959288',X'2E78E480405D7B98'
         05 000D0                QDATA,8  X'2F4B8ED0283A6D3C',X'302F394219248446'
         05 000D4                QDATA,8  X'311D83C94FB6D2AC',X'3212725DD1D243AC'
         05 000D8                QDATA,8  X'32B877AA3236A4B8',X'33734ACA5F6226F0'
         05 000DC                QDATA,8  X'34480EBE7B9D5858',X'352D09370D425736'
         05 000E0                QDATA,8  X'361C25C268497682',X'37119799812DEA11'
         05 000E4                QDATA,8  X'37AFEBFF0BCB24A8',X'386DF37F675EF6EC'
```

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05 000E8          QDATA,8  X'3944B82FA09B5A54',X'3A2AF31DC4611874'
05 000EC          QDATA,8  X'3B1AD7F29ABCAF48',X'3C10C6F7A0B5ED8D'
05 000F0          QDATA,8  X'3CA7C5AC471B4788',X'3D68DB8BAC710CB4'
05 000F4          QDATA,8  X'3E4189374BC6A7F0',X'3F28F5C28F5C28F6'
05 000F8          QDATA,8  X'401999999999999A',X'4110000000000000'
05 000FC          QDATA,8  X'41A0000000000000',X'4264000000000000'
05 00100          QDATA,8  X'433E800000000000',X'4427100000000000'
05 00104          QDATA,8  X'45186A0000000000',X'45F4240000000000'
05 00108          QDATA,8  X'4698968000000000',X'475F5E1000000000'
05 0010C          QDATA,8  X'483B9ACA00000000',X'492540BE40000000'
05 00110          QDATA,8  X'4A174876E8000000',X'4AE8D4A510000000'
05 00114          QDATA,8  X'4B9184E72A000000',X'4C5AF3107A400000'
05 00118          QDATA,8  X'4D38D7EA4C680000',X'4E2386F26FC10000'
05 0011C          QDATA,8  X'4F16345785D8A000',X'4FDE0B6B3A764000'
05 00120          QDATA,8  X'508AC7230489E800',X'5156BC75E2D63100'
05 00124          QDATA,8  X'523635C9ADC5DEA0',X'5321E19E0C9BAB24'
05 00128          QDATA,8  X'54152D02C7E14AF6',X'54D3C21BCECCEDA0'
05 0012C          QDATA,8  X'5584595161401488',X'5652B7D2DCC80CD4'
05 00130          QDATA,8  X'5733B2E3C9FD0804',X'58204FCE5E3E2502'
05 00134          QDATA,8  X'591431E0FAE6D721',X'59C9F2C9CD046750'
05 00138          QDATA,8  X'5A7E37BE2022C090',X'5B4EE2D6D415B85C'
05 0013C          QDATA,8  X'5C314DC6448D9338',X'5D1ED09BEAD87C03'
05 00140          QDATA,8  X'5E13426172C74D82',X'5EC097CE7BC90718'
05 00144          QDATA,8  X'5F785EE10D5DA46C',X'604B3B4CA85A86C4'
05 00148          QDATA,8  X'612F050FE938943A',X'621D6329F1C35CA5'
05 0014C          QDATA,8  X'63125DFA371A19E7',X'63B7ABC627050308'
05 00150          QDATA,8  X'6472CB5BD86321E4',X'6547BF19673DF530'
05 00154          QDATA,8  X'662CD76FE086B93C',X'671C06A5EC5433C6'
05 00158          QDATA,8  X'68118427B3B4A05C',X'68AF298D050E4398'
05 0015C          QDATA,8  X'696D79F82328EA3C',X'6A446C3B15F99268'
05 00160          QDATA,8  X'6B2AC3A4EDBBFB80',X'6C1ABA4714957D30'
05 00164          QDATA,8  X'6D10B46C6CDD6E3E',X'6DA70C3C40A64E70'
05 00168          QDATA,8  X'6E6867A5A867F104',X'6F4140C78940F6A4'
05 0016C          QDATA,8  X'7028C87CB5C89A26',X'71197D4DF19D6057'
05 00170          QDATA,8  X'71FEE50B7025C368',X'729F4F2726179A20'
05 00174          QDATA,8  X'7363917877CEC054',X'743E3AEB4AE13836'
05 00178          QDATA,8  X'7526E4D30ECCC322',X'76184F03E93FF9F5'
05 0017C          QDATA,8  X'76F316271C7FC390',X'7797EDD871CFDA38'
05 00180          QDATA,8  X'785EF4A74721E864',X'793B58E88C75313E'
```

```
05 00184            QDATA,8  X'7A25179157C93EC8',X'7B172EBAD6DDC73D'
05 00188            QDATA,8  X'7BE7D34C64A9C860',X'7C90E40FBEEA1D38'
05 0018C            QDATA,8  X'7D5A8E89D7525244',X'7E3899162693736A'
05 00190            QDATA,8  X'7F235FADD81C2822'
                   **
05 000F8           FPZERO  EQU     FPP10+152
05 000FA           FPONE   EQU     FPP10+154
05 000FC           FPTEN   EQU     FPP10+156
05 00118           FP10P15 EQU     FPONE+30
05 00136           FP10P30 EQU     FPONE+60
                   **
05 00192           FPNZ QDATA,8 X'8000000000000000'
```

```
                                **
                                ** DOUBLE PRECISION DIGITS
                                **
         05 00194               FPDIGITS EQU    $
         05 00194                QDATA,8  X'0000000000000000',X'4110000000000000'
         05 00198                QDATA,8  X'4120000000000000',X'4130000000000000'
         05 0019C                QDATA,8  X'4140000000000000',X'4150000000000000'
         05 001A0                QDATA,8  X'4160000000000000',X'4170000000000000'
         05 001A4                QDATA,8  X'4180000000000000',X'4190000000000000'
         05 0019C               FP4     EQU     FPDIGITS+8
                                **
                                ** FP:POW10: RETURN A POWER OF 10 FROM THE TABLE
                                ** R1: LINK
                                ** R5: EXPONENT
                                ** RETURNS: R4/5 10^EXPONENT
                                **
         05 001A8               FP:POW10 EQU    $
   05 001A8    12400192                LD,4    FPNZ            INVALID RESULT
   05 001A9    215FFFB3 A              CI,5    -77
   05 001AA    69120000 A              BL      0,1             TOO SMALL
   05 001AB    2150004B A              CI,5    +75
   05 001AC    69220000 A              BG      0,1             TOO BIG
   05 001AD    124A00F8                LD,4    FPZERO,5        GET RESULT
   05 001AE    68020000 A              B       0,1             RETURN IT
```

```
                          ** FNDLOWER: FIND THE LARGEST SMALLER (OR EQUAL) FLOAT VALUE IN THE TABLE
                          ** R1: LINK
                          ** R2/3: VALUE WE ARE SEARCHING
                          ** R4: # TABLE ENTRIES
                          ** R5: TABLE ADDRESS
                          ** RETURNS: R6 INDEX TO ENTRY IN THE TABLE (-1 IF OUTSIDE)
                          **
         05 001AF         FNDLOWER EQU    $
   05 001AF    02200050 A          LCI     5
   05 001B0    8B700000 A          PSM,7   *0
   05 001B1    32800004 A          LW,8    4               SAVE ENTRY COUNT
   05 001B2    22600000 A          LI,6    0               BASE INDEX
   05 001B3    912C0005 A          CD,2    *5,6            CHECK FIRST ENTRY
   05 001B4    691001CD            BL      FNDOUT          OUTSIDE LOW
   05 001B5    683001CE            BE      FNDDONE
   05 001B6    32700004 A          LW,7    4               TOTAL ENTRIES
   05 001B7    207FFFFF A          AI,7    -1              LAST ENTRY
   05 001B8    912E0005 A          CD,2    *5,7            CHECK LAST ENTRY
   05 001B9    691001BC            BL      FNDCHK          NOT LUCKY HERE
   05 001BA    32600007 A          LW,6    7               FOUND IT!
   05 001BB    680001CE            B       FNDDONE
   05 001BC    31700006 A   FNDCHK  CW,7    6               ANY LEFT?
   05 001BD    682001CE            BLE     FNDDONE         ONLY ONE
                          ** COMPUTE PROBE INDEX
   05 001BE    32400006 A          LW,4    6
   05 001BF    30400007 A          AW,4    7
   05 001C0    2540007F A          SLS,4   -1              PROBE INDEX
   05 001C1    31400006 A          CW,4    6               SAME AS LOW?
   05 001C2    682001CE            BLE     FNDDONE         MUST BE DONE
                          ** CHECK PROBED VALUE
   05 001C3    92A80005 A          LD,10   *5,4            FOR EASIER DEBUG
   05 001C4    91280005 A          CD,2    *5,4            CHECK PROBED VALUE
   05 001C5    691001C9            BL      FNDLOW
   05 001C6    692001CB            BG      FNDHIGH
   05 001C7    32600004 A          LW,6    4               EQUAL, DONE
   05 001C8    680001CE            B       FNDDONE
                          **
   05 001C9    32700004 A   FNDLOW  LW,7    4               TAKE LOW HALF OF RANGE
   05 001CA    680001BC            B       FNDCHK          CHECK AGAIN
```

```
05 001CB    32600004 A   FNDHIGH LW,6    4               TAKE HIGH HALF OF RANGE
05 001CC    680001BC             B       FNDCHK
                         **
05 001CD    226FFFFF A   FNDOUT  LI,6    -1
05 001CE    32400008 A   FNDDONE LW,4    8               RESTORE ENTRY COUNT
05 001CF    02200050 A           LCI     5
05 001D0    8A700000 A           PLM,7   *0              RESTORE REGS
05 001D1    21600000 A           CI,6    0               SET CC
05 001D2    68020000 A           B       0,1             RETURN
```

```
                          ** DETERMINE FLOATING POINT COMPONENTS FOR PRINTING
                          ** R1: LINK
                          ** R2/3: FP VALUE
                          ** RETURNS: R4: SIGN, R5: EXP (POWER OF 10),
                          **         R14/15: PACKED DECIMAL MANTISSA.
                          **
        05 001D3          FP:INTERPRET EQU $
05 001D3    89600000 A           PSW,6   *0
05 001D4    89700000 A           PSW,7   *0
05 001D5    02200030 A           LCI     3
05 001D6    8B100000 A           PSM,1   *0
                          **
05 001D7    22E00000 A           LI,14   0               CLEAR DECIMAL DIGITS
05 001D8    22F00000 A           LI,15   0
05 001D9    22400000 A           LI,4    0               ASSUME POSITIVE
                          **
05 001DA    11200192             CD,2    FPNZ            BOUNDARY CASE IS NEGATIVE ZERO
05 001DB    68300207             BE      IFNEGZ          HANDLE SEPARATELY
05 001DC    11200052             CD,2    D0              CHECK FOR TRUE ZERO
05 001DD    68300208             BE      IFZERO
05 001DE    692001E1             BG      IFPOS           B/POSITIVE
05 001DF    224FFFFF A           LI,4    -1              REMEMBER NEGATIVE
05 001E0    1A200002 A           LCD,2   2               GET COMPLEMENT
05 001E1    89400000 A   IFPOS   PSW,4   *0              SAVE SIGN
05 001E2    22400099 A           LI,4    77+76           ENTRY COUNT FOR POWERS TABLE
05 001E3    22500060             LI,5    FPP10
05 001E4    6A1001AF             BAL,1   FNDLOWER        FIND NEXT LOWEST NUMBER
05 001E5    68300202             BE      IFPOW           B/IF POWER OF 10 EXACTLY
05 001E6    691001E9             BL      IFSCV           B/LOWER THAN FIRST
05 001E7    1E2C0060             FDL,2   FPP10,6         GET SIGNIFICAND
05 001E8    680001EF             B       IFADJ           GO ADJUST EXPONENT
                          ** LOWER THAN FIRST ENTRY...
05 001E9    1F2000FC     IFSCV   FML,2   FPTEN           SCALE VALUE
05 001EA    11200060             CD,2    FPP10           UNTIL BIGGER THAN FIRST
05 001EB    681001EE             BGE     IFSCD
05 001EC    206FFFFF A           AI,6    -1              COMPENSATE FOR SCALING
05 001ED    680001E9             B       IFSCV
05 001EE    1E200060     IFSCD   FDL,2   FPP10           SIGNIFICAND
                          ** DETERMINE EXPONENT FROM INDEX
```

```
05 001EF    225FFFB3 A   IFADJ   LI,5    -77
05 001F0    30500006 A           AW,5    6               GET EXPONENT
05 001F1    89500000 A           PSW,5   *0              SAVE EXPONENT
05 001F2    22700010 A           LI,7    16              GENERATE 16 DIGITS
05 001F3    2240000A A   IFDIG   LI,4    10
05 001F4    22500194             LI,5    FPDIGITS
05 001F5    6A1001AF             BAL,1   FNDLOWER        GET A DIGIT IN R6
05 001F6    25E00104 A           SLD,14  4               MAKE SPACE IN ACCUM
05 001F7    49F00006 A           OR,15   6               ACCUMULATE DIGIT
05 001F8    1C2C0194             FSL,2   FPDIGITS,6      REMOVE CURRENT DIGIT
05 001F9    683001FD             BEZ     IFTZ            ALL THE REST ARE ZERO
05 001FA    1F2000FC             FML,2   FPTEN           PROMOTE ANOTHER
05 001FB    647001F3             BDR,7   IFDIG           DO NEXT DIGIT
                          ** TODO: IF THERE IS ANOTHER DIGIT, MAYBE USE IT TO ROUND THE RESULT??
05 001FC    6800020A             B       IFFIN           DONE, RETURN
                          ** FILL ANY TRAILING ZEROS
05 001FD    207FFFFF A   IFTZ    AI,7    -1              COUNT DIGIT JUST ADDED
05 001FE    25700002 A           SLS,7   2               REMAINING NUMBER OF BITS
05 001FF    20700100 A           AI,7    X'100'          LEFT,DOUBLE
05 00200    A5E00007 A           S,14    *7              4 BITS FOR EVERY ZERO
05 00201    6800020A             B       IFFIN
                          ** SOME SPECIAL CASES
05 00202    206FFFB3 A   IFPOW   AI,6    -77             GET EXPONENT
05 00203    89600000 A           PSW,6   *0              SAVE IT
05 00204    32E00001 A           LW,14   1
05 00205    25E0001C A           SLS,14  28              1.0000...
05 00206    6800020A             B       IFFIN
05 00207    22200000 A   IFNEGZ  LI,2    0               CLEAR SIGN FOR NEG ZERO
05 00208    89200000 A   IFZERO  PSW,2   *0              POSITIVE SIGN
05 00209    89200000 A           PSW,2   *0              ZERO EXPONENT
                          ** COME HERE TO RESTORE REGS AND RETURN
05 0020A    02200050 A   IFFIN   LCI     5
05 0020B    8A100000 A           PLM,1   *0              RESTORE 1-3, 4 AND 5 FROM COMPUTATION
05 0020C    88700000 A           PLW,7   *0
05 0020D    88600000 A           PLW,6   *0
05 0020E    E8000001 A           B       *1              DONE
```

```
                          ** CONVERT FLOAT COMPONENTS TO FP DOUBLE
                          ** R1: LINK
                          ** R4: SIGN, R5: EXP (POWER OF 10),
                          ** R14/15: PACKED DECIMAL SIGNIFICAND.
                          ** RETURNS SKIPPING: R2/3: FP VALUE
                          **
        05 0020F          FP:COMPUTE EQU  $
05 0020F    215FFFA2 A           CI,5    -94             SMALLEST POSSIBLE
05 00210    69120000 A           BL      0,1             RETURN ERROR
05 00211    2150004B A           CI,5    +75             HIGHEST POSSIBLE EXPONENT
05 00212    69220000 A           BG      0,1             RETURN ERROR
05 00213    12200052             LD,2    D0              ACCUMULATE RESULT HERE
05 00214    11E00052             CD,14   D0              CHECK FOR ZERO SIGNIFICAND
05 00215    6830022A             BE      CFDONE          RETURN TRUE ZERO
05 00216    02200060 A           LCI     6
05 00217    8B400000 A           PSM,4   *0              SAVE 4-9
05 00218    22700010 A           LI,7    16              UNPACK 16 DIGITS
05 00219    25E0037C A   CFDIGIT SCD,14  -4              DIGIT TO LOW ORDER
05 0021A    2260000F A           LI,6    X'F'
05 0021B    4B60000F A           AND,6   15              GET DIGIT
05 0021C    2160000A A           CI,6    10              VALIDATE
05 0021D    6920022B             BG      CFFAIL          BAD DIGIT, RETURN
05 0021E    1F2000FC             FML,2   FPTEN
05 0021F    1D2C0194             FAL,2   FPDIGITS,6
05 00220    64700219             BDR,7   CFDIGIT         DO NEXT DIGIT
                          ** NOW HAVE SIGNIFICAND, BUT IT IS SCALED BY 10^15
05 00221    205FFFF1 A           AI,5    -15             SO ADJUST THIS
05 00222    215FFFB3 A           CI,5    -77             CHECK FOR VERY SMALL
05 00223    68100226             BGE     CFSCALE         B/NO
05 00224    1E200136             FDL,2   FP10P30         REMOVE THE 15, AND TAKE 15 MORE
05 00225    2050001E A           AI,5    30
05 00226    1F2A0060     CFSCALE FML,2   FPP10,5         SCALE BY POWER OF 10
05 00227    20400000 A           AI,4    0               CHECK SIGN
05 00228    6810022A             BGEZ    CFDONE          POSITIVE
05 00229    1A200002 A           LCD,2   2               NEGATIVE
05 0022A    20100001 A   CFDONE  AI,1    +1              RETURN SKIPPING
05 0022B    02200060 A   CFFAIL  LCI     6
05 0022C    8A400000 A           PLM,4   *0              RESTORE INPUTS
05 0022D    68020000 A           B       0,1             RETURN
```

```
                                ** CONVERT TEXT (L, BA) IN R8 TO FLOATING POINT DOUBLE (R10/11)
                                **
         05 0022E               FP:FROMTEXT EQU $
   05 0022E    02200070 A               LCI     7
   05 0022F    8B100000 A               PSM,1   *0              SAVE 1-7
   05 00230    22700000 A               LI,7    0               NO DECIMAL ENCOUNTERED YET
   05 00231    12A00052                 LD,10   D0              INITIAL RETURN VALUE
   05 00232    3220005E                 LW,2    MFFFFFF
   05 00233    4B200008 A               AND,2   8               R2: BA(TEXT)
   05 00234    72300008 A               LB,3    8               R3: LENGTH
   05 00235    6830027F                 BEZ     CVFNDX          ZERO IS ERROR
                                ** NOW PROCESS FIRST CHARACTER
   05 00236    72140000 A               LB,1    0,2             FIRST CHARACTER
   05 00237    201FFFB2 A               AI,1    -X'4E'          -PLUS
   05 00238    6930023A                 BNEZ    CVFNCP          B/NOT PLUS
   05 00239    6A10023D                 BAL,1   CVFNDP          BRANCH, SETTING R1 NONZERO
                                ** NOT MINUS, SEE IF PLUS
   05 0023A    201FFFEE A       CVFNCP  AI,1    -X'12'          -(MINUS-PLUS)
   05 0023B    69300240                 BNEZ    CVFND0          NOT A MINUS OR A PLUS
   05 0023C    22100000 A               LI,1    0               NEGATIVE
                                ** SKIP OVER LEADING SIGN
   05 0023D    20200001 A       CVFNDP  AI,2    1               NEXT CHAR
   05 0023E    64300240                 BDR,3   CVFND0          COUNT IT, BRANCH IF ANYTHING MORE
   05 0023F    6800027F                 B       CVFNDX          SIGN WITH NO DIGITS, ERROR
                                ** R1 = 0 IF VALUE IS NEGATIVE
         05 00240               CVFND0  EQU     $
   05 00240    72540000 A       CVFNXT  LB,5    0,2             GET A DIGIT
   05 00241    21500010 04              CI,5    FCHARD          IS IT THE DECIMAL POINT
   05 00242    69300247                 BNE     CVFNDEC         B/YES
   05 00243    20700000 A               AI,7    0               SEEN A DECIMAL ALREADY?
   05 00244    6910027F                 BLZ     CVFNDX          B/YES, ERROR
   05 00245    227FFFFF A               LI,7    -1              NEXT DIGIT IS SCALED BY 10^-1
   05 00246    68000258                 B       CVFDD           DONE WITH THIS
   05 00247    4950005A         CVFNDEC OR,5    MCASE           UPSHIFT
   05 00248    215000C5 A               CI,5    'E'             EXPONENT?
   05 00249    6830025B                 BE      CVFEXP          B/YES
   05 0024A    215000F0 A               CI,5    '0'
   05 0024B    6910027F                 BL      CVFNDX          ERROR
   05 0024C    215000F9 A               CI,5    '9'
```

```
05 0024D    6920027F            BG      CVFNDX          ERROR
05 0024E    205FFF10 A          AI,5    -X'F0'          GET DIGIT
05 0024F    20700000 A          AI,7    0               IN FRACTION?
05 00250    68100256            BGEZ    CVFNFR          B/NO
05 00251    124A0194            LD,4    FPDIGITS,5      GET DIGIT VALUE
05 00252    1F4E00FA            FML,4   FPONE,7         FRACTIONAL ADDEND
05 00253    1DA00004 A          FAL,10  4               ACCUMULATE
05 00254    207FFFFF A          AI,7    -1              ONE LESS POWER
05 00255    68000258            B       CVFDD           DONE THIS DIGIT
05 00256    1FA000FC     CVFNFR FML,10  FPTEN           BY RADIX
05 00257    1DAA0194            FAL,10  FPDIGITS,5      ADD THE DIGIT
05 00258    20200001 A   CVFDD  AI,2    +1              POINT AT NEXT
05 00259    64300240            BDR,3   CVFNXT
05 0025A    68000279            B       CVFNDN          NO EXPONENT, FINISH UP
                         ** HANDLE THE EXPONENT
05 0025B    20200001 A   CVFEXP AI,2    +1              POINT AT NEXT
05 0025C    203FFFFF A          AI,3    -1              UPDATE REMAINING COUNT
05 0025D    6820027F            BLEZ    CVFNDX          MISSING EXPONENT
05 0025E    72540000 A          LB,5    0,2             NEXT DIGIT
05 0025F    22400000 A          LI,4    0               ASSUME POSITIVE EXPONENT
05 00260    22700000 A          LI,7    0               ACCUMULATE EXPONENT IN R7
05 00261    2150004E A          CI,5    '+'             POSITIVE EXPONENT?
05 00262    68300266            BE      CVFPOSX         YES
05 00263    21500060 A          CI,5    '-'             NEGATIVE?
05 00264    6930026A            BNE     CVFXDIG         B/NO
05 00265    224FFFFF A          LI,4    -1              REMEMBER
05 00266    20200001 A   CVFPOSX AI,2   +1              SKIP SIGN
05 00267    203FFFFF A          AI,3    -1              COUNT IT
05 00268    6820027F            BLEZ    CVFNDX          ERROR
05 00269    72540000 A          LB,5    0,2             FIRST DIGIT OF EXPONENT
05 0026A    215000F0 A   CVFXDIG CI,5   '0'
05 0026B    6910027F            BL      CVFNDX          ERROR
05 0026C    215000F9 A          CI,5    '9'
05 0026D    6920027F            BG      CVFNDX          ERROR
05 0026E    205FFF10 A          AI,5    -X'F0'          GET DIGIT
05 0026F    2370000A A          MI,7    10
05 00270    30700005 A          AW,7    5               ADD IN DIGIT
05 00271    2170004E A          CI,7    78              TOO BIG?
05 00272    6920027F            BG      CVFNDX          B/YES, ERROR
```

```
05 00273    20200001 A          AI,2    +1              POINT AT NEXT
05 00274    6430026A            BDR,3   CVFXDIG         GO PROCESS
05 00275    20400000 A          AI,4    0               WAS EXPONENT NEGATIVE?
05 00276    68300278            BEZ     $+2             NO
05 00277    3A700007 A          LCW,7   7               NEGATE IT
                            ** FINISHED, ADJUST ACCUMULATE VALUE BY POWER OF 10.
05 00278    1FAE0060            FML,10  FPP10,7
                            ** DONE ACCUMULATING, ADJUST IF NEGATIVE
05 00279    20100000 A  CVFNDN  AI,1    0               NEGATIVE?
05 0027A    6930027C            BNEZ    CVFNNN          B/NO
05 0027B    1AA0000A A          LCD,10  10              FLIP IT
                            ** RETURN THE VALUE
05 0027C    02200070 A  CVFNNN  LCI     7
05 0027D    8A100000 A          PLM,1   *0              RESTORE 1-7
05 0027E    68020001 A          B       1,1             DONE
                            ** ERROR RETURN, RESULTS UNDEFINED.
05 0027F    02200070 A  CVFNDX  LCI     7
05 00280    8A100000 A          PLM,1   *0              RESTORE 1-7
05 00281    68020000 A          B       0,1             ERROR RETURN
```

```
                          **
                          ** CONVERT DECIMAL TEXT (L,BA) IN R8 TO INTEGER
                          ** RETURNS:     INTEGER IN R10/11
                          **
        05 00282          INT:FROMTEXT EQU $
05 00282    02200090 A           LCI     9
05 00283    8B100000 A           PSM,1   *0              SAVE 1-9
05 00284    2270000A A           LI,7    10              ASSUME RADIX 10
05 00285    12A00052             LD,10   D0              INITIAL RETURN VALUE
05 00286    3220005E             LW,2    MFFFFFF
05 00287    4B200008 A           AND,2   8               R2: BA(TEXT)
05 00288    72300008 A           LB,3    8               R3: LENGTH
05 00289    683002C7             BEZ     CVINDX          ZERO IS ERROR
                          ** NOW PROCESS FIRST CHARACTER
05 0028A    72140000 A           LB,1    0,2             FIRST CHARACTER
05 0028B    201FFFB2 A           AI,1    -X'4E'          -PLUS
05 0028C    6930028E             BNEZ    CVINCP          B/NOT PLUS
05 0028D    6A100291             BAL,1   CVINDP          R1 NONZERO, BRANCH
                          ** NOT MINUS, SEE IF PLUS
05 0028E    201FFFEE A   CVINCP  AI,1    -X'12'          -(MINUS-PLUS)
05 0028F    69300294             BNEZ    CVIND0          NOT A MINUS OR A PLUS
05 00290    22100000 A           LI,1    0               NEGATIVE
                          ** SKIP OVER LEADING SIGN
05 00291    20200001 A   CVINDP  AI,2    1               NEXT CHAR
05 00292    64300294             BDR,3   CVIND0          COUNT IT, BRANCH IF ANYTHING MORE
05 00293    680002C7             B       CVINDX          SIGN WITH NO DIGITS, ERROR
                          ** R1 = 0 IF VALUE IS NEGATIVE
05 00294    22400000 A   CVIND0  LI,4    0
05 00295    72540000 A           LB,5    0,2             GET A DIGIT
05 00296    215000F0 A           CI,5    '0'
05 00297    693002A3             BNE     CVINXT          NOT, ZERO, RADIX IS 10
05 00298    20200001 A           AI,2    1               BA(NEXT CHAR)
05 00299    203FFFFF A           AI,3    -1              DECREMENT COUNT
05 0029A    683002C4             BEZ     CVINNN          NONE, JUST ZERO, RETURN
05 0029B    72540000 A           LB,5    0,2
05 0029C    215000A7 A           CI,5    X'A7'           LITTLE X
05 0029D    683002A0             BE      CVIHEX          YES
05 0029E    215000E7 A           CI,5    X'E7'           BIG X
05 0029F    693002A3             BNE     CVINXT          NO, IS DECIMAL
```

```
05 002A0    20200001 A   CVIHEX  AI,2    1              SKIP X CHARACTER
05 002A1    203FFFFF A           AI,3    -1             COUNT IT
05 002A2    22700010 A           LI,7    16             NEW RADIX
                         ** PROCESS EACH DIGIT
05 002A3    72540000 A   CVINXT  LB,5    0,2            GET A DIGIT
05 002A4    215000F0 A           CI,5    '0'
05 002A5    691002AC             BL      CVINCX         CHECK HEXITS
05 002A6    215000F9 A           CI,5    '9'
05 002A7    692002C7             BG      CVINDX         ERROR
05 002A8    205FFF10 A           AI,5    -X'F0'         GET DIGIT
05 002A9    21700010 A           CI,7    16
05 002AA    693002B6             BNE     CVINMD         B/NOT HEX
05 002AB    680002B4             B       CVINH
05 002AC    21700010 A   CVINCX  CI,7    16             IS HEX ALLOWED?
05 002AD    693002C7             BNE     CVINDX         B/NO
05 002AE    4950005A             OR,5    MCASE          MAKE UPPERCASE
05 002AF    205FFF3F A           AI,5    -X'C1'         SUBTRACT 'A'
05 002B0    691002C7             BL      CVINDX         ERROR
05 002B1    21500005 A           CI,5    5              >F?
05 002B2    692002C7             BG      CVINDX         YES, ERROR
05 002B3    2050000A A           AI,5    10             MAKE HEXIT
05 002B4    25A00104 A   CVINH   SLD,10  4              MAKE ROOM
05 002B5    680002BE             B       CVIAC          ACCUMULATE DIGIT
05 002B6    3280000A A   CVINMD  LW,8    10             MULTI-STEP MULTIPLY REQUIRED?
05 002B7    683002BD             BEZ     CVISM          B/NO
05 002B8    3290000B A           LW,9    11             COPY LOW WORD
05 002B9    25A00103 A           SLD,10  3              *8
05 002BA    25800101 A           SLD,8   1              *2
05 002BB    10A00008 A           AD,10   8              ...*10
05 002BC    680002BE             B       CVIAC          ACCUMULATE DIGIT
05 002BD    37A00007 A   CVISM   MW,10   7              BY RADIX
05 002BE    10A00004 A   CVIAC   AD,10   4              ADD THE DIGIT
05 002BF    20200001 A           AI,2    +1             POINT AT NEXT
05 002C0    643002A3             BDR,3   CVINXT
                         ** DONE ACCUMULATING, ADJUST IF NEGATIVE
05 002C1    20100000 A   CVINDN  AI,1    0              NEGATIVE?
05 002C2    693002C4             BNEZ    CVINNN         B/NO
05 002C3    1AA0000A A           LCD,10  10             FLIP IT
                         ** RETURN VALUE
```

```
579   05 002C4    02200090 A   CVINNN  LCI     9
580   05 002C5    8A100000 A           PLM,1   *0              RESTORE 1-7
581   05 002C6    68020001 A           B       1,1             DONE
582                           ** ERROR RETURN, RESULTS UNDEFINED.
583   05 002C7    02200090 A   CVINDX  LCI     9
584   05 002C8    8A100000 A           PLM,1   *0              RESTORE 1-7
585   05 002C9    68020000 A           B       0,1             ERROR RETURN
```

```
                                **
                                ** CONVERT 64-BIT INTEGER IN R10/11 TO TEXT
                                ** R1: LINK
                                ** R8: BUFFER LEN AND BYTE ADDRESS
                                ** R9: OPTIONS (16 BITS), DECIMALS (8 BITS), S (8 BITS)
                                **      S IS FORMAT WIDTH, OR ZERO FOR VARIABLE
                                **      OPTIONS:
                                **          BIT 0: UNSIGNED VALUE
                                **          BIT 1: PRINT SIGN (EVEN IF +)
                                **          BIT 2: LEADING ZEROS
                                **          BIT 3: EURO (INVERT , AND .)
                                **          BIT 4: THOUSANDS SEPARATOR     **** NOT IMPLEMENTED YET
                                **          BIT 5: TRUNCATE RIGHT IF TOO BIG (OTHERWISE LEFT)
                                **          BIT 6: PRINT *'S IF TOO BIG
                                **
         05 002CA               INT:TOTEXT EQU  $
   05 002CA    022000B0 A                LCI     11
   05 002CB    8B100000 A                PSM,1   *0              SAVE R1-11
   05 002CC    04800000 02               M:TRAP  ,(IGNORE,BOTH)
   02 00000    14000000 A
   02 00001    00000003 A
   05 002CD    89800000 A                PSW,8   *0              SAVE PREV TRAP STATUS
                                **
   05 002CE    22300040 A                LI,3    FORMATW
   05 002CF    71300008 A                CB,3    8               CHECK FORMAT WIDTH
   05 002D0    682002D2                  BLE     $+2             USE MAXIMUM AVAILABLE
   05 002D1    72300008 A                LB,3    8               USE FIELD WIDTH
                                **
   05 002D2    32500009 A                LW,5    9               COPY FORMAT WORD
   05 002D3    25500070 A                SLS,5   -16             ISOLATE OPTIONS BITS
   05 002D4    22400000 A                LI,4    0               ASSUME POSITIVE
   05 002D5    1260000A A                LD,6    10              COPY VALUE
   05 002D6    681002DB                  BGEZ    DDIGIT
   05 002D7    21508000 A                CI,5    INTO_UNSIGNED
   05 002D8    694002DB                  BANZ    DDIGIT
   05 002D9    73100004 A                MTB,+1  4               REMEMBER NEGATIVE
   05 002DA    1A600006 A                LCD,6   6               GET ABSOLUTE VALUE
   05 002DB    21600000 A   DDIGIT       CI,6    0               HIGH WORD STILL RELEVANT?
   05 002DC    683002EC                  BEZ     DDIGITL         NO, REST IS EASIER
```

```
05 002DD    89700000 A          PSW,7   *0
05 002DE    25600160 A          SLD,6   -32             HIGH ORDER TO 7
05 002DF    36600054            DW,6    I10
05 002E0    B2100000 A          LW,1    *0              GET TOP OF STACK
05 002E1    C6700001 A          XW,7    *1              R6-7 = (BORROW, LOW ORDER)
                          **
                          ** Division by 10 here can overflow, because the quotient is signed,
                          ** So we'll divide by 20 and then multiply by 2, fixing up the
                          ** remainder after
                          **
05 002E2    36600055            DW,6    I20             REMAINDER(0-19) IN R8
05 002E3    25700001 A          SLS,7   1               MULTIPLY QUOTIENT BY 2
05 002E4    2160000A A          CI,6    10              CHECK REMAINDER
05 002E5    691002E8            BL      GODIGIT         OK, USE IT
05 002E6    20700001 A          AI,7    1               ONE MORE FOR THE QUOTIENT
05 002E7    206FFFF6 A          AI,6    -10             CORRECT REMAINER
05 002E8    206000F0 A   GODIGIT AI,6   '0'             GET A CHARACTER VALUE
05 002E9    75660000 04         STB,6   FORMAT,3
05 002EA    88600000 A          PLW,6   *0              RETRIEVE HIGH QUOTIENT
05 002EB    643002DB            BDR,3   DDIGIT
                          *
05 002EC    21700000 A   DDIGITL CI,7   0               DONE YET?
05 002ED    683002F3            BEZ     DDECLX          YES/ GO FINISH
05 002EE    22600000 A          LI,6    0
05 002EF    36600054            DW,6    I10             DIGIT TO R6, REST TO R7
05 002F0    206000F0 A          AI,6    '0'             GET CHAR VALUE
05 002F1    75660000 04         STB,6   FORMAT,3
05 002F2    643002EC            BDR,3   DDIGITL
        05 002F3            DDECLX  EQU     $
05 002F3    226000F0 A          LI,6    '0'
05 002F4    2220003F A          LI,2    FORMATW-1
05 002F5    38200003 A          SW,2    3               NUMBER OF DIGITS
05 002F6    692002FA            BGZ     DDECNZ          MORE THAN ZERO
05 002F7    75660000 04         STB,6   FORMAT,3        PUT IN A ZERO
05 002F8    203FFFFF A          AI,3    -1
05 002F9    20200001 A          AI,2    +1              COUNT DIGIT
        05 002FA            DDECNZ  EQU     $
05 002FA    2190FF00 A          CI,9    X'FF00'         ANY FRACTION?
05 002FB    6840031D            BAZ     DDECALGN        NO
```

```
05 002FC    32100009 A          LW,1    9
05 002FD    25100078 A          SLS,1   -8              GET FRACTION DIGITS
05 002FE    31200001 A   DDECCF  CW,2    1               ARE THERE ENOUGH ?
05 002FF    69200311            BG      DDECIF          MORE THAN ENOUGH, DECIMAL MUST BE INSERTED
05 00300    68300304            BE      DDECGF
                         ** ADD LEADING ZEROS SO THERE ARE ENOUGH FOR THE FRACTION.
05 00301    75660000 04         STB,6   FORMAT,3
05 00302    20200001 A          AI,2    +1
05 00303    643002FE            BDR,3   DDECCF
      05 00304           DDECGF  EQU     $
05 00304    22600010 04         LI,6    FCHARD
05 00305    75660000 04         STB,6   FORMAT,3
05 00306    20200001 A          AI,2    +1
05 00307    203FFFFF A          AI,3    -1
05 00308    4B90005C            AND,9   MFF             DESTINATION LENGTH
05 00309    6830030C            BEZ     DDECAZ          VARIABLE
05 0030A    31600009 A          CW,6    9               CAN WE TOLERATE A LEADING ZERO
05 0030B    6810031D            BGE     DDECALGN        B/NO, WILL START WITH DECIMAL
05 0030C    226000F0 A   DDECAZ  LI,6    '0'
05 0030D    75660000 04         STB,6   FORMAT,3
05 0030E    20200001 A          AI,2    +1
05 0030F    203FFFFF A          AI,3    -1
05 00310    6800031D            B       DDECALGN        READY FOR ALIGNMENT
                         ** INSERT A DECIMAL
05 00311    89200000 A   DDECIF  PSW,2   *0              SAVE TOTAL DIGIT COUNT
05 00312    38200009 A          SW,2    9               NUMBER TO MOVE
05 00313    30300002 A          AW,3    2
05 00314    20200001 A          AI,2    +1              INCLUDE DECIMAL
05 00315    22600010 04         LI,6    FCHARD
      05 00316           ANLZFX3 EQU     $               NB/ REFERENCED BY ANLZ INST.
05 00316    72160000 04  DDECIN  LB,1    FORMAT,3
05 00317    75660000 04         STB,6   FORMAT,3
05 00318    32600003 A          LW,6    3
05 00319    203FFFFF A          AI,3    -1
05 0031A    64200316            BDR,2   DDECIN
05 0031B    88200000 A          PLW,2   *0              RESTORE TOTAL
05 0031C    20200001 A          AI,2    +1              COUNT THE DECIMAL
                         ** ALIGN A DECIMAL STRING
      05 0031D           DDECALGN EQU    $
```

```
05 0031D    22600060 A          LI,6    '-'
                         ** IF NEGATIVE INSERT SIGN NOW.
05 0031E    73000004 A          MTB,0   4               IS IT?
05 0031F    69300323            BNEZ    DDECNEG         YES
05 00320    21504000 A          CI,5    INTO_PRTSIGN    PRINT SIGN ANYWAY?
05 00321    68400326            BAZ     DDECLEN         NO
05 00322    2260004E A          LI,6    '+'             ADJUST SIGN
05 00323    75660000 04  DDECNEG STB,6  FORMAT,3
05 00324    20200001 A          AI,2    +1
05 00325    203FFFFF A          AI,3    -1
      05 00326          DDECLEN EQU     $
05 00326    4B90005C            AND,9   MFF             DESIRED FORMAT WIDTH
05 00327    6830034C            BEZ     DDECMOV         DONE, MOVE TO LINE BUFFER
                         ** EASIER TO USE BUFFER FOR THIS, BUT LIMITS MAX SIZE OF DESTINATION.
05 00328    21900040 A          CI,9    FORMATW         CHECK BUFFER LIMITS
05 00329    6820032B            BLE     $+2             OK
05 0032A    22900040 A          LI,9    FORMATW         USE MAXIUM
05 0032B    31900002 A          CW,9    2
05 0032C    69100345            BL      DDTRUNC         OOPS, TOO BIG
05 0032D    6830034C            BE      DDECMOV         JUST RIGHT
                         ** PREPEND BLANKS OR ZEROS
05 0032E    38900002 A          SW,9    2               NUMBER TO FILL
05 0032F    21502000 A          CI,5    INTO_LEADZEROS
05 00330    69400337            BANZ    DDECZ
05 00331    22600040 A          LI,6    ' '
05 00332    F5660000 04  DDECFIB STB,6  *FORMAT,3
05 00333    20200001 A          AI,2    +1
05 00334    203FFFFF A          AI,3    -1
05 00335    64900332            BDR,9   DDECFIB
05 00336    6800034C            B       DDECMOV         MOVE IT
                         ** PREPEND LEADING ZEROS
05 00337    73000004 A   DDECZ  MTB,0   4               NEGACTIVE?
05 00338    6830033A            BEZ     DDECZP          NO
05 00339    20300001 A          AI,3    1               START FILL OVER SIGN
05 0033A    226000F0 A   DDECZP LI,6    '0'
05 0033B    75660000 04  DDECFI0 STB,6  FORMAT,3
05 0033C    20200001 A          AI,2    +1
05 0033D    203FFFFF A          AI,3    -1
05 0033E    6490033B            BDR,9   DDECFI0
```

```
05 0033F    73000004 A          MTB,0   4               NEGATIVE
05 00340    6830034C            BEZ     DDECMOV         NO DONE
05 00341    22600060 A          LI,6    '-'
05 00342    75660000 04         STB,6   FORMAT,3        STICK IN FRONT
05 00343    203FFFFF A          AI,3    -1
05 00344    6800034C            B       DDECMOV         GO MOVE IT
                            **
                            ** COME HERE WHEN FORMATTED NUMBER IS TOO BIG
      05 00345          DDTRUNC EQU     $
05 00345    2230003F A          LI,3    FORMATW-1
05 00346    4B90005C            AND,9   MFF
05 00347    32200009 A          LW,2    9
05 00348    2260005C A          LI,6    '*'             MY GOD IT'S FULL OF STARS
05 00349    75660000 04 DDECFIS STB,6   FORMAT,3
05 0034A    203FFFFF A          AI,3    -1
05 0034B    64900349            BDR,9   DDECFIS
                            **
                            ** FINALLY, WE HAVE A STRING (IN FORMAT BUFFER):
                            ** R2=LENGTH, R3=INDEX OF FIRST CHAR LESS ONE
                            ** FILL WITH BLANKS IN FRONT IF DESTINATION STRING IS LONGER.
                            **
05 0034C    72100008 A  DDECMOV LB,1    8               DESTINATION BUFFER LENGTH
05 0034D    38100002 A          SW,1    2
05 0034E    68200356            BLEZ    DDECMNF         NO FILL REQUIRED
05 0034F    22600040 A          LI,6    ' '
05 00350    75600000 04         STB,6   FORMAT          SET FIRST CHARACTER
05 00351    22600001 04         LI,6    FORMAT+1
05 00352    32700006 A          LW,7    6
05 00353    75100007 A          STB,1   7               SET LENGTH
05 00354    616FFFFF A          MBS,6   -1              FILL WITH BLANKS
05 00355    68000357            B       DDECMVD         MOVE THE NUMBER
                            **
05 00356    32700008 A  DDECMNF LW,7    8               DESTINATION AND LENGTH
05 00357    20300001 A  DDECMVD AI,3    +1              INDEX FIRST BYTE
05 00358    44600316            ANLZ,6  ANLZFX3         GET ADDRESS OF STRING IN BUF
05 00359    61600000 A          MBS,6   0               MOVE FORMATTED NUMBER
05 0035A    88800000 A          PLW,8   *0              GET PRV TRAP STATE
05 0035B    04800002 02         M:TRAP  (RESTORE,8)
02 00002    14000008 A
```

```
          02 00003    80000000 A
    779   05 0035C    022000B0 A           LCI     11              RESTORE CALLERS REGS
    780   05 0035D    68020000 A           B       0,1             DONE
```

```
                          **
                          ** CONVERT 32-BIT SIGNED INTEGER IN R5 TO TEXT
                          ** R1: LINK
                          ** R8: W (4 BITS), S (4 BITS), DESTINATION BA
                          **      IF S IS NONZERO, IT IS BUFFER SIZE AND FORMAT WIDTH
                          **      ELSE W IS BUFFER SIZE, AND FORMAT WIDTH IS VARIABLE
                          ** R8: HIGH BYTE RETURNS LENGTH
                          **
        05 0035E          INT32:TOTEXT EQU $
05 0035E    02200070 A            LCI     7
05 0035F    8B100000 A            PSM,1   *0              SAVE 1-7
05 00360    3260005E              LW,6    MFFFFFF         24 BITS OF
05 00361    4B600008 A            AND,6   8               BYTE ADDRESS
05 00362    72700008 A            LB,7    8
05 00363    68000366              B       ITX$
                          **
                          ** ALTERNATE ENTRY:
                          ** R6: DESTINATION
                          ** R7: W/S AS ABOVE
                          ** R8: HIGH BYTE RETURNS LENGTH IF VARIABLE
                          **
        05 00364          INT:TEXT$ EQU   $
05 00364    02200070 A            LCI     7
05 00365    8B100000 A            PSM,1   *0              SAVE 1-7
                          **
        05 00366          ITX$    EQU     $
05 00366    22200000 A            LI,2    0               R2: BIT 31: NEGATIVE, BIT 30: VARIABLE
05 00367    20500000 A            AI,5    +0              POSITIVE?
05 00368    6810036B              BGE     ITXP            B/YES
05 00369    20200001 A            AI,2    1               REMEMBER
05 0036A    3A500005 A            LCW,5   5
05 0036B    4B70005B      ITXP    AND,7   MF              IS FORMAT WIDTH SPECIFIED?
05 0036C    69300370              BNEZ    ITXF            B/YES
05 0036D    20200002 A            AI,2    2               REMEMBER
05 0036E    72700008 A            LB,7    8
05 0036F    2570007C A            SLS,7   -4              GET MAXIMUM SIZE
05 00370    32300007 A    ITXF    LW,3    7               COPY WIDTH
05 00371    30700006 A            AW,7    6               FIRST BYTE PAST END
05 00372    207FFFFF A    ITXLP   AI,7    -1              PREVIOUS BYTE
```

```
05 00373    31700006 A           CW,7    6
05 00374    6910038E             BL      ITXDN           B/DONE
05 00375    22400000 A           LI,4    0               CLEAR UPPER WORD
05 00376    36400054             DW,4    I10             GET A DIGIT
05 00377    204000F0 A           AI,4    X'F0'           EBCIDIC
05 00378    754E0000 A           STB,4   0,7
05 00379    21200002 A           CI,2    2               VARIABLE?
05 0037A    68400372             BAZ     ITXLP           B/NO
05 0037B    20500000 A           AI,5    +0              ANY MORE NUMBER?
05 0037C    69300372             BNEZ    ITXLP           B/YES
05 0037D    21200001 A           CI,2    1               NEGATIVE?
05 0037E    68400384             BAZ     ITXVF           B/NO
05 0037F    31700006 A           CW,7    6               IS THERE SPACE FOR SIGN?
05 00380    69100398             BL      ITXDNF          NO
05 00381    207FFFFF A           AI,7    -1
05 00382    22400060 A           LI,4    '-'
05 00383    754E0000 A           STB,4   0,7             SET SIGN
05 00384    38700006 A   ITXVF   SW,7    6               BYTES TO SHIFT LEFT
05 00385    6820039C             BLEZ    ITXEX           DONE
05 00386    38300007 A           SW,3    7               # BYTES TO MOVE
05 00387    75300008 A           STB,3   8               SET FOR RETURN
05 00388    724E0000 A   ITXMV   LB,4    0,7
05 00389    754C0000 A           STB,4   0,6
05 0038A    20700001 A           AI,7    +1
05 0038B    20600001 A           AI,6    +1
05 0038C    64300388             BDR,3   ITXMV
05 0038D    6800039C             B       ITXEX
                         **
05 0038E    20500000 A   ITXDN   AI,5    +0              ANY MORE NUMBER?
05 0038F    69300398             BNEZ    ITXDNF          YES, DID NOT FIT
05 00390    21200001 A           CI,2    1               WAS IT NEGATIVE?
05 00391    6840039C             BAZ     ITXEX           NO, DONE
05 00392    724C0000 A           LB,4    0,6             CHECK FIRST DIGIT
05 00393    214000F0 A           CI,4    X'F0'           IS IT ZERO
05 00394    69300398             BNE     ITXDNF          DOES NOT FIT
05 00395    22400060 A           LI,4    '-'
05 00396    754C0000 A           STB,4   0,6             SET THE SIGN
05 00397    6800039C             B       ITXEX           DONE
                         ** COME HERE WHEN TOO BIG FOR BUFFER
```

```
860   05 00398    2240005C A   ITXDNF  LI,4    '*'
861   05 00399    754C0000 A           STB,4   0,6            FILL ENTIRE BUFFER
862   05 0039A    20600001 A           AI,6    +1
863   05 0039B    64300398             BDR,3   ITXDNF
864   05 0039C    02200070 A   ITXEX   LCI     7
865   05 0039D    8A100000 A           PLM,1   *0             GET R1-7 BACK
866   05 0039E    68020000 A           B       0,1            DONE, RETURN
```

```
                          ** DATES
                          ** VARIOUS INTERVALS IN YEARS
05 0039F    00000064 A    YPCENTURY  DATA 100
05 003A0    00000190 A    YPQUADCENT DATA 400
                          ** VARIOUS INTERVALS IN DAYS
05 003A1    00000007 A    DPWEEK     DATA 7
05 003A2    0000016D A    DPYEAR     DATA 365
05 003A3    000005B5 A    DPQUADYEAR DATA 1461           365 * 3 + 366
05 003A4    00008EAC A    DPCENTURY  DATA 36524
05 003A5    00008EAD A    DPLEAPCENT DATA 36525          1461 * 25
05 003A6    00023AB1 A    DPQUADCENT DATA 146097         36524 * 3 + 36525
                          **
                          ** JDMAX ALLOWS FOR A 64-BIT 100-NANOSECOND TIMESTAMP.
                          ** CONSIDER IF WORTH REDUCING THIS TO 23 BITS, I.E. 8388607
05 003A7    00A2E3FE A    JDMAX   DATA    10675198       SEPT 13, 29227
       0000722B           YEARMAX EQU     29227
                          **
                          **
       05 003A8           MONTHDAY QDATA,1        31,28,31,30,31,30,31,31,30,31,30,31
       05 003AB           MONTHDLY QDATA,1        31,29,31,30,31,30,31,31,30,31,30,31
                                  BOUND   4
```

```
                          **
                          ** DETERMINE IF YEAR IS A LEAP YEAR
                          ** R1: LINK
                          ** R2: YEAR
                          ** RETURNS SKIPPING IF LEAP YEAR
                          **
         05 003AE         ISLEAPYEAR EQU  $
   05 003AE    02200050 A          LCI     5
   05 003AF    8B100000 A          PSM,1   *0
   05 003B0    22100003 A          LI,1    3
   05 003B1    4B100002 A          AND,1   2               DOES IT DIVIDE BY 4?
   05 003B2    693003BE            BNEZ    ILYF            NO, NOT A LEAP YEAR
   05 003B3    22400000 A          LI,4    0
   05 003B4    32500002 A          LW,5    2               GET YEAR
   05 003B5    3640039F            DW,4    YPCENTURY
   05 003B6    20400000 A          AI,4    0               TEST REMAINER
   05 003B7    693003BB            BNE     ILYT            NOT A CENTURY SO IS A LEAP
   05 003B8    22100003 A          LI,1    3
   05 003B9    4B100005 A          AND,1   5               IS CENTURY DIVISIBLE BY 4?
   05 003BA    693003BE            BNEZ    ILYF            NO, NOT A LEAP
   05 003BB    02200050 A   ILYT   LCI     5
   05 003BC    8A100000 A          PLM,1   *0
   05 003BD    68020001 A          B       1,1             IS A LEAP YEAR
   05 003BE    02200050 A   ILYF   LCI     5
   05 003BF    8A100000 A          PLM,1   *0
   05 003C0    68020000 A          B       0,1             NOT A LEAP YEAR
```

```
                          **
                          ** COMPUTE JULIAN DAY OF YEAR
                          ** R1: LINK
                          ** R2: YEAR, R3: MONTH, R4: DAY
                          ** RETURNS R5: JULIAN DAY OF YEAR (0-365)
                          ** **** DOES NOT VALIDATE PARAMETERS *****
                          **
         05 003C1         CALCDOY EQU     $
   05 003C1    02200030 A          LCI     3
   05 003C2    8B600000 A          PSM,6   *0              SAVE WORKING REGISTERS
                          **
   05 003C3    32500004 A          LW,5    4               START WITH DAY OF MONTH
   05 003C4    205FFFFF A          AI,5    -1              MAKE ZERO BASE
   05 003C5    32600003 A          LW,6    3               COPY MONTH
   05 003C6    646003C8            BDR,6   NOTJAN
   05 003C7    680003D4            B       DOYDONE         JULIAN DAY = DAY OF JANUARY
   05 003C8    22700000 A   NOTJAN LI,7    0
   05 003C9    728E03A8     DOYMON LB,8    MONTHDAY,7
   05 003CA    30500008 A          AW,5    8               ACCUMULATE MONTH
   05 003CB    20700001 A          AI,7    1
   05 003CC    646003C9            BDR,6   DOYMON          CONTINUE FOR MONTHS < MONTH
   05 003CD    32700001 A          LW,7    1               COPY LINK
   05 003CE    6A1003AE            BAL,1   ISLEAPYEAR
   05 003CF    680003D3            B       NOTLEAP
   05 003D0    21300002 A          CI,3    2               FEBRUARY?
   05 003D1    682003D3            BLE     NOTLEAP         YES, DONE
   05 003D2    20500001 A          AI,5    1               COUNT FEB 29
   05 003D3    32100007 A   NOTLEAP LW,1   7               RETRIEVE RETURN ADDRESS
   05 003D4    02200030 A   DOYDONE LCI    3
   05 003D5    8A600000 A          PLM,6   *0              RESTORE WORKING REGISTERS
   05 003D6    E8000001 A          B       *1              DONE
```

```
                          **
                          ** GET DAY OF WEEK FROM JULIAN DAY
                          ** R1: LINK
                          ** R5: JULIAN DAY
                          ** RETURNS R4: DAY OF WEEK(0-6, MON-SUN)
                          ** RETURNS SKIPPING IF VALID INPUT
                          **
         05 003D7          JD:DOW  EQU     $
   05 003D7    22400000 A          LI,4    0               MAKE IT 64 BITS
   05 003D8    20500000 A          AI,5    0               CHECK PARAM
   05 003D9    68220000 A          BLEZ    0,1             INVALID
   05 003DA    315003A7            CW,5    JDMAX
   05 003DB    69220000 A          BG      0,1             INVALID
   05 003DC    89500000 A          PSW,5   *0              SAVE JD
   05 003DD    364003A1            DW,4    DPWEEK          GET DOW IN R4
   05 003DE    88500000 A          PLW,5   *0              RESTORE JD
   05 003DF    68020001 A          B       1,1             DONE
```

```
                              **
                              ** INTERPRET JULIAN DAY
                              ** R1: LINK
                              ** R5: JULIAN DAY SINCE JAN 1, 0001
                              ** RETURNS R2,R3,R4: YEAR, MONTH, DAY
                              **
         05 003E0             JD:INTERPRET EQU $
                              ** CHECK PARAMETERS
   05 003E0    22200001 A           LI,2    1
   05 003E1    22300000 A           LI,3    0
   05 003E2    22400000 A           LI,4    0
   05 003E3    20500000 A           AI,5    0
   05 003E4    E9100001 A           BLZ     *1              NOT VALID
   05 003E5    315003A7             CW,5    JDMAX
   05 003E6    E9200001 A           BG      *1              NOT VALID
   05 003E7    89100000 A           PSW,1   *0              SAVE RETURN
   05 003E8    89500000 A           PSW,5   *0              SAVE INPUT
   05 003E9    364003A6             DW,4    DPQUADCENT
   05 003EA    375003A0             MW,5    YPQUADCENT
   05 003EB    30200005 A           AW,2    5
   05 003EC    32500004 A           LW,5    4               REMAINING DAYS
   05 003ED    22400000 A           LI,4    0               CLEAR TOP OF DIVIDEND
   05 003EE    364003A4             DW,4    DPCENTURY
                              ** The preceding calculation is 4 only for the last day of a quad century,
                              ** (eg. 1600, 2000), these centuries have an extra day.
   05 003EF    21500004 A           CI,5    4               CHECK
   05 003F0    693003F5             BNE     INTNLD          B/NO
   05 003F1    20200190 A           AI,2    400             RETURN SPECIAL CASE
   05 003F2    2230000C A           LI,3    12
   05 003F3    2240001F A           LI,4    31
   05 003F4    6800040D             B       INTDONE
   05 003F5    3750039F     INTNLD  MW,5    YPCENTURY
   05 003F6    30200005 A           AW,2    5               NOW HAVE BASE CENTURY
   05 003F7    32500004 A           LW,5    4               REMAINING DAYS
   05 003F8    22400000 A           LI,4    0               CLEAR TOP OF DIVIDEND
   05 003F9    364003A3             DW,4    DPQUADYEAR
   05 003FA    25500002 A           SLS,5   2               QUADYEARS
   05 003FB    30200005 A           AW,2    5
   05 003FC    32500004 A           LW,5    4               REMAINING DAYS
```

```
   1006   05 003FD    22400000 A          LI,4    0
   1007   05 003FE    364003A2            DW,4    DPYEAR
   1008   05 003FF    30200005 A          AW,2    5               NOW HAVE YEAR IN R2
   1009   05 00400    725603A8     INTMON LB,5    MONTHDAY,3
   1010   05 00401    21300001 A          CI,3    1               FEBRUARY?
   1011   05 00402    69300406            BNE     NOTFEB          B/NO
   1012   05 00403    6A1003AE            BAL,1   ISLEAPYEAR      HOW MANY DAYS?
   1013   05 00404    68000406            B       $+2             28
   1014   05 00405    20500001 A          AI,5    1               29
   1015   05 00406    31400005 A   NOTFEB CW,4    5               IS THIS THE MONTH?
   1016   05 00407    6910040B            BL      GOTMON          B/YES
   1017   05 00408    38400005 A          SW,4    5               REMOVE THOSE DAYS
   1018   05 00409    20300001 A          AI,3    +1
   1019   05 0040A    68000400            B       INTMON
   1020   05 0040B    20300001 A   GOTMON AI,3    +1              1-BASE
   1021   05 0040C    20400001 A          AI,4    +1              1-BASE
   1022   05 0040D    88500000 A   INTDONE PLW,5   *0              RESTORE ORIGINAL INPUT
   1023   05 0040E    88100000 A          PLW,1   *0              AND RETURN ADDRESS
   1024   05 0040F    E8000001 A          B       *1              DONE
```

```
                          **
                          **
                          ** CONVERT DATE TO JULIAN DAY
                          ** R1: LINK
                          ** R2: YEAR, R3: MONTH, R4: DAY
                          ** RETURNS R5: JULIAN DAY STARTING WITH JAN 1, 0001.
                          **         CC: SET FOR BL ON ERROR
                          **
        05 00410          JD:COMPUTE EQU $
                          ** CHECK PARAMETERS
05 00410    225FFFFF A          LI,5    -1              INVALID DATE
05 00411    21200000 A          CI,2    0
05 00412    6820042B            BLE     JDCX            YEAR TOO SMALL
05 00413    2120722B A          CI,2    YEARMAX
05 00414    6920042B            BG      JDCX            YEAR TOO BIG
05 00415    21300000 A          CI,3    0
05 00416    6820042B            BLE     JDCX            MONTH TOO SMALL
05 00417    2130000C A          CI,3    12
05 00418    6920042B            BG      JDCX            MONTH TOO BIG
05 00419    21400000 A          CI,4    0
05 0041A    6820042B            BLE     JDCX            DAY TOO SMALL
                          **
05 0041B    89100000 A          PSW,1   *0              SAVE LINK
05 0041C    32100003 A          LW,1    3               COPY MONTH
05 0041D    21100002 A          CI,1    2               FEBRUARY?
05 0041E    69300427            BNE     JDCNF           B/NO
05 0041F    6A1003AE            BAL,1   ISLEAPYEAR
05 00420    68000424            B       JDCNL           B/NO
                          **
05 00421    2140001D A          CI,4    29
05 00422    6820042D            BLE     JDPOK           B/OK
05 00423    6800042A            B       JDCDE           RETURN ERROR
                          **
05 00424    2140001C A   JDCNL  CI,4    28
05 00425    6820042D            BLE     JDPOK           B/OK
05 00426    6800042A            B       JDCDE           RETURN ERROR
                          **
05 00427    201FFFFF A   JDCNF  AI,1    -1              =>ZERO BASE
05 00428    714203A8            CB,4    MONTHDAY,1      VALID DAY OF MONTH?
```

```
05 00429    6820042D            BLE     JDPOK           B/YES
                          ** DAY IS TOO LARGE FOR THE MONTH/YEAR:
05 0042A    88100000 A   JDCDE   PLW,1   *0              RETRIEVE LINK
05 0042B    225FFFFF A   JDCX    LI,5    -1              SETTING CC
05 0042C    E8000001 A           B       *1              GET OUT
                          **
                          ** SAVE PARAMETERS
05 0042D    02200030 A   JDPOK   LCI     3
05 0042E    8B200000 A           PSM,2   *0
                          **
                          ** SAVE WORKING REGISTERS
05 0042F    02200060 A           LCI     6
05 00430    8B600000 A           PSM,6   *0
05 00431    22600000 A           LI,6    0
05 00432    32700002 A           LW,7    2
                          ** ADJUST TO YEAR 0001 BASE
05 00433    207FFFFF A           AI,7    -1
05 00434    366003A0             DW,6    YPQUADCENT      6: YEARS IN QUADCENT, 7: QUADCENT
05 00435    22800000 A           LI,8    0
05 00436    32900006 A           LW,9    6
05 00437    3680039F             DW,8    YPCENTURY       8: YEAR IN CENTURY, 9: C(0-3) OF QC
05 00438    22B00003 A           LI,11   3
05 00439    4BB00008 A           AND,11  8               YEARS SINCE LEAP
05 0043A    2580007E A           SLS,8   -2              QUAD-YEARS
                          ** FIRST, GET JD OF TARGET YEAR
05 0043B    6A1003C1             BAL,1   CALCDOY
05 0043C    37B003A2             MW,11   DPYEAR
05 0043D    3050000B A           AW,5    11              DAYS SINCE LAST LEAP
05 0043E    32B00008 A           LW,11   8               COPY QUADS-YEARS IN CENTURY
05 0043F    37B003A3             MW,11   DPQUADYEAR
05 00440    3050000B A           AW,5    11
05 00441    379003A4             MW,9    DPCENTURY
05 00442    30500009 A           AW,5    9
05 00443    377003A6             MW,7    DPQUADCENT
05 00444    30500007 A           AW,5    7
                          ** RESTORE WORKING REGISTERS
05 00445    02200060 A           LCI     6
05 00446    8A600000 A           PLM,6   *0
                          ** RESTORE PARAMETERS
```

```
   1104   05 00447    02200040 A          LCI     4
   1105   05 00448    8A100000 A          PLM,1   *0
   1106   05 00449    21500000 A          CI,5    0               SET CONDITION CODE
   1107   05 0044A    E8000001 A          B       *1              DONE
```

```
                          **
                          ** DCOMP IS CALLED BY CVDATE TO FETCH THE NEXT COMPONENT
                          ** THIS CODE IS AWFUL
                          ** R11: LINK
                          ** RETURNS: NO SKIP: NO MORE DATA,
                          **      SKIP 1: NUMERIC (R5),
                          **      SKIP 2: TEXT: EITHER SINGLE CHAR; OR 3-BYTE TEXTC
                          **
05 0044B    08406160 A    TXCPUN  TEXTC   ' /-+,()'''
05 0044C    4E6B4D5D A
05 0044D    7D404040 A
                          **
05 0044E    20600000 A    DCOMP   AI,6    0               CHECK LENGTH
05 0044F    E820000B A            BLEZ    *11             NOTHING LEFT
05 00450    89200000 A            PSW,2   *0
05 00451    89300000 A            PSW,3   *0
05 00452    22300000 A            LI,3    0
05 00453    722E0000 A            LB,2    0,7             GET A CHAR
05 00454    212000F0 A            CI,2    X'F0'           NUMBER?
05 00455    69100465              BL      DCOMPNN         NO
05 00456    212000F9 A    DCOMPCN CI,2    X'F9'
05 00457    6920047A              BG      DCOMPX          INVALID CHAR
05 00458    202FFF10 A            AI,2    -X'F0'          BINARY
05 00459    2330000A A            MI,3    10              RADIX
05 0045A    30300002 A            AW,3    2               ACCUMULATE
05 0045B    20700001 A            AI,7    +1              POINT AT NEXT
05 0045C    206FFFFF A            AI,6    -1              COUNT CHAR
05 0045D    68200463              BLEZ    DCOMPR1         RETURN NUMERIC
05 0045E    21300059              CI,3    I10000000       >7 DIGITS?
05 0045F    68100463              BGE     DCOMPR1         YES/RETURN WHAT WE HAVE
05 00460    722E0000 A            LB,2    0,7
05 00461    212000F0 A            CI,2    X'F0'           MORE NUMBER?
05 00462    68100456              BGE     DCOMPCN         B/YES
05 00463    20B00001 A    DCOMPR1 AI,11   +1              RETURN SKIPPING 1
05 00464    68000479              B       DCOMPR
                          ** COME HERE IF NOT NUMERIC
05 00465    4920005A      DCOMPNN OR,2    MCASE           UPSHIFT
05 00466    20700001 A            AI,7    +1
05 00467    206FFFFF A            AI,6    -1
```

```
05 00468    7250044B            LB,5    TXCPUN          GET NUMBER OF PUNCTUATION CHARS
05 00469    712A044B     DCOMPP CB,2    TXCPUN,5        IS IT PUNCTUATION?
05 0046A    68300477            BE      DCOMPCH         YES/RETURN CHAR
05 0046B    64500469            BDR,5   DCOMPP          CHECK NEXT
05 0046C    22300003 A   DCOMP3 LI,3    3
05 0046D    22500002 A          LI,5    2               DO TWO CHARS
05 0046E    25300008 A   DCOMPT SLS,3   8
05 0046F    49300002 A          OR,3    2
05 00470    722E0000 A          LB,2    0,7             GET NEXT
05 00471    4920005A            OR,2    MCASE           SHIFT
05 00472    20700001 A          AI,7    +1
05 00473    206FFFFF A          AI,6    -1
05 00474    6910047A            BLZ     DCOMPX
05 00475    6450046E            BDR,5   DCOMPT
05 00476    25300008 A          SLS,3   8
                         ** SINGLE BYTE PUNCTUATION
05 00477    49300002 A   DCOMPCH OR,3   2
05 00478    20B00002 A          AI,11   +2              RETURN SKIPPING 2
05 00479    32500003 A   DCOMPR LW,5    3               GET RETURN VALUE
05 0047A    88300000 A   DCOMPX PLW,3   *0
05 0047B    88200000 A          PLW,2   *0
05 0047C    E800000B A          B       *11
```

```
                          **
                          ** CONVERT DATE TEXT TO JULIAN DAY
                          ** R1: LINK
                          ** R8: TEXT: MM/DD/[YY]YY, DD[-]MMM[-][YY]YY, YYYY[-]MM[-]DD,
                          **          YYYYMMDD, JD[:|+|-]N
                          ** RETURNS R11: JULIAN DAY
                          **
05 0047D    03D1C47A A    TXCJD   TEXTC   'JD:'
05 0047E    03D1C460 A    TXCJDM  TEXTC   'JD-'
05 0047F    03D1C44E A    TXCJDP  TEXTC   'JD+'
                          **
      05 00480            JD:FROMTEXT EQU $
05 00480    022000A0 A            LCI     10
05 00481    8B100000 A            PSM,1   *0              SAVE R1-R10
05 00482    22200000 A            LI,2    0
05 00483    22300000 A            LI,3    0
05 00484    22400000 A            LI,4    0
05 00485    72600008 A            LB,6    8               LENGTH
05 00486    3270005E              LW,7    MFFFFFF         24 BITS
05 00487    4B700008 A            AND,7   8               ADDRESS
05 00488    6AB0044E      CVDGC   BAL,11  DCOMP           GET A COMPONENT IN R5
05 00489    680004E7              B       CVDFAIL         NOTHING LEFT
05 0048A    680004A0              B       CVDNUM          NUMBER
                          ** COME HERE WITH UP TO 3 BYTES OF TEXTC IN R5
05 0048B    73000005 A            MTB,+0  5               SINGLE CHARACTER PUNCTUATION ?
05 0048C    68300488              BEZ     CVDGC           B/YES, IGNORE
05 0048D    22A00000 A            LI,10   0               JULIAN DAY TYPE
05 0048E    3150047D              CW,5    TXCJD           JULIAN DAY#?
05 0048F    683004C9              BE      CVDJD           B/YES
05 00490    20A00001 A            AI,10   +1
05 00491    3150047E              CW,5    TXCJDM          TODAY LESS N?
05 00492    683004C9              BE      CVDJD           B/YES
05 00493    20A00001 A            AI,10   +1
05 00494    3150047F              CW,5    TXCJDP          TODAY PLUS N?
05 00495    683004C9              BE      CVDJD           B/YES
                          ** NOT JULIAN DAY FORM:
05 00496    20300000 A            AI,3    0               ALREADY HAVE MONTH?
05 00497    6820049B              BLEZ    CVDMMM          NO, GET IT
05 00498    20400000 A            AI,4    0               ALREADY HAVE DAY?
```

```
05 00499    692004E7            BGZ     CVDFAIL         YES, FAIL
05 0049A    32400003 A          LW,4    3               MUST BE DD-MMM-YY
05 0049B    2230000C A   CVDMMM  LI,3    12
05 0049C    3157FFFF S   CVDFM   CW,5    UT:MONTH3-1,3
05 0049D    683004B3            BE      CVDNX           GOT MONTH
05 0049E    6430049C            BDR,3   CVDFM           KEEP LOOKING
05 0049F    680004E7            B       CVDFAIL         NOT FOUND
                         ** COME HERE WITH A NUMBER IN R5
05 004A0    31500058     CVDNUM  CW,5    I1000000        7+ DIGITS?
05 004A1    681004DB            BGE     CVDISO
05 004A2    21500064 A          CI,5    100             >2 DIGITS?
05 004A3    681004B0            BGE     CVDY            YES, MUST BE YEAR
05 004A4    20300000 A          AI,3    0               HAVE MONTH?
05 004A5    692004A8            BGZ     CVDN2           YES
05 004A6    32300005 A          LW,3    5               ASSUME MONTH
05 004A7    680004B3            B       CVDNX           GO GET ANOTHER COMPNENT
05 004A8    20400000 A   CVDN2   AI,4    0               HAVE DAY?
05 004A9    692004AC            BGZ     CVDYY           YES, MUST BE YEAR
05 004AA    32400005 A          LW,4    5               NOW HAVE MONTH AND DAY, MAYBE YEAR
05 004AB    680004B3            B       CVDNX
05 004AC    2050076C A   CVDYY   AI,5    1900
05 004AD    215007B2 A          CI,5    1970            >=70?
05 004AE    681004B0            BGE     CVDY            YES, 20TH
05 004AF    20500064 A          AI,5    100             NO,21ST
05 004B0    20200000 A   CVDY    AI,2    0               HAVE YEAR?
05 004B1    692004E7            BGZ     CVDFAIL         YES, FAIL
05 004B2    32200005 A          LW,2    5               COPY YEAR
                         **      B       CVDNX
05 004B3    20600000 A   CVDNX   AI,6    0               ANYTHING LEFT?
05 004B4    69200488            BGZ     CVDGC           YES, GET NEXT
                         ** VERIFY ALL COMPONENTS
05 004B5    21200000 A          CI,2    0
05 004B6    682004E7            BLEZ    CVDFAIL         MISSING YEAR  ** MAYBE USE CURRENT ?? **
05 004B7    21300000 A          CI,3    0
05 004B8    682004E7            BLEZ    CVDFAIL         MISSING MONTH
05 004B9    2130000C A          CI,3    12
05 004BA    692004E7            BG      CVDFAIL         INVALID MONTH
05 004BB    21400000 A          CI,4    0
05 004BC    682004E7            BLEZ    CVDFAIL         MISSING DAY
```

```
05 004BD    32500003 A          LW,5    3
05 004BE    205FFFFF A          AI,5    -1              ZERO BASE
05 004BF    725A03A8            LB,5    MONTHDAY,5
05 004C0    31400005 A          CW,4    5               CHECK MAX DAYS
05 004C1    682004E1            BLE     CVDCC           GOOD
05 004C2    21300002 A          CI,3    2               IS IT FEB?
05 004C3    693004E7            BNE     CVDFAIL
05 004C4    6A1003AE            BAL,1   ISLEAPYEAR
05 004C5    680004E7            B       CVDFAIL         NOT A LEAP YEAR
05 004C6    2140001D A          CI,4    29
05 004C7    682004E1            BLE     CVDCC           B/OK
05 004C8    680004E7            B       CVDFAIL
                          **
05 004C9    20200000 A  CVDJD   AI,2    +0
05 004CA    693004E7            BNEZ    CVDFAIL
05 004CB    20300000 A          AI,3    +0
05 004CC    693004E7            BNEZ    CVDFAIL
05 004CD    20400000 A          AI,4    +0
05 004CE    693004E7            BNEZ    CVDFAIL
05 004CF    6AB0044E            BAL,11  DCOMP           GET MORE
05 004D0    680004E7            B       CVDFAIL         NOTHING LEFT
05 004D1    680004D3            B       CVDJDN          JD NUMBER
05 004D2    680004E7            B       CVDFAIL         NO GOOD
05 004D3    3220000A A  CVDJDN  LW,2    10              TYPE OF JD
05 004D4    683004E2            BEZ     CVDGJD          ABSOLUTE
05 004D5    6A100526            BAL,1   DT:NOW          GET TODAY'S DATE IN 11
05 004D6    21200001 A          CI,2    1               RELATIVE MINUS?
05 004D7    693004D9            BNE     $+2             NO, ADD OFFSET
05 004D8    3A500005 A          LCW,5   5               NEGATE OFFSET
05 004D9    30B00005 A          AW,11   5               CALCULATE DATE
05 004DA    680004E3            B       CVDJDR          DONE, RETURN
                          **
      05 004DB          CVDISO  EQU     $
05 004DB    22400000 A          LI,4    0
05 004DC    3640039F            DW,4    YPCENTURY       I.E. 100
05 004DD    32300005 A          LW,3    5
05 004DE    22200000 A          LI,2    0
05 004DF    3620039F            DW,2    YPCENTURY
05 004E0    46200003 A          XW,2    3               2=YEAR,3=MON,4=DAY
```

```
                                **
05 004E1    6A100410     CVDCC   BAL,1   JD:COMPUTE      GET JULIAN DAY
05 004E2    32B00005 A   CVDGJD  LW,11   5               COPY THAT
05 004E3    022000A0 A   CVDJDR  LCI     10
05 004E4    8A100000 A           PLM,1   *0              RESTORE R1-R10
05 004E5    20B00000 A           AI,11   0               SET CC
05 004E6    68020000 A           B       0,1             RETURN
                                **
05 004E7    022000A0 A   CVDFAIL LCI     10
05 004E8    8A100000 A           PLM,1   *0              RESTORE R1-R10
05 004E9    22BFFFFF A           LI,11   -1
05 004EA    68020000 A           B       0,1             RETURN
```

```
                          **
                          ** COMPUTE BINARY VALUE FROM 2 EBCDIC DIGITS
                          ** R1: LINK
                          ** R4: 2 EBCDIC DIGITS IN LOW 16 BITS
                          ** RETURNS R4 BINARY VALUE
                          **
        05 004EB          TWODIG  EQU     $
05 004EB    89300000 A          PSW,3   *0              SAVE R3
05 004EC    32300004 A          LW,3    4               COPY DIGITS
05 004ED    4B40005B            AND,4   MF              2ND DIGIT
05 004EE    25300078 A          SLS,3   -8
05 004EF    4B30005B            AND,3   MF              GET 1ST DIGIT
05 004F0    2330000A A          MI,3    10
05 004F1    30400003 A          AW,4    3               RESULT IN 4
05 004F2    88300000 A          PLW,3   *0              RESTORE R3
05 004F3    E8000001 A          B       *1              RETURN
                          **
                          ** CONVERT A CPV FILE SYSTEM DATE TO JULIAN DAY
                          ** R1: LINK
                          ** R4/5: EBCDIC MMDDHHYY
                          ** RETURNS R5: JULIAN DAY, R4: DAY OF WEEK
                          **
        05 004F4          JD:FROMCPV EQU  $
05 004F4    02200030 A          LCI     3
05 004F5    8B100000 A          PSM,1   *0              SAVE R1-R3
05 004F6    32300004 A          LW,3    4               SAVE THIS FOR NOW
05 004F7    32400005 A          LW,4    5               YYMM
05 004F8    4B40005D            AND,4   MFFFF           YY
05 004F9    6A1004EB            BAL,1   TWODIG          GET BINARY YEAR OF CENTURY
05 004FA    32200004 A          LW,2    4
05 004FB    2020076C A          AI,2    1900            START WITH 20TH CENTURY
05 004FC    212007B2 A          CI,2    1970            IF > 70
05 004FD    681004FF            BGE     $+2             YES, 1970-1999
05 004FE    20200064 A          AI,2    100             2000-2069
                          **
05 004FF    32400003 A          LW,4    3               MMDD
05 00500    25400070 A          SLS,4   -16             MM
05 00501    6A1004EB            BAL,1   TWODIG          GET BINARY MONTH
05 00502    46300004 A          XW,3    4
```

```
   1338   05 00503    6A1004EB            BAL,1   TWODIG         GET BINARY DAY
   1339   05 00504    6A100410            BAL,1   JD:COMPUTE     R5 <- JD
   1340   05 00505    6A1003D7            BAL,1   JD:DOW         R4 <- DOW
   1341   05 00506    02000000 A          NOP                    *IMPOSSIBLE*
   1342   05 00507    02200030 A          LCI     3
   1343   05 00508    8A100000 A          PLM,1   *0             RESTORE R1-R3
   1344   05 00509    E8000001 A          B       *1             DONE
```

```
                          **
                          ** CONVERT JD TO THE DATE PART OF THE ISO TEXT, IE YYYY-MM-DD
                          ** R1: LINK
                          ** R2: WA(TEXT BUFFER)
                          ** R5: JULIAN DAY
                          ** RETURNS TEXTC YYYY-MM-DD IN TEXT BUFFER (*R2)
                          **
         05 0050A          JD:TOTEXT EQU   $
   05 0050A    02200090 A          LCI     9
   05 0050B    8B100000 A          PSM,1   *0
   05 0050C    32600002 A          LW,6    2               COPY BUFFER WA
   05 0050D    25600002 A          SLS,6   2               BA
   05 0050E    2290000A A          LI,9    10
   05 0050F    759C0000 A          STB,9   0,6             SET LENGTH
   05 00510    20600001 A          AI,6    +1
   05 00511    6A1003E0            BAL,1   JD:INTERPRET
   05 00512    32500002 A          LW,5    2               GET YEAR
   05 00513    22700004 A          LI,7    4               4 DIGITS PLEASE
   05 00514    6A100364            BAL,1   INT:TEXT$       CONVERT YEAR
   05 00515    20600004 A          AI,6    4
   05 00516    22900060 A          LI,9    '-'
   05 00517    759C0000 A          STB,9   0,6             PUT IN SEPARATOR
   05 00518    20600001 A          AI,6    +1
   05 00519    22700002 A          LI,7    2               2 DIGITS
   05 0051A    32500003 A          LW,5    3               GET MONTH
   05 0051B    6A100364            BAL,1   INT:TEXT$       CONVERT MONTH
   05 0051C    20600002 A          AI,6    +2
   05 0051D    759C0000 A          STB,9   0,6             ANOTHER SEPARATOR
   05 0051E    20600001 A          AI,6    +1
   05 0051F    32500004 A          LW,5    4               GET DAY
   05 00520    6A100364            BAL,1   INT:TEXT$       CONVERT DAY
   05 00521    02200090 A          LCI     9
   05 00522    8A100000 A          PLM,1   *0              GET CALLER'S REGS BACK
   05 00523    68020000 A          B       0,1             RETURN
```

```
                          **
                          ** TIMESTAMPS AND INTERVALS
                          **
                          ** A TIMESTAMP IS A DOUBLEWORD MILLISECONDS SINCE 0:00 JAN 1, 0001.
                          ** AN INTERVAL IS A DOUBLEWORD NUMBER OF MILLISECONDS.
                          **
05 00524    05265C00 A    MSPERDAY DATA   86400000        X'05265C00'
05 00525    0036EE80 A    MSPERHOUR DATA   3600000
                          **
                          ** GET CURRENT JULIAN DAY IN R11, MILLISECOND OF DAY IN R10
                          **
      05 00526            DT:NOW  EQU     $
05 00526    02200090 A            LCI     9
05 00527    8B100000 A            PSM,1   *0              SAVE R1-R9
05 00528    221010C0 A            LI,1    X'10C0'         FPT CODE 10, BITS 8 AND 9
05 00529    55100004 02           STH,1   MTFPT           FIX UP FPT FOR EXTENDED RESULT
05 0052A    04800004 02  ,MTFPT   M:TIME  TIMEBUF,TMS
02 00004    10800000 04
                          ** PROCESS TIME OF DAY FIRST...
05 0052B    72300009 A            LB,3    9               HOUR
05 0052C    2240001F A            LI,4    X'1F'
05 0052D    4B400009 A            AND,4   9               TWO-MILLI'S
05 0052E    25900078 A            SLS,9   -8
05 0052F    2250003F A            LI,5    X'3F'
05 00530    4B500009 A            AND,5   9               SECOND
05 00531    25900078 A            SLS,9   -8
05 00532    2260003F A            LI,6    X'3F'
05 00533    4B600009 A            AND,6   9               MINUTE
                          ** HAVE TIME COMPONENTS, NOW CALCULATE
05 00534    2330003C A            MI,3    60
05 00535    30300006 A            AW,3    6               MINUTE OF DAY
05 00536    2330003C A            MI,3    60
05 00537    30300005 A            AW,3    5               SECOND OF DAY
05 00538    233003E8 A            MI,3    1000            MILLISECONDS
05 00539    3250000A A            LW,5    10              1000TH'S OF MINUTE
05 0053A    2350003C A            MI,5    60              MILLISECONDS
05 0053B    30300005 A            AW,3    5               ...ALMOST THERE
05 0053C    25400001 A            SLS,4   1               MILLI'S SINCE LAST 1000TH
05 0053D    30300004 A            AW,3    4               MILLISECOND OF DAY
```

```
05 0053E    89300000 A          PSW,3   *0              SAVE FOR LATER
                          ** NOW GET JULIAN DAY
05 0053F    52200008 A          LH,2    8               YEAR (0-99)
05 00540    2020076C A          AI,2    1900
05 00541    212007B2 A          CI,2    1970
05 00542    68100544            BGE     $+2             1970-1999
05 00543    20200064 A          AI,2    100             2000-2069
05 00544    4B80005D            AND,8   MFFFF           1-BASED JULIAN DAY OF YEAR
05 00545    208FFFFF A          AI,8    -1              MAKE ZERO-BASED
05 00546    224003AB            LI,4    MONTHDLY        ASSUME LEAPYEAR
05 00547    6A1003AE            BAL,1   ISLEAPYEAR      CHECK
05 00548    224003A8            LI,4    MONTHDAY        NO
05 00549    22300000 A          LI,3    0
05 0054A    F2560004 A   JDTFM  LB,5    *4,3            DAYS IN MONTH
05 0054B    31800005 A          CW,8    5               IS THIS THE ONE?
05 0054C    69100550            BL      JDTGM           B/YES
05 0054D    38800005 A          SW,8    5               REMOVE THOSE DAYS
05 0054E    20300001 A          AI,3    +1              NEXT MONTH
05 0054F    6800054A            B       JDTFM           KEEP LOOKING
05 00550    32400008 A   JDTGM  LW,4    8               DAY OF MONTH (0-N-1)
05 00551    20300001 A          AI,3    +1              1-BASE
05 00552    20400001 A          AI,4    +1              1-BASE
05 00553    6A100410            BAL,1   JD:COMPUTE      GET THE ANSWER
05 00554    32B00005 A          LW,11   5
05 00555    022000A0 A          LCI     10
05 00556    8A100000 A          PLM,1   *0              GET REGS BACK
05 00557    68020000 A          B       0,1             RETURN
```

```
                          **
                          ** GET THE CURRENT TIME
                          ** R1: LINK
                          ** RETURNS: R10,11: TIMESTAMP
                          **
        05 00558          TS:NOW  EQU     $
05 00558    02200030 A            LCI     3
05 00559    8B100000 A            PSM,1   *0              SAVE R1-R3
05 0055A    6A100526              BAL,1   DT:NOW          GET JD(R11) / MS OF DAY(R10)
05 0055B    12200052              LD,2    D0
05 0055C    4630000A A            XW,3    10              CLEAR 10, COPY MS OF DAY
05 0055D    37A00524              MW,10   MSPERDAY        BASE TS FOR TODAY
05 0055E    10A00002 A            AD,10   2               TIMESTAMP TO 2 MILLI RESOLUTION.
05 0055F    02200030 A            LCI     3
05 00560    8A100000 A            PLM,1   *0              RESTORE REGS
05 00561    68020000 A            B       0,1             RETURN
```

```
                          **
                          ** INTERPRET TIMESTAMP
                          ** R1: LINK
                          ** R10,11: TIMESTAMP
                          ** RETURNS: R2,3,4: Y,M,D; R5: JULIAN DAY OF YEAR;
                          **         R6,7,8,9: H,M,S,MS
                          **         R10: MILLISECOND OF DAY; R11: JULIAN DAY.
                          **         CC: SET FOR BL ON ERROR
                          **
        05 00562          TS:INTERPRET EQU $
05 00562    89100000 A          PSW,1   *0              SAVE LINK REGISTER
05 00563    36A00524            DW,10   MSPERDAY        R10: MS IN DAY; R11: JD
05 00564    3250000B A          LW,5    11              COPY JULIAN DAY
05 00565    6A1003E0            BAL,1   JD:INTERPRET    R2-4: Y/M/D
05 00566    6A1003C1            BAL,1   CALCDOY         R5: JULIAN DAY OF YEAR
05 00567    20500001 A          AI,5    +1              MAKE 1-BASED
05 00568    22800000 A          LI,8    0
05 00569    3290000A A          LW,9    10              MS OF DAY
05 0056A    36800057            DW,8    I1000           R8: MS OF SECOND, R9: SECOND OF DAY
05 0056B    32700008 A          LW,7    8
05 0056C    22800000 A          LI,8    0               CLEAR HIGH WORD
05 0056D    36800056            DW,8    I60             R8: SECOND OF MINUTE, R9: MINUTE OF DAY
05 0056E    46700009 A          XW,7    9               R7: MINUTE OF DAY, R9: MS OF SECOND
05 0056F    22600000 A          LI,6    0               CLEAR HIGH WORD
05 00570    36600056            DW,6    I60             R6: MINUTE OF HOUR, R7: HOUR OF DAY
05 00571    46600007 A          XW,6    7               R6-8: H:M:S
05 00572    88100000 A          PLW,1   *0              GET LINK BACK
05 00573    E8000001 A          B       *1              RETURN
```

```
                          **
                          ** COMPUTE TIMESTAMP
                          ** R1: LINK
                          ** R2,3,4: Y,M,D;
                          ** R6,7,8,9: H,M,S,MS
                          ** RETURNS: R10,11: TIMESTAMP
                          **
         05 00574         TS:COMPUTE EQU $
   05 00574    22AFFFFF A          LI,10   -1
   05 00575    22BFFFFF A          LI,11   -1
                          ** CHECK TIME COMPONENTS
   05 00576    21900000 A          CI,9    0
   05 00577    E9100001 A          BL      *1              INVALID
   05 00578    219003E8 A          CI,9    1000
   05 00579    E8100001 A          BGE     *1              INVALID
   05 0057A    21800000 A          CI,8    0
   05 0057B    E9100001 A          BL      *1              INVALID
   05 0057C    2180003C A          CI,8    60
   05 0057D    E8100001 A          BGE     *1              INVALID
   05 0057E    21700000 A          CI,7    0
   05 0057F    E9100001 A          BL      *1              INVALID
   05 00580    2170003C A          CI,7    60
   05 00581    E8100001 A          BGE     *1              INVALID
   05 00582    21600000 A          CI,6    0
   05 00583    E9100001 A          BL      *1              INVALID
   05 00584    21600018 A          CI,6    24
   05 00585    E8100001 A          BGE     *1              INVALID
                          **
   05 00586    02200090 A          LCI     9
   05 00587    8B100000 A          PSM,1   *0              SAVE R1-9
   05 00588    6A100410            BAL,1   JD:COMPUTE      GET JULIAN DAY IN R5
   05 00589    20500000 A          AI,5    0               IS IT OK?
   05 0058A    6910060E            BLZ     TSX             B/NO
   05 0058B    32B00005 A          LW,11   5               STASH IN R11
   05 0058C    2250003C A          LI,5    60
   05 0058D    37500006 A          MW,5    6               MINUTES FROM HOURS
   05 0058E    30500007 A          AW,5    7
   05 0058F    2350003C A          MI,5    60              SECONDS FROM H:M
   05 00590    30500008 A          AW,5    8               SECOND OF DAY
```

```
   1532   05 00591    235003E8 A          MI,5    1000
   1533   05 00592    30500009 A          AW,5    9               MILLISECOND OF DAY
   1534   05 00593    32A00005 A          LW,10   5               STASH IN R10
   1535   05 00594    02200060 A          LCI     6
   1536   05 00595    8A400000 A          PLM,4   *0              RESTORE R4-9
   1537   05 00596    680005A1            B       TSJDMS
```

```
                          **
                          ** CONSTRUCT TIMESTAMP FROM JULIAN DAY AND MILLISECOND
                          ** R1: LINK
                          ** R10: MILLISECOND OF DAY
                          ** R11: JULIAN DAY
                          ** RETURNS: R10,11: TIMESTAMP. CC SET FOR BL ON ERROR
                          **
        05 00597          TS:COMPOSE EQU $
                          ** CHECK PARAMS
05 00597    21A00000 A          CI,10   0
05 00598    691005AA            BL      TSCX            INVALID
05 00599    31A00524            CW,10   MSPERDAY
05 0059A    681005AA            BGE     TSCX            INVALID
05 0059B    21B00000 A          CI,11   0
05 0059C    691005AA            BL      TSCX            INVALID
05 0059D    21B003A7            CI,11   JDMAX
05 0059E    692005AA            BG      TSCX            INVALID
                          **
05 0059F    02200030 A          LCI     3
05 005A0    8B100000 A          PSM,1   *0              SAVE R1-3
05 005A1    3230000A A   TSJDMS  LW,3    10
05 005A2    22200000 A          LI,2    0
05 005A3    22A00000 A          LI,10   0
05 005A4    37A00524            MW,10   MSPERDAY        BASE TS FOR DAY
05 005A5    10A00002 A          AD,10   2
05 005A6    02200030 A          LCI     3
05 005A7    8A100000 A          PLM,1   *0              RESTORE R1-3
05 005A8    21A00000 A          CI,10   0               SET CONDITION CODE
05 005A9    E8000001 A          B       *1              RETURN
                          **
05 005AA    22AFFFFF A   TSCX    LI,10   -1
05 005AB    22BFFFFF A          LI,11   -1
05 005AC    E8000001 A          B       *1              RETURN ERROR VALUE
```

```
                          **
                          ** TS:FROMTEXT
                          ** R1: LINK
                          ** R8: TEXT: DATE AND TIME SEPARATED BY A SPACE, OR,
                          **          TS[:|+|-]N
                          ** RETURNS: R10,11: TIMESTAMP. CC SET FOR BL ON ERROR.
                          **
        05 005AC          TSSFRM  EQU     $-1
05 005AD    00E3E27A A           DATA    'TS:'
05 005AE    00E3E260 A           DATA    'TS-'
05 005AF    00E3E24E A           DATA    'TS+'
                          **
        05 005B0          TS:FROMTEXT EQU $
05 005B0    02200090 A           LCI     9
05 005B1    8B100000 A           PSM,1   *0              SAVE R1-9
                          **
05 005B2    3250005E             LW,5    MFFFFFF
05 005B3    4B500008 A           AND,5   8               BA(INPUT)
05 005B4    22600040 A           LI,6    ' '
05 005B5    72700008 A           LB,7    8               LENGTH OF DATE+TIME
05 005B6    223000F0 A           LI,3    X'F0'           ALSO CHECK FOR NUMERICS
05 005B7    22900000 A           LI,9    0
05 005B8    716A0000 A   TSFNS   CB,6    0,5             GOT SPACE?
05 005B9    683005D3             BE      TSSPC           B/FOUND ONE
05 005BA    713A0000 A           CB,3    0,5             NUMERIC?
05 005BB    692005BD             BG      $+2             B/NO
05 005BC    20900001 A           AI,9    +1              COUNT NUMERICS
05 005BD    20500001 A           AI,5    +1
05 005BE    647005B8             BDR,7   TSFNS           B/MORE CHARS
                          ** NO SPACE IN INPUT, CHECK SPECIAL FORMS
05 005BF    71900008 A           CB,9    8               ALL NUMERIC?
05 005C0    683005D4             BE      TSNSF           B/YES
05 005C1    3250005E             LW,5    MFFFFFF
05 005C2    4B500008 A           AND,5   8               GET ADDRESS OF START
05 005C3    22600000 A           LI,6    0
05 005C4    22700003 A           LI,7    3
05 005C5    724A0000 A   TSSFA   LB,4    0,5             GET ONE
05 005C6    4940005A             OR,4    MCASE           UPPERCASE IT
05 005C7    25600008 A           SLS,6   8               MAKE SPACE
```

```
05 005C8    49600004 A          OR,6    4               ADD CHAR
05 005C9    20500001 A          AI,5    +1              NEXT
05 005CA    647005C5            BDR,7   TSSFA           B/MORE TO GET
05 005CB    22700003 A          LI,7    3
05 005CC    316E05AC     TSSFF  CW,6    TSSFRM,7
05 005CD    683005F9            BE      TSHSF           B/FOUND ONE
05 005CE    647005CC            BDR,7   TSSFF           CHECK NEXT
                         ** NOT A SPECIAL FORM, ASSUME JUST DATE
05 005CF    22A00000 A          LI,10   0               MIDNIGHT
05 005D0    6A100480            BAL,1   JD:FROMTEXT     DATE
05 005D1    6910060E            BL      TSX             B/INVALID INPUT
05 005D2    6800060B            B       TSRET
                         **
05 005D3    72700008 A   TSSPC  LB,7    8               GET LENGTH AGAIN
                         **
                         ** TOTALLY NUMERIC. ACCEPT YYYYMMDD[HH[MM[SS[MMM]]]]
                         ** R9 = LENGTH
05 005D4    22700008 A   TSNSF  LI,7    8
05 005D5    75700008 A          STB,7   8
05 005D6    6A100480            BAL,1   JD:FROMTEXT     GET JULIAN DAY
05 005D7    6910060E            BL      TSX             B/INVALID
05 005D8    22200000 A          LI,2    0               INITIALIZE TOD
05 005D9    3230000B A          LW,3    11              COPY JD
05 005DA    209FFFF8 A          AI,9    -8
05 005DB    682005F7            BLEZ    TSNC            DONE, COMPOSE TS
05 005DC    22700002 A          LI,7    2
05 005DD    75700008 A          STB,7   8
05 005DE    6A100282            BAL,1   INT:FROMTEXT    GET HOUR
05 005DF    6800060E            B       TSX             B/INVALID
05 005E0    23B00525            MI,11   MSPERHOUR       CONVERT TO MS
05 005E1    3020000B A          AW,2    11              ACCUMULATE
05 005E2    209FFFFE A          AI,9    -2
05 005E3    682005F7            BLEZ    TSNC            DONE, COMPOSE
05 005E4    20800002 A          AI,8    +2
05 005E5    6A100282            BAL,1   INT:FROMTEXT    GET MINUTE
05 005E6    6800060E            B       TSX             B/INVALID
05 005E7    23B0EA60 A          MI,11   60000           CONVERT TO MS
05 005E8    3020000B A          AW,2    11              ACCUMULATE
05 005E9    209FFFFE A          AI,9    -2
```

```
05 005EA    682005F7            BLEZ    TSNC            DONE, COMPOSE
05 005EB    20800002 A          AI,8    +2
05 005EC    6A100282            BAL,1   INT:FROMTEXT    GET SECOND
05 005ED    6800060E            B       TSX             B/INVALID
05 005EE    23B003E8 A          MI,11   1000            CONVERT TO MS
05 005EF    3020000B A          AW,2    11              ACCUMULATE
05 005F0    209FFFFE A          AI,9    -2
05 005F1    682005F7            BLEZ    TSNC            DONE, COMPOSE
05 005F2    20800002 A          AI,8    +2
05 005F3    73100008 A          MTB,+1  8               THREE BYTES LONG
05 005F4    6A100282            BAL,1   INT:FROMTEXT    GET MILLISECOND
05 005F5    6800060E            B       TSX             B/INVALID
05 005F6    3020000B A          AW,2    11              ACCUMULATE
05 005F7    12A00002 A   TSNC   LD,10   2               COPY TS AND JD
05 005F8    6800060B            B       TSRET           COMPOSE AND RETURN
                         **
                         ** OTHER SPECIAL FORMS
                         ** R7: 1=ABSOLUTE, 2=RELATIVE MINUS, 3=RELATIVE PLUS
05 005F9    22A00000 A   TSHSF  LI,10   0
05 005FA    22B00000 A          LI,11   0
05 005FB    20800003 A          AI,8    3               POINT AT 4TH CHAR
05 005FC    73D00008 A          MTB,-3  8               ADJUST COUNT
05 005FD    68300600            BEZ     TSREL           NONE, USE ZERO
05 005FE    6A100282            BAL,1   INT:FROMTEXT    GET INPUT INTEGER PART
05 005FF    6800060E            B       TSX             INVALID
05 00600    1220000A A   TSREL  LD,2    10              COPY NUMERIC PART
05 00601    207FFFFE A          AI,7    -2
05 00602    69100607            BLZ     TSGTS           HAVE TIMESTAMP
05 00603    69200605            BGZ     $+2
05 00604    1A200002 A          LCD,2   2               MAKE NEGATIVE
05 00605    6A100558            BAL,1   TS:NOW          GET CURRENT TS
05 00606    10A00002 A          AD,10   2               ADJUST BY - OR + OFFSET
05 00607    02200090 A   TSGTS  LCI     9
05 00608    8A100000 A          PLM,1   *0
05 00609    21A00000 A          CI,10   0               SET CC
05 0060A    E8000001 A          B       *1              RETURN
                         **
05 0060B    02200060 A   TSRET  LCI     6
05 0060C    8A400000 A          PLM,4   *0              RESTORE R2-9
```

```
05 0060D    680005A1            B       TSJDMS          GO COMPOSE TS
                           **
05 0060E    02200090 A   TSX     LCI     9
05 0060F    8A100000 A           PLM,1   *0              RESTORE ALL REGS
05 00610    22AFFFFF A           LI,10   -1
05 00611    22BFFFFF A           LI,11   -1              SETTING CC
05 00612    E8000001 A           B       *1
```

```
                          **
                          ** TS:TOTEXT
                          ** R1: LINK
                          ** R2: WA(10 WORD BUFFER FOR TEXTC RESULT)
                          **     HH:MM:SS, WEDNESDAY, SEPTEMBER 29, 2026
                          ** R10,11: TIMESTAMP
                          **
                          **
         05 00613         TS:TOTEXT EQU   $
   05 00613    022000D0 A          LCI     13
   05 00614    8B100000 A          PSM,1   *0              SAVE R1-13
   05 00615    32D00002 A          LW,13   2               COPY BUFFER WA
                          **
   05 00616    36A00524            DW,10   MSPERDAY        R11=JD, R10=MS OF DAY
   05 00617    3250000A A          LW,5    10
   05 00618    36500057            DW,5    I1000           LOSE MS
   05 00619    22400000 A          LI,4    0
   05 0061A    36400056            DW,4    I60             R4=SS
   05 0061B    32C00004 A          LW,12   4               SAVE THAT
   05 0061C    22400000 A          LI,4    0
   05 0061D    36400056            DW,4    I60
   05 0061E    32600002 A          LW,6    2               WA(BUFFER)
   05 0061F    4B60005E            AND,6   MFFFFFF         CLEAN IT
   05 00620    25600002 A          SLS,6   2               BA(BUFFER)
   05 00621    20600001 A          AI,6    +1              SKIP LENGTH BYTE
   05 00622    22700002 A          LI,7    2
   05 00623    6A100364            BAL,1   INT:TEXT$       CONVERT HOUR
   05 00624    20600002 A          AI,6    +2
   05 00625    2290007A A          LI,9    ':'
   05 00626    759C0000 A          STB,9   0,6
   05 00627    20600001 A          AI,6    +1
   05 00628    32500004 A          LW,5    4               MM
   05 00629    6A100364            BAL,1   INT:TEXT$       CONVERT MINUTE
   05 0062A    20600002 A          AI,6    +2
   05 0062B    759C0000 A          STB,9   0,6
   05 0062C    20600001 A          AI,6    +1
   05 0062D    3250000C A          LW,5    12              SS
   05 0062E    6A100364            BAL,1   INT:TEXT$
   05 0062F    20600002 A          AI,6    +2
```

```
05 00630    22200144            LI,2    BA(COMMASPACE)
05 00631    32300006 A          LW,3    6               DESTINATION
05 00632    73200003 A          MTB,+2  3               SET LENGTH
05 00633    61200000 A          MBS,2   0               MOVE COMMA, SPACE
                            ** DESTINATION TEXTC LENGTH IS NOT YET SET
                            ** TIME IS DONE, NOW DO DATE
05 00634    3250000B A          LW,5    11              JD
05 00635    6A1003D7            BAL,1   JD:DOW          R4=DOW
05 00636    22400007 A          LI,4    7               ERROR
05 00637    32280049            LW,2    UT:DAYNAME,4
05 00638    F2900002 A          LB,9    *2              LENGTH
05 00639    75900003 A          STB,9   3               SET LENGTH
05 0063A    25200002 A          SLS,2   2
05 0063B    61200001 A          MBS,2   1               MOVE DAY NAME TEXT
05 0063C    22200144            LI,2    BA(COMMASPACE)
05 0063D    73200003 A          MTB,+2  3               SET LENGTH
05 0063E    61200000 A          MBS,2   0               MOVE THAT
05 0063F    2090000C A          AI,9    +12             TOTAL LENGTH OF OUTPUT SO FAR
05 00640    32700003 A          LW,7    3               DESTINATION BA
05 00641    3250000B A          LW,5    11              JULIAN DAY
05 00642    6A1003E0            BAL,1   JD:INTERPRET
05 00643    32660026            LW,6    UT:MONTHNAME-1,3 GET WA(TEXTC MONTHNAME)
05 00644    25600002 A          SLS,6   2
05 00645    721C0000 A          LB,1    0,6             LENGTH
05 00646    75100007 A          STB,1   7               SET MOVE LENGTH
05 00647    61600001 A          MBS,6   1               MOVE MONTHNAME
05 00648    30900001 A          AW,9    1               UPDATE LENGTH SO FAR
05 00649    20900009 A          AI,9    +9              FINAL LENGTH WILL BE...
05 0064A    F590000D A          STB,9   *13             PUT IN DESTINATION COUNT
05 0064B    22900040 A          LI,9    ' '             NEED A SPACE AFTER MONTH
05 0064C    759E0000 A          STB,9   0,7
05 0064D    32600007 A          LW,6    7               COPY DESTINATION
05 0064E    20600001 A          AI,6    +1
05 0064F    32500004 A          LW,5    4               GET DAY OF MONTH
05 00650    22700002 A          LI,7    2               2 DIGITS
05 00651    6A100364            BAL,1   INT:TEXT$
05 00652    20600002 A          AI,6    +2
05 00653    2210006B A          LI,1    ','
05 00654    751C0000 A          STB,1   0,6
```

```
   1776   05 00655    20600001 A          AI,6    +1
   1777   05 00656    759C0000 A          STB,9   0,6
   1778   05 00657    20600001 A          AI,6    +1
   1779   05 00658    32500002 A          LW,5    2               GET YEAR
   1780   05 00659    22700004 A          LI,7    4               4 DIGITS PLEASE
   1781   05 0065A    6A100364            BAL,1   INT:TEXT$       CONVERT YEAR
   1782                            ** DONE
   1783   05 0065B    022000D0 A          LCI     13
   1784   05 0065C    8A100000 A          PLM,1   *0              RESTORE REGS
   1785   05 0065D    E8000001 A          B       *1              RETURN
   1786                            **
   1787                                   END

CONTROL SECTION SUMMARY: 01 00000   PT 0     02 00005   PT 0     03 00000   PT 1     04 00012   PT 0
                         05 0065E   PT 1
```

```
*    SYMBOL VALUES
        ANLZFX3*05 00316         BUFFER*04 00000          BUFSIZE*00000010         CALCDOY*05 003C1
        CFDIGIT*05 00219         CFDONE*05 0022A          CFFAIL*05 0022B          CFSCALE*05 00226
        COMMASPACE*05 00051      CVDCC*05 004E1           CVDFAIL*05 004E7         CVDFM*05 0049C
        CVDGC*05 00488           CVDGJD*05 004E2          CVDISO*05 004DB          CVDJD*05 004C9
        CVDJDN*05 004D3          CVDJDR*05 004E3          CVDMMM*05 0049B          CVDNUM*05 004A0
        CVDNX*05 004B3           CVDN2*05 004A8           CVDY*05 004B0            CVDYY*05 004AC
        CVFDD*05 00258           CVFEXP*05 0025B          CVFNCP*05 0023A          CVFNDEC*05 00247
        CVFNDN*05 00279          CVFNDP*05 0023D          CVFNDX*05 0027F          CVFND0*05 00240
        CVFNFR*05 00256          CVFNNN*05 0027C          CVFNXT*05 00240          CVFPOSX*05 00266
        CVFXDIG*05 0026A         CVIAC*05 002BE           CVIHEX*05 002A0          CVINCP*05 0028E
        CVINCX*05 002AC          CVINDN*05 002C1          CVINDP*05 00291          CVINDX*05 002C7
        CVIND0*05 00294          CVINH*05 002B4           CVINMD*05 002B6          CVINNN*05 002C4
        CVINXT*05 002A3          CVISM*05 002BD           DCOMP*05 0044E           DCOMPCH*05 00477
        DCOMPCN*05 00456         DCOMPNN*05 00465         DCOMPP*05 00469          DCOMPR*05 00479
        DCOMPR1*05 00463         DCOMPT*05 0046E          DCOMPX*05 0047A          DCOMP3*05 0046C
        DDECALGN*05 0031D        DDECAZ*05 0030C          DDECCF*05 002FE          DDECFIB*05 00332
        DDECFIS*05 00349         DDECFI0*05 0033B         DDECGF*05 00304          DDECIF*05 00311
        DDECIN*05 00316          DDECLEN*05 00326         DDECLX*05 002F3          DDECMNF*05 00356
        DDECMOV*05 0034C         DDECMVD*05 00357         DDECNEG*05 00323         DDECNZ*05 002FA
        DDECZ*05 00337           DDECZP*05 0033A          DDIGIT*05 002DB          DDIGITL*05 002EC
        DDTRUNC*05 00345         DNFRI*05 0003F           DNMON*05 00035           DNSAT*05 00041
        DNSUN*05 00033           DNTHU*05 0003C           DNTUE*05 00037           DNWED*05 00039
        DNXXX*05 00044           DOYDONE*05 003D4         DOYMON*05 003C9          DPCENTURY*05 003A4
        DPLEAPCENT*05 003A5      DPQUADCENT*05 003A6      DPQUADYEAR*05 003A3      DPWEEK*05 003A1
        DPYEAR*05 003A2          DT:NOW-05 00526          D0*05 00052              FCHARD*04 00010
        FCHART*04 00011          FNDCHK*05 001BC          FNDDONE*05 001CE         FNDHIGH*05 001CB
        FNDLOW*05 001C9          FNDLOWER*05 001AF        FNDOUT*05 001CD          FORMAT*04 00000
        FORMATW*00000040         FP:COMPUTE-05 0020F      FP:FROMTEXT-05 0022E
        FP:INTERPRET-05 001D3    FP:POW10-05 001A8        FPDIGITS*05 00194        FPNZ*05 00192
        FPONE*05 000FA           FPP10*05 00060           FPTEN*05 000FC           FPZERO*05 000F8
        FP10P15*05 00118         FP10P30*05 00136         FP4*05 0019C             GODIGIT*05 002E8
        GOTMON*05 0040B          IFADJ*05 001EF           IFDIG*05 001F3           IFFIN*05 0020A
        IFNEGZ*05 00207          IFPOS*05 001E1           IFPOW*05 00202           IFSCD*05 001EE
        IFSCV*05 001E9           IFTZ*05 001FD            IFZERO*05 00208          ILYF*05 003BE
        ILYT*05 003BB            INT:FROMTEXT-05 00282    INT:TEXT$*05 00364       INT:TOTEXT-05 002CA
        INTDONE*05 0040D         INTMON*05 00400          INTNLD*05 003F5          INTO_EURO*00001000
        INTO_LEADZEROS*00002000  INTO_PRTSIGN*00004000    INTO_SEPARATE*00000800   INTO_STARS*00000200
        INTO_TRUNCLEFT*00000400  INTO_UNSIGNED*00008000   INT32:TOTEXT-05 0035E    ISLEAPYEAR*05 003AE
```

```
        ITX$*05 00366           ITXDN*05 0038E          ITXDNF*05 00398         ITXEX*05 0039C
        ITXF*05 00370           ITXLP*05 00372          ITXMV*05 00388          ITXP*05 0036B
        ITXVF*05 00384          I10*05 00054            I1000*05 00057          I1000000*05 00058
        I10000000*05 00059      I20*05 00055            I60*05 00056            JD:COMPUTE-05 00410
        JD:DOW-05 003D7         JD:FROMCPV-05 004F4     JD:FROMTEXT-05 00480
        JD:INTERPRET-05 003E0   JD:TOTEXT-05 0050A      JDCDE*05 0042A          JDCNF*05 00427
        JDCNL*05 00424          JDCX*05 0042B           JDMAX*05 003A7          JDPOK*05 0042D
        JDTFM*05 0054A          JDTGM*05 00550          MCASE*05 0005A          MF*05 0005B
        MFF*05 0005C            MFFFF*05 0005D          MFFFFFF*05 0005E        MNAPR*05 00013
        MNAUG*05 0001A          MNDEC*05 00024          MNFEB*05 0000E          MNJAN*05 0000C
        MNJUL*05 00018          MNJUN*05 00016          MNMAR*05 00011          MNMAY*05 00015
        MNNOV*05 00021          MNOCT*05 0001F          MNSEP*05 0001C          MONTHDAY*05 003A8
        MONTHDLY*05 003AB       MSPERDAY-05 00524       MSPERHOUR-05 00525      MTFPT*02 00004
        NOTFEB*05 00406         NOTJAN*05 003C8         NOTLEAP*05 003D3        S:UTILS*05 00000
        TIMEBUF*04 00000        TS:COMPOSE-05 00597     TS:COMPUTE-05 00574
        TS:FROMTEXT-05 005B0    TS:INTERPRET-05 00562   TS:NOW-05 00558         TS:TOTEXT-05 00613
        TSCX*05 005AA           TSFNS*05 005B8          TSGTS*05 00607          TSHSF*05 005F9
        TSJDMS*05 005A1         TSNC*05 005F7           TSNSF*05 005D4          TSREL*05 00600
        TSRET*05 0060B          TSSFA*05 005C5          TSSFF*05 005CC          TSSFRM*05 005AC
        TSSPC*05 005D3          TSX*05 0060E            TWODIG*05 004EB         TXCJD*05 0047D
        TXCJDM*05 0047E         TXCJDP*05 0047F         TXCPUN*05 0044B         UT:DATA-04 00000
        UT:DAYNAME-05 00049     UT:MONTHNAME-05 00027   UT:MONTH3-05 00000      YEARMAX*0000722B
        YPCENTURY*05 0039F      YPQUADCENT*05 003A0
*    EXTERNAL SYMBOLS
        DT:NOW-DEF     FP:COMPUTE-DEF               FP:FROMTEXT-DEF              FP:INTERPRET-DEF
        FP:POW10-DEF   INT:FROMTEXT-DEF             INT:TOTEXT-DEF               INT32:TOTEXT-DEF
        JD:COMPUTE-DEF               JD:DOW-DEF     JD:FROMCPV-DEF               JD:FROMTEXT-DEF
        JD:INTERPRET-DEF             JD:TOTEXT-DEF  MSPERDAY-DEF   MSPERHOUR-DEF TS:COMPOSE-DEF
        TS:COMPUTE-DEF               TS:FROMTEXT-DEF              TS:INTERPRET-DEF
        TS:NOW-DEF     TS:TOTEXT-DEF  UT:DATA-DEF    UT:DAYNAME-DEF               UT:MONTHNAME-DEF
        UT:MONTH3*DEF
* NO UNDEFINED SYMBOLS
*    ERROR SEVERITY LEVEL: 0
* NO ERROR LINES
```

```
SET M:SI/S:LA
SET M:BO/R:LA
AP SI,SD,BO,LO
```

```
                                 TITLE   'S:LA - LIST FILES, SELECTING BY ATTRIBUTES'
                                 PCC     0
                         *
                         * LA    list files matching a pattern, selected by various optional
                         *       attributes, and optionally delete the matched files.
                         *
                                 SYSTEM  SIG7
                                 SYSTEM  BPM
                                 SYSTEM  SYSPRINT
                                 SYSTEM  SYSUTILS
                         *
                                 REF     J:ACCN
                                 REF     J:CCBUF
                                 REF     J:JIT
                                 REF     JB:CCARS
                                 REF     MATCH
                                 REF     INT:FROMTEXT
                                 REF     JD:FROMTEXT
                                 REF     JD:FROMCPV
                                 REF     TS:NOW
                                 REF     TS:TOTEXT
                                 REF     MSPERDAY
                         *
                         * REGISTER NAMES
                         *
        0000000A         SR3     EQU     10
                         *
                         * STACK
                         *
        0000012C         SSIZE   EQU     300
   01 00000              STACK   RES     SSIZE
                                 BOUND   8
   01 0012C    00000000  SPD     DATA    STACK
   01 0012D    012C     A        DATA,2  SSIZE,0
   01 0012D 2  0000     A
                         *
                         * COUNTERS, ETC.
                         *
                                 BOUND   8
```

```
     39         01 0012E          RUNTIME QDATA,8 0
     40   01 00130                RUNTEXT RES     10
     41         01 0013A          EXPYJD  QDATA,8 0
     42         01 0013C          EXPYDATE QDATA,8 0
     43         01 0013E          SAVEDATE QDATA,8 0
     44   01 00140    00000000 A  TOTFILES DATA   0
     45   01 00141    00000000 A  ERRFILES DATA   0
     46   01 00142    00000000 A  DELFILES DATA   0
     47   01 00143    00000000 A  TOTACS  DATA    0
     48   01 00144    00000000 A  ACFILES DATA    0
     49   01 00145    00000000 A  ACTGRANS DATA   0
     50   01 00146    00000000 A  FILGRANS DATA   0
     51   01 00147    00000000 A  FILRECS DATA    0
     52   01 00148    00000000 A  ORGTYPE DATA    0
     53   01 00149    00000000 A  ISEXPIRED DATA  0
     54   01 0014A                RDACCTS RES     40              MAXIMUM 19 READ ACCOUNTS
     55   01 00172                WRACCTS RES     40              MAXIMUM 19 WRITE ACCOUNTS
     56   01 0019A                EXACCTS RES     40              MAXIMUM 19 EXECUTE ACCOUNTS
     57   01 001C2    00000000 A  ERRCODE DATA    0
     58   01 001C3    00000000 A  SUBCODE DATA    0
     59   01 001C4    00000000 A  BREAK   DATA    0               USER HIT BREAK (NOT IMPLEMENTED YET)
     60                           *
     61   01 001C5    00000000 A  LISTOBC DATA    0               OPEN BRACKET COUNT.
     62   01 001C6    00000000 A  CMDLEN  DATA    0
     63   01 001C7    00000000 A  CMDPOS  DATA    0
     64   01 001C8                CMDBUF  RES     33
     65                           *
     66   01 001E9    034DC35D A  CHARTEXT TEXTC  '(C)'
     67   01 001EA    10C1C3C3 A  ACTEXT  TEXTC   'ACCOUNT ........'
          01 001EB    D6E4D5E3 A
          01 001EC    404B4B4B A
          01 001ED    4B4B4B4B A
          01 001EE    4B404040 A
     68                                  BOUND   8
     69         01 001F0          FPATTERN        QDATA   0,0,0,0,0,0,0,0
     70         01 001F8          FPREFIX QDATA   0,0,0,0,0,0,0,0
     71                                  BOUND   8
     72         01 00200          APATTERN QDATA  0,0,0
     73         01 00203          APREFIX QDATA   0,0,0
```

```
                                  BOUND   8
        01 00206          CURRENTA QDATA  0,0
        01 00208          DATBUF  QDATA   0,0
                          *
01 0020A                  FPARAM  RES     256
01 0030A    FFFFFFFF A    LFRETURN DATA   -1
                          **
01 0030B    14000000 08   ADJUST  GEN,8,24 X'14',F:LF
01 0030C    0000E000 A            DATA    X'E000'
01 0030D    00000000 A            DATA    0
01 0030E                  ADJVLP  RES     24
                          **
                          ** USED TO COUNT RECORDS.
        01 00326          PRECFPT EQU $
01 00326    1D000000 08           GEN,8,24 X'1D',F:LF
01 00327    C0000000 A            DATA    X'C0000000'
01 00328    00000064 A            DATA    100               100 RECS (TO PERMIT BREAKS)
01 00329    0000061F 09           DATA    PRECABN           ABNORMAL ADR
```

```
                                *
                                * OPTIONS
                                *
                                        OPEN    ODATA,OSTATIC
        01 0032A                ODATA   SET     $
06 00000                                CSECT   1
        06 00000                OSTATIC SET     $
01 0032A                                ORG     ODATA
01 0032A
                                *
        00000000                OPTION  CNAME
                                        PROC
                                        LIST    0
                                        OPEN    T,L,C,A
                                ODATA   SET     $
                                  DO NUM(AF)>3
                                AF(NUM(AF)) EQU $
                                  FIN
                                        ORG     OSTATIC
                                  DO NUM(LF)>0
                                LF(1)   EQU     $
                                  FIN
                                  DO NUM(AF)>2
                                T  SET  SCOR(AF(3),BOOL,INTEGER,TEXT,DATE)
                                        DATA    T
                                  ELSE
                                        DATA    0
                                  FIN
                                  DO NUM(AF)>5
                                L       SET     AF(5)
                                  ELSE
                                    DO T>1
                                L       SET     4
                                    ELSE
                                L       SET     1
                                    FIN
                                  FIN
                                  DO NUM(AF)>4
                                C       SET     AF(4)
```

```
                              ELSE
                            C       SET     1
                              FIN
                              DO NUM(AF)>1
                                    DATA    AF(NUM(AF))
                                    DATA,2   L,C
                                    TEXTC   AF(1)
                                    TEXTC   AF(2)
                              ELSE
                                    DATA    0,0,0,0
                              FIN
                                    BOUND   4
                            OSTATIC SET     $
                                    ORG     ODATA
                                    DATA    0               BOOL VALUE, OR COUNT
                              DO T>1
                                DO C
                                    DATA,2      0,0         FLAGS(COMPARISON OPERATOR) AND LENGTH
                                  DO (L+3)/4
                                    DATA    0
                                  FIN
                                FIN
                              FIN
                                    CLOSE   T,L,C,A
                                    LIST    1
                                    PEND
```

```
                            *
                            *         |         |         |         |         |         |         |
                            OPTIONS OPTION  'ACCESSED',;
                                'a specific last open date range',;
                                DATE,4,O:ACCESSED
                             OPTION 'CREATED',;
                                'a specific creation date range',;
                                DATE,4,O:CREATED
                             OPTION 'DELETE!','ANY MATCHED FILES WILL BE DELETED',BOOL,O:DELETE
                             OPTION 'DETAIL',;
                                'Print more file information (RECORDS, R, W, E, etc)',;
                                BOOL,O:DETAIL
                             OPTION 'EXPIRED!','Files that are expired',BOOL,O:EXPIRED
                             OPTION 'EXPIRES',;
                                'a specific expiry date range',;
                                DATE,4,O:EXPIRES
                             OPTION 'GRANULES','the specified size range',;
                                INTEGER,4,4,O:GRAN
                             OPTION 'IGNORE','Ignore any access errors',BOOL,O:IGNORE
                             OPTION 'KEYLENGTH','the specified key size',;
                                INTEGER,4,4,O:KEYLEN
                             OPTION 'MODIFIED',;
                                'a specific modification date range',;
                                DATE,4,O:MODIFIED
                             OPTION 'ORG',;
                                'one of the specified ORGs (C,K, or R)',;
                                TEXT,3,4,O:ORG
                             OPTION 'RECORDS','the specified number of records',;
                                INTEGER,4,8,O:RECORDS
                             OPTION 'SAVED',;
                                'a specific saved date range',;
                                DATE,4,O:SAVED
                             OPTION 'UNSAVED','Files that have not been saved',BOOL,O:UNSAVED
                             OPTION 'VERBOSE','Print debugging information',BOOL,O:VERBOSE
                            ENDOPTS OPTION
                                    CLOSE   ODATA,OSTATIC
```

```
                          **
                          **
07 00000                  F:LA    DSECT   0
      07 00000            F:LA    Q:DCB   FILE,PASS,SYNON
      07 00020            F:LA:AC EQU     F:LA+X'20'
                          **
08 00000                  F:LF    DSECT   0
      08 00000            F:LF    Q:DCB   FILE,PASS,SYNON
                          **
                          **
09 00000                          CSECT   1
                                  BOUND   8
      09 00000            FIRSTFILE EQU   $
09 00000    01010101 A            DATA    X'01010101'             FILENAME
09 00001    01000000 A            DATA    X'01000000'
                          **
                          ** MASKS
                          **
09 00002    08000000 A    M08000000 DATA  X'08000000'
09 00003    0000000F A    M0F     DATA    X'0000000F'
09 00004    0000007F A    M7F     DATA    X'0000007F'
09 00005    000000FF A    MFF     DATA    X'000000FF'
                          **
09 00006    02020202 A    W02020202 DATA  X'02020202'
09 00007    40404040 A    WBLANK  DATA    X'40404040'
```

```
                          **
                          **
                          ** VARIOUS TEXT CONSTANTS
                          **
                          **              |123456789|123456789|123456789|123456789|
                          ACHEAD  QTEXTC  'ORG  GRAN   RECORDS  CREATED     MODIFIE',;
         09 00008                         'D     ACCESSED XS NAME'
         09 00018         CCKR    QTEXT   'CCKR'
   09 00019    08404040 A   BLANK8  TEXTC   '        '
   09 0001A    40404040 A
   09 0001B    40404040 A
   09 0001C    026B4040 A   COMMA   TEXTC   ', '
   09 0001D    01604040 A   DASH    TEXTC   '-'
   09 0001E    034B4B4B A   DOTS    TEXTC   '...'
   09 0001F    04405C5C A   STARS   TEXTC   ' ** '
   09 00020    40404040 A
   09 00021    0840C6C9 A   TXFILES TEXTC   ' FILES, '
   09 00022    D3C5E26B A
   09 00023    40404040 A
         09 00024         TXGRANS QTEXTC  ' TOTAL GRANULES'
         09 00028         TXDONEAC QTEXTC ' DONE AT FILE '
         09 0002C         TXNOMATCH QTEXTC ' NO MATCH '
         09 0002F         RDTEXTC QTEXTC  '  READ  '
         09 00032         WRTEXTC QTEXTC  '  WRITE '
         09 00035         EXTEXTC QTEXTC  '  EXEC  '
         09 00038         ERRPARSE QTEXTC 'UNEXPECTED COMMAND DATA AT POSITION '
         09 00042         FINPARSE QTEXTC 'COMMAND LENGTH '
         09 00046         FILEMUST QTEXTC 'Select files with '
                          **
                          **
```

```
                                **
                                ** OPERATORS - PRETTY HOKEY
                                **
09 0004B    0006     A   OPERTAB DATA,2  6,'<','<=','=','!=','>=','>'
09 0004B 2  004C     A
09 0004C    4C7E     A
09 0004C 2  007E     A
09 0004D    5A7E     A
09 0004D 2  6E7E     A
09 0004E    006E     A


                                **
                                ** CHARACTER TYPE TABLES
         00000000               SPC     EQU     0               SPACE
         00000001               UCA     EQU     1               UPPPER CASE ALPHA
         00000002               LCA     EQU     2               LOWER CASE ALPHA
         00000003               NAT     EQU     3               NATIONAL (E.G. _ $ : ! )
         00000004               DIG     EQU     4               DECIMAL DIGIT
         00000005               AOP     EQU     5               ARITHMETIC OPERATOR
         00000006               ROP     EQU     6               RELATIONAL OPERATOR
         00000007               PUN     EQU     7               PUNCTUATION (INCLUDES .)
         00000008               CTL     EQU     8               CONTROL CHARACTER
                                **
                                        BOUND   8
         09 00050               CHARTYPE EQU    $
09 00050                                DO1     8               FIRST 64 ARE CONTROL CHARACTERS
         09 00050                       QDATA,1 CTL,CTL,CTL,CTL,CTL,CTL,CTL,CTL
         09 00052
         09 00054
         09 00056
         09 00058
         09 0005A
         09 0005C
         09 0005E
                                **
         09 00060                       QDATA,1 SPC,CTL,CTL,CTL,CTL,CTL,CTL,CTL         40-47
         09 00062                       QDATA,1 CTL,CTL,PUN,PUN,ROP,AOP,AOP,AOP         48-4F
         09 00064                       QDATA,1 AOP,CTL,CTL,CTL,CTL,CTL,CTL,CTL         50-57
```

```
    272         09 00066                   QDATA,1 CTL,CTL,NAT,NAT,AOP,AOP,AOP,PUN         58-5F
    273         09 00068                   QDATA,1 AOP,AOP,CTL,CTL,CTL,CTL,CTL,CTL         60-67
    274         09 0006A                   QDATA,1 CTL,PUN,PUN,PUN,AOP,NAT,ROP,PUN         68-6F
    275         09 0006C                   QDATA,1 CTL,CTL,CTL,CTL,CTL,CTL,CTL,CTL         70-77
    276         09 0006E                   QDATA,1 CTL,PUN,NAT,PUN,AOP,PUN,ROP,PUN         78-7F
    277         09 00070                   QDATA,1 CTL,LCA,LCA,LCA,LCA,LCA,LCA,LCA         80-87
    278         09 00072                   QDATA,1 LCA,LCA,CTL,CTL,CTL,CTL,CTL,CTL         88-8F
    279         09 00074                   QDATA,1 CTL,LCA,LCA,LCA,LCA,LCA,LCA,LCA         90-97
    280         09 00076                   QDATA,1 LCA,LCA,CTL,CTL,CTL,CTL,CTL,CTL         98-9F
    281         09 00078                   QDATA,1 CTL,PUN,LCA,LCA,LCA,LCA,LCA,LCA         A0-A7
    282         09 0007A                   QDATA,1 LCA,LCA,CTL,CTL,CTL,CTL,CTL,CTL         A8-AF
    283         09 0007C                   QDATA,1 PUN,CTL,CTL,CTL,CTL,CTL,CTL,CTL         B0-B7
    284         09 0007E                   QDATA,1 CTL,CTL,PUN,PUN,CTL,CTL,CTL,CTL         B8-BF
    285         09 00080                   QDATA,1 PUN,UCA,UCA,UCA,UCA,UCA,UCA,UCA         C0-C7
    286         09 00082                   QDATA,1 UCA,UCA,CTL,CTL,CTL,CTL,CTL,CTL         C8-CF
    287         09 00084                   QDATA,1 PUN,UCA,UCA,UCA,UCA,UCA,UCA,UCA         D0-D7
    288         09 00086                   QDATA,1 UCA,UCA,CTL,CTL,CTL,CTL,CTL,CTL         D8-DF
    289         09 00088                   QDATA,1 PUN,CTL,UCA,UCA,UCA,UCA,UCA,UCA         E0-E7
    290         09 0008A                   QDATA,1 UCA,UCA,CTL,CTL,CTL,CTL,CTL,CTL         E8-EF
    291         09 0008C                   QDATA,1 DIG,DIG,DIG,DIG,DIG,DIG,DIG,DIG         F0-F7
    292         09 0008E                   QDATA,1 DIG,DIG,CTL,CTL,CTL,CTL,CTL,CTL         F8-FF
```

```
                                  BOUND   8
        09 00090          START   EQU     $
                          **      MTW,+1  O:VERBOSE       *** OFF ***
                          ** INITIALIZE THE STACK
09 00090    2200012C 01           LI,0    SPD
                          ** INITIALIZE PRINT MODULE
09 00091    6A100000 X            BAL,1   PB:INIT
09 00092    6A100000 X            BAL,1   PI:SPACE        PRINT A BLANK LINE
09 00093    6A100000 X            BAL,1   TS:NOW          R10/11=TIMESTAMP
09 00094    15A0012E 01           STD,10  RUNTIME         REMEMBER THIS
09 00095    22200130 01           LI,2    RUNTEXT         DESTINATION
09 00096    6A100000 X            BAL,1   TS:TOTEXT       MAKE TEXTC
                          ** SET UP FOR COMMAND SCAN
09 00097    22200000 N            LI,2    BA(J:CCBUF)
09 00098    22300720 01           LI,3    BA(CMDBUF)
09 00099    72700000 X            LB,7    JB:CCARS        COMMAND LENGTH (ONLINE)
09 0009A    33000000 X            MTW,0   J:JIT
09 0009B    6910009E              BLZ     SONLINE
09 0009C    2270004F A            LI,7    79              COMMAND LENGTH (BATCH)
09 0009D    20200001 A            AI,2    +1              SKIP THE BANG
09 0009E    357001C6 01  SONLINE  STW,7   CMDLEN
09 0009F    75700003 A            STB,7   3
09 000A0    61200000 A            MBS,2   0               MOVE COMMAND FROM JIT
09 000A1    207FFFFF A            AI,7    -1              INDEX LAST CHARACTER
09 000A2    721E01C8 01           LB,1    CMDBUF,7        GET IT
09 000A3    2110000D A            CI,1    X'0D'           RETURN CHARACTER
09 000A4    683000A7              BE      CMDCR           B/YES
09 000A5    21100015 A            CI,1    X'15'           NEWLINE?
09 000A6    693000A8              BNE     NOTCR           NO
09 000A7    357001C6 01  CMDCR    STW,7   CMDLEN          TRUNCATE IT
09 000A8    22700000 A   NOTCR    LI,7    0
09 000A9    357001C7 01           STW,7   CMDPOS          SET CURRENT POSITION
09 000AA    357001C5 01           STW,7   LISTOBC         CLEAR BRACKET COUNT
09 000AB    6A1003BF              BAL,1   CMDNXS          GET NEXT COMMAND CHAR, SKIPPING BLANKS
09 000AC    68000195              B       CURACC          NOTHING THERE
                          ** SCAN PAST PROGRAM NAME
                          ** SCAN CI,8    X'0D'           CR?
                          **      BE      CURACC          NO ARGUMENTS, SO LIST CURRENT ACCOUNT
09 000AD    21800040 A   SCAN     CI,8    ' '             BLANK TERMINATES PROGRAM NAME
```

```
09 000AE    683000B2            BE      SCANFNP
09 000AF    6A1003A5            BAL,1   CMDNX           GET NEXT CHARACTER
09 000B0    68000195            B       CURACC          NONE, NO ARGUMENTS
09 000B1    680000AD            B       SCAN
                          * START FILENAME PATTERN
09 000B2    6A1003B2    SCANFNP BAL,1   CMDSKB          SKIP TO NEXT NON-BLANK
09 000B3    68000192            B       SCANDONE        JUST PROGRAM NAME, LIST CURRENT ACCT.
09 000B4    2230001F A          LI,3    31              MAX LENGTH
09 000B5    224001F0 01         LI,4    FPATTERN        DESTINATION
09 000B6    6AB0043D            BAL,11  GETPATTERN
09 000B7    69100192            BCS,1   SCANDONE        FINISHED
09 000B8    2180004B A          CI,8    '.'             ACCOUNT SPECIFIED?
09 000B9    693000BF            BNE     SCANOPT         NO, SEE IF OTHER OPTIONS
09 000BA    6A1003BF            BAL,1   CMDNXS          EAT THE .
09 000BB    22300008 A          LI,3    8               MAX LENGTH
09 000BC    22400200 01         LI,4    APATTERN        DESTINATION
09 000BD    6AB0043D            BAL,11  GETPATTERN
09 000BE    69100192            BCS,1   SCANDONE        FINISHED
                          *
09 000BF    6A1003BF    SCANOPT BAL,1   CMDNXS          NEXT SIGNIFICANT CHARACTER
09 000C0    68000192            B       SCANDONE        NOTHING THERE
09 000C1    2180006B A  SCANCMA CI,8    ','             INTRODUCTION TO OPTIONS?
09 000C2    69300172            BNE     SCANOE          NO, ERROR
09 000C3    33000382 01         MTW,0   O:VERBOSE
09 000C4    683000DA            BEZ     SCANPOP         NO DEBUG
09 000C5    321001C7 01         LW,1    CMDPOS
09 000C6    72B201C8 01         LB,11   CMDBUF,1
09 000C7    22100002 A          LI,1    2
09 000C8    75B201E9 01         STB,11  CHARTEXT,1
                                PB:PRINT (TEXTC,'COMMA POSITION '),(WORD,CMDPOS,PBF(DECIMAL)),;
        09 000C9                        (TEXTC,', CHAR '),(WORD,8,PBF(HEX)),(TEXTC,CHARTEXT)
```

```
                          ** PROCESS ONE OPTION
09 000DA    6A1003B2     SCANPOP BAL,1   CMDSKB          SKIP TO NEXT SIGNIFICANT CHARACTER
09 000DB    68000172             B       SCANOE          NOTHING THERE
09 000DC    2180004D A           CI,8    '('             START OF LIST?
09 000DD    693000E1             BNE     SCANOKW         NO SHOULD BE KEYWORD
09 000DE    331001C5 01          MTW,+1  LISTOBC
09 000DF    6A1003BF             BAL,1   CMDNXS          GO GET NEXT CHAR
09 000E0    68000192             B       SCANDONE
09 000E1    2230000F A   SCANOKW LI,3    15              MAX LENGTH
                          *
                          **  FIND KEYWORD, THEN PROCESS ARGUMENTS
                          **  FINISH OPTION MATCHING ) IF REQUIRED.
                          *
09 000E2    6A100409             BAL,1   GETKEYWORD
09 000E3    68000180             B       SCANKER         UNRECOGNIZED
                          ** R7=WA(OPTION TABLE ENTRY)
09 000E4    322E0000 A           LW,2    0,7             GET TYPE OF OPTION
09 000E5    324E0001 A           LW,4    1,7             OPTION ADDRESS
09 000E6    325E0002 A           LW,5    2,7             LENGTH AND MAX
09 000E7    52600005 A           LH,6    5               LENGHT
09 000E8    22307FFF A           LI,3    X'7FFF'
09 000E9    4B500003 A           AND,5   3               MAX COUNT
09 000EA    32300007 A           LW,3    7
09 000EB    20300003 A           AI,3    3               OPTION NAME
09 000EC    21200004 A           CI,2    4               MAX TYPE
09 000ED    69200172             BG      SCANOE
                          ** DISPATCH R2=TYPE,R3=OPTION NAME,R4=BUFFER WA,R5=MAX COUNT,R6=LENGTH
09 000EE    680400EF             B       OPTDISP,2       DISPATCH
09 000EF    68000172     OPTDISP B       SCANOE          NOT FOUND
09 000F0    680000F4             B       OPTBOOL         A BOOLEAN FLAG
09 000F1    6800010C             B       OPTVAL          AN INTEGER
09 000F2    6800010C             B       OPTVAL          A STRING
09 000F3    6800010C             B       OPTVAL          A DATE
                          *
09 000F4    33180000 A   OPTBOOL MTW,+1  0,4             THIS IS REALLY A COUNTER.
09 000F5    33000382 01          MTW,0   O:VERBOSE
09 000F6    68300154             BEZ     OPTNEXT         NO
       09 000F7                  PB:PRINT (TEXTC,*3),(TEXTC,COMMA),(WORD,*4,PBF(DECIMAL))
09 00103    68000154             B       OPTNEXT
```

```
                            **
                            ** FOR NON BOOLEANS, THERE ARE A LIST OF VALUES
                            **
                            **  0: [ COUNT ]
                            **  REPEAT COUNT TIMES:
                            **  1:  [ OP,LN ] LN IN BYTES
                            **  2:  [ VALUE ]
                            **      ...
                            **  ...:[ OP,LN ] LN IN BYTES
                            **  N:  [ VALUE ]
                            **
                            ** NXENT IS USED TO SKIP THROUGH THE ENTRIES IN THE VALUE LIST
                            **   R1 = LINK, R4 = WA(CURRENT VALUE)
                            **
09 00104    89100000 A   NXENT   PSW,1   *0
09 00105    32180000 A           LW,1    0,4             OP | LEN
09 00106    4B100005             AND,1   MFF             LEN
09 00107    20100007 A           AI,1    7               INCLUDE OP/LEN WORD
09 00108    2510007E A           SLS,1   -2              LENGTH IN WORDS
09 00109    30400001 A           AW,4    1               POINT AT NEXT
09 0010A    88100000 A           PLW,1   *0
09 0010B    68020000 A           B       0,1             RETURN
                            ** PROCESS A COMPARISON OPERATOR AND VALUE
                            ** R2 = 2: INTEGER, 3: STRING, 4:DATE
09 0010C    22900080 A   OPTVAL  LI,9    X'80'           BEGIN BLOCK MARK
09 0010D    31580000 A           CW,5    0,4             FULL ALREADY?
09 0010E    68200169             BLE     ERRFULL
09 0010F    32700004 A           LW,7    4               COUNTER ADDRESS
09 00110    20400001 A           AI,4    1               FIRST VALUE BLOCK
09 00111    328E0000 A           LW,8    0,7             GET NUMBER OF BLOCKS IN USE
09 00112    68300115             BEZ     OPTGO           NONE
09 00113    6A100104     OPTIND  BAL,1   NXENT           SKIP ONE
09 00114    64800113             BDR,8   OPTIND          COUNT ONE
09 00115    F5900004 A   OPTGO   STB,9   *4
09 00116    6A1003BF     OPTNXI  BAL,1   CMDNXS          SHOULD BE COMPARISON OPERATOR
09 00117    68000192             B       SCANDONE        SHOULD CHECK AT LEAST ONE COMPARISON?
09 00118    2180006B A           CI,8    ','             END OF RELATIVE EXPN
09 00119    69300120             BNE     OPTCKC          CHECK CLOSURE
09 0011A    330001C5 01          MTW,0   LISTOBC
```

```
09 0011B    683000C1            BEZ     SCANCMA         NO, IS END OF OPTION
09 0011C    22C00003 A          LI,12   3               TREAT AS =
09 0011D    20500000 A          AI,5    0               ANY SPACE LEFT?
09 0011E    68200169            BLEZ    ERRFULL         B/NO
09 0011F    68000126            B       OPTEQ
09 00120    2180005D A   OPTCKC CI,8    ')'             CHECK END OF LIST
09 00121    6830015B            BE      OPTCKB          GO DEAL WITH END OF LIST
09 00122    20500000 A          AI,5    0               ANY SPACE LEFT?
09 00123    68200169            BLEZ    ERRFULL         B/NO, ERROR
09 00124    6A10045A            BAL,1   GETOPER         GET THE OPERATION NUMBER IN R12
09 00125    68000172            B       SCANOE          B/ERROR
09 00126    F2100004 A   OPTEQ  LB,1    *4              PRESERVE TOP BIT
09 00127    B5600004 A          STW,6   *4              SAVE LENGTH
09 00128    D5C00004 A          STH,12  *4              SAVE OPERATION
09 00129    F5100004 A          STB,1   *4
09 0012A    6A1003C5            BAL,1   GETTOKEN        SHOULD BE APPROPRIATE TYPE
09 0012B    68000172            B       SCANOE          B/ERROR
                         ** CONVERT IF APPROPRIATE, AND STORE
09 0012C    21200003 A          CI,2    3               TEXT?
09 0012D    68300140            BE      OPTTXT          B/YES
09 0012E    21200004 A          CI,2    4               DATE?
09 0012F    69300133            BNE     OPTVINT         B/NO
09 00130    6A100000 X          BAL,1   JD:FROMTEXT     CONVERT DATE: JD=>R11
09 00131    69100172            BL      SCANOE          INVALID DATE
09 00132    68000137            B       OPCK4
                         ** MUST BE INTEGER
09 00133    6A100000 X   OPTVINT BAL,1  INT:FROMTEXT    =>R10/11
09 00134    68000172            B       SCANOE          B/CONVERSION ERROR
09 00135    21600008 A          CI,6    8               VALUE SIZE MUST BE 8 OR 4
09 00136    6830013C            BE      OPTI8
09 00137    21600004 A   OPCK4  CI,6    4
09 00138    69300172            BNE     SCANOE          ERROR
09 00139    35B80001 A   OPTST4 STW,11  1,4             TRUNCATE TO 4 BYTES
09 0013A    20400002 A          AI,4    +2              NEXT SLOT
09 0013B    68000151            B       OPTICL
09 0013C    35A80001 A   OPTI8  STW,10  1,4             SAVE VALUE - MIGHT NOT BE ALIGNED
09 0013D    35B80002 A          STW,11  2,4
09 0013E    20400003 A          AI,4    +3              POINT AT NEXT
09 0013F    68000151            B       OPTICL
```

```
                          ** PROCESS STRING
        09 00140          OPTTXT  EQU     $
09 00140    71600008 A            CB,6    8               CHECK LENGTH
09 00141    6910015F              BL      ERRTEXT         B/TOO BIG...
09 00142    32900004 A            LW,9    4               CALCULATE
09 00143    20900001 A            AI,9    1
09 00144    25900002 A            SLS,9   2               DESTINATION
09 00145    72A00008 A            LB,10   8               GET ACTUAL LENGTH
09 00146    D2B00004 A            LH,11   *4              BEGIN BLOCK MARK AND OPERATOR
09 00147    B5A00004 A            STW,10  *4              SET ACTUAL LENGTH
09 00148    D5B00004 A            STH,11  *4              PUT TOP HW BACK
09 00149    75A00009 A            STB,10  9               SET LENGTH FOR MOVE
09 0014A    20A00003 A            AI,10   3
09 0014B    25A0007E A            SLS,10  -2              WORDS REQUIRED
09 0014C    20A00001 A            AI,10   +1              2 WORDS HEADER
09 0014D    3040000A A            AW,4    10              UPDATE BLOCK POINTER
09 0014E    22A00000 A            LI,10   0
09 0014F    35A9FFFF A            STW,10  -1,4            CLEAR LAST WORD
09 00150    61800000 A            MBS,8   0               COPY STRING
                          ** COUNT THIS VALUE ENTRY
09 00151    B3100007 A    OPTICL  MTW,+1  *7              UPDATE COUNT OF VALUES
09 00152    205FFFFF A            AI,5    -1
09 00153    68000116              B       OPTNXI
                          **      MTW,0   LISTOBC         IS THIS A LIST ?
                          **      BLEZ    OPTNXI          NO, SEE IF ANOTHER COMPARISON
                          **      B       OPTCKB
                          **
                          **      OPTCKR  LW,4    7               RESTORE VALUE TABLE PTR
                          **              LI,9    0               CONTINUATION VALUE
                          **              B       OPTNXV
                          **
09 00154    6A1003BF      OPTNEXT BAL,1   CMDNXS          GET NEXT CHARACTER
09 00155    68000192              B       SCANDONE        FINISHED
09 00156    2180005D A            CI,8    ')'             END OF LIST?
09 00157    6830015B              BE      OPTCKB          YES
09 00158    2180000D A            CI,8    X'D'            END OF LINE?
09 00159    693000C1              BNE     SCANCMA         NO MUST BE COMMA
09 0015A    68000172              B       SCANOE          MUST BE ERROR
                          **
```

```
                          ** COME HERE TO PROCESS ')'
09 0015B    330001C5 01  OPTCKB  MTW,0   LISTOBC         IS IT VALID?
09 0015C    68200172             BLEZ    SCANOE          NO
09 0015D    33F001C5 01          MTW,-1  LISTOBC         COUNT IT
09 0015E    680000BF             B       SCANOPT         COMMA SHOULD BE NEXT
                          *
                          ERRTEXT PB:PRINT (TEXTC,'VALUE TOO BIG FOR '),;
      09 0015F                            (TEXTC,*3)
09 00168    68000172             B       SCANOE
                          *
                          ERRFULL PB:PRINT (TEXTC,'TOO MANY COMPARISONS FOR '),;
      09 00169                            (TEXTC,*3)
                          *
09 00172    33000382 01  SCANOE  MTW,0   O:VERBOSE
09 00173    68300175             BEZ     SCANPE
      09 00174                    Q:SNAP  'SCANOE',(RUNTIME,CURRENTA)
      09 00175            SCANPE  PB:PRINT (TEXTC,ERRPARSE),(WORD,CMDPOS,PBF(DECIMAL))
09 0017F    68000370             B       PRINTHELP
                          *
      09 00180            SCANKER EQU     $               UNRECOGNISED KEYWORD
      09 00180                    PB:PRINT,NF (TEXTC,'UNRECOGNIZED KEYWORD: "')
09 00186    6A100000 X           BAL,1   PB:TEXT8
                                  PB:PRINT (TEXTC,'" AT POSITION '),;
      09 00187                            (WORD,CMDPOS,PBF(DECIMAL))
09 00191    68000370             B       PRINTHELP
                          *
      09 00192            SCANDONE EQU    $
09 00192    722001F0 01          LB,2    FPATTERN        ANY FILENAME PATTERN?
09 00193    692004F3             BGZ     GOLIST          YES
09 00194    68000370             B       PRINTHELP       SOMETHING ON COMMAND LINE,
                          *
09 00195    2220015C A   CURACC  LI,2    X'015C'         TEXTC('*')
09 00196    552001F0 01          STH,2   FPATTERN        USE THAT.
09 00197    680004F3             B       GOLIST
```

```
                          *         |         |         |         |         |         |
         09 00198         HELPTEXT QTEXTC ' '
         09 00199          QTEXTC 'LA: List all files with name and account matching pattern'
         09 001A8          QTEXTC 'Version date: AUGUST, 2026'
         09 001AF          QTEXTC 'LA filepattern[.accountpattern][,selection-options]'
         09 001BC          QTEXTC ' '
         09 001BD          QTEXTC 'One file is listed one per line in a manner similar to the '
         09 001CC          QTEXTC 'output of the PCL L(A) command.  If the DETAIL option is '
         09 001DB          QTEXTC 'supplied, then the information for each file is split into '
         09 001EA          QTEXTC 'multiple lines.'
         09 001EE          QTEXTC ' '
         09 001EF          QTEXTC 'File and account patterns can contain wildcard characters:'
         09 001FE          QTEXTC '   *   matches any sequence of characters'
         09 00209          QTEXTC '   ?   matches any single character'
         09 00212          QTEXTC '   +   matches any single digit'
         09 0021A          QTEXTC '   \   naturalizes the following character'
         09 00225          QTEXTC ' '
         09 00226          QTEXTC '   !pattern matches if pattern does not'
         09 00230          QTEXTC ' '
         09 00231          QTEXTC 'e.g.'
         09 00233          QTEXTC ' LA * lists all files in the current account'
         09 0023F          QTEXTC ' LA *.* lists all files on the system'
                           QTEXTC ' LA J:*+ lists all files beginning with J:,',;
         09 00249                 ' and ending with a digit'
         09 0025A          QTEXTC ' LA ABC.9* lists files named ABC in accounts beginning with 9'
         09 0026A          QTEXTC ' LA \* lists the file "*"'
         09 00271          QTEXTC ' LA !A* lists files that do not begin with A'
         09 0027D          QTEXTC ' '
         09 0027E          QTEXTC ' Selection options are separated by commas. Except for simple'
         09 0028E          QTEXTC ' boolean options and arguments can be specified as a list, '
         09 0029D          QTEXTC ' e.g:'
         09 0029F          QTEXTC '   (ORG,''K'',''C'')'
         09 002A4          QTEXTC ' Dates and integers can use comparators rather than the comma:'
         09 002B4          QTEXTC '   ACCESSED<29FEB96'
         09 002B9          QTEXTC '   (CREATED>01JAN97<31DEC97)'
         09 002C1          QTEXTC '   (GRAN<5>10)'
         09 002C5          QTEXTC ' '
         09 002C6          QTEXTC ' Dates can take various forms including: DDMMMYY, YYYY-MM-DD.'
         09 002D6          QTEXTC '   The form JD-n or JD+n express dates relative to the current'
```

```
    595         09 002E6            QTEXTC '   day, e.g. ACCESSED>JD-7 selects files accessed in the last'
    596         09 002F6            QTEXTC '   week.'
    597         09 002F9            QTEXTC ' '
    598         09 002FA            QTEXTC ' Using a list of values constructs a condition which is'
    599         09 00308            QTEXTC ' satisifed if any of the terms match,  e.g.'
    600         09 00313            QTEXTC '   GRAN<5>10 selects files of less than 5 granules OR more'
    601         09 00322            QTEXTC '   than 10'
    602         09 00325            QTEXTC ' Using multiple entries implies an AND, e.g.'
    603         09 00331            QTEXTC '   GRAN>5,GRAN<10 selects files with greater than 5 AND less'
    604         09 00341            QTEXTC '   than 10 granules'
    605         09 00346            QTEXTC ' '
    606         09 00347            QTEXTC 'Selection Options:'
    607         09 0034C            QTEXTC ' At least 3 characters of the option name are required.  When'
    608         09 0035C            QTEXTC ' the option ends in ! (e.g. DELETE!), abbreviation is not'
    609         09 0036B            QTEXTC ' allowed.'
    610         09 0036E            QTEXTC ' '
    611   09 0036F    00000000 A    DATA   0
```

```
        09 00370          PRINTHELP EQU   $
09 00370    22200198            LI,2    HELPTEXT
        09 00371          MOREHELP EQU    $
09 00371    6A100000 X          BAL,1   PI:TEXTC
                                TXINC,3 2              POINT AT NEXT
09 00376    F2300002 A          LB,3    *2
09 00377    69300371            BNE     MOREHELP
                          * DUMP OPTION TABLE
09 00378    22800000 06         LI,8    OPTIONS
09 00379    B2700008 A   OPTHELP LW,7   *8             OPTION TYPE
09 0037A    682003A2            BLEZ    HELPX          DONE
09 0037B    20800003 A          AI,8    3              SKIP ADDRESS, LENGTH, AND COUNT
        09 0037C                PB:PRINT (TEXTC,*8)    OPTION NAME
        09 00383                PB:PRINT,NF (TAB,4)
                                TXINC,1 8              OPTION DESCRIPTION
09 0038D    21700001 A          CI,7    1              BOOL
09 0038E    68300395            BE      OPTDESC
        09 0038F                PB:PRINT,NF (TEXTC,FILEMUST)
        09 00395          OPTDESC PB:PRINT (TEXTC,*8),(INDENT,4)
                                TXINC,1 8
09 003A1    68000379            B       OPTHELP
09 003A2    04900001 A   HELPX   CAL1,9  1
```

```
                              **
                              ** CMDCH: RETURN NEXT COMMAND LINE CHARACTER, DO NOT UPDATE POS
                              ** CMDNX: GET THE NEXT COMMAND LINE CHARACTER, UPDATE POS
                              ** R1 = LINK
                              ** RETURNS R8 CHARACTER.  SKIPPING IF OK.
                              **
09 003A3    22800000 A   CMDCH   LI,8    0
09 003A4    680003A6             B       $+2
09 003A5    22800001 A   CMDNX   LI,8    1
09 003A6    89100000 A           PSW,1   *0
09 003A7    321001C7 01          LW,1    CMDPOS
09 003A8    311001C6 01          CW,1    CMDLEN
09 003A9    681003AF             BGE     CMDR0           NOTHING LEFT
09 003AA    668001C7 01          AWM,8   CMDPOS
09 003AB    728201C8 01  CMDANLZ LB,8    CMDBUF,1
09 003AC    88100000 A           PLW,1   *0
09 003AD    68020001 A           B       1,1             RETURN SKIPPING
                              *
09 003AE    351001C7 01  CMDS1   STW,1   CMDPOS          SAVE BUFFER POSITION
09 003AF    22800000 A   CMDR0   LI,8    0               SET RETURN VALUE
09 003B0    88100000 A   CMDRX   PLW,1   *0              GET RETURN ADDRESS
09 003B1    68020000 A           B       0,1             EMPTY RETURN
```

```
                                **
                                ** CMDSKB: SKIP BLANK CHARACTERS IN COMMAND
                                ** R1 = LINK
                                ** RETURNS R8 CHARACTER.  SKIPPING IF OK.
                                **
09 003B2    89100000 A   CMDSKB  PSW,1   *0
09 003B3    321001C7 01          LW,1    CMDPOS
09 003B4    311001C6 01  CMDCKN  CW,1    CMDLEN
09 003B5    681003AE             BGE     CMDS1           NOTHING LEFT
09 003B6    22800040 A           LI,8    ' '
09 003B7    718201C8 01          CB,8    CMDBUF,1        IS IT BLANK?
09 003B8    693003BB             BNE     CMDRC           NO, RETURN IT
09 003B9    20100001 A           AI,1    1
09 003BA    680003B4             B       CMDCKN          CHECK NEXT
09 003BB    351001C7 01  CMDRC   STW,1   CMDPOS
09 003BC    728201C8 01          LB,8    CMDBUF,1        GET THE CHARACTER
09 003BD    88100000 A           PLW,1   *0
09 003BE    68020001 A           B       1,1             RETURN SKIPPING
                                **
                                ** CMDNXS: GET NEXT NON-BLANK COMMAND CHARACTER
                                ** R1 = LINK
                                ** RETURNS R8 CHARACTER.  SKIPPING IF OK.
                                **
09 003BF    89100000 A   CMDNXS  PSW,1   *0
09 003C0    6A1003B2             BAL,1   CMDSKB          SKIP BLANKS
09 003C1    680003AF             B       CMDR0           EMPTY RETURN
09 003C2    331001C7 01          MTW,+1  CMDPOS          UPDATE FOR CURRENT CHARACTER
09 003C3    88100000 A   CMDR1   PLW,1   *0
09 003C4    68020001 A           B       1,1             RETURN SKIPPING
```

```
                          **
                          ** GET A STRING, OR A DELIMITED INTEGER, DATE OR KEYWORD
                          **
                          **
                          ** RETURNS SKIPPING R8=LEN/BA OF INPUT TOKEN, R9=VOLATILE
                          **
        09 003C5          GETTOKEN EQU    $
09 003C5    02200050 A           LCI     5
09 003C6    8B100000 A           PSM,1   *0              SAVE R1-5
09 003C7    6A1003BF             BAL,1   CMDNXS
09 003C8    68000406             B       GETTFAIL        NOTHING
09 003C9    321001C7 01          LW,1    CMDPOS          UPDATED POSITION
09 003CA    22400000 A           LI,4    0               CURRENT COUNT
09 003CB    32900008 A           LW,9    8               COPY INITIAL CHARACTER
09 003CC    2180007D A           CI,8    X'7D'           QUOTE?
09 003CD    683003D3             BE      GETTQ
09 003CE    2180007F A           CI,8    X'7F'           DOUBLE?
09 003CF    683003D3             BE      GETTQ
09 003D0    201FFFFF A           AI,1    -1              BACK UP TO INITIAL CHARACTER
09 003D1    20400001 A           AI,4    +1              COUNT IT
09 003D2    22900000 A           LI,9    0               NO QUOTE
09 003D3    443003AB     GETTQ   ANLZ,3  CMDANLZ         GET BA(START CHARACTER)
09 003D4    69D00406             BCS,13  GETTFAIL        OOPS
09 003D5    20900000 A           AI,9    0               QUOTED?
09 003D6    693003DB             BNE     GETTLP          B/YES
09 003D7    32100008 A           LW,1    8               COPY CHAR
09 003D8    72120050             LB,1    CHARTYPE,1      GET TYPE
09 003D9    21100005 A           CI,1    AOP
09 003DA    681003EF             BGE     GETTDN          SINGLE CHARACTER TOKEN
09 003DB    6A1003A5     GETTLP  BAL,1   CMDNX           FETCH NEXT
09 003DC    680003EF             B       GETTDN          DONE
09 003DD    32100008 A           LW,1    8               COPY CHAR
09 003DE    72120050             LB,1    CHARTYPE,1      GET ITS TYPE
09 003DF    683003E6             BEZ     GETTQS
09 003E0    21100005 A           CI,1    AOP
09 003E1    691003EC             BL      GETTLPI
                          ** FOR DATES, PERMIT ARITH OPERATORS (-, /, ETC),
                          ** BUT NOT RELATIVE (<, >, =) OR PUNCTUATION
09 003E2    21200004 A           CI,2    4               DATE?
```

```
09 003E3    693003E6            BNE     GETTQS          NO
09 003E4    21100005 A          CI,1    AOP
09 003E5    683003EC            BE      GETTLPI         ACCEPT
09 003E6    20900000 A   GETTQS  AI,9    0               QUOTED STRING?
09 003E7    693003EA            BNEZ    GETTQC          YES, CHECK CHARACTER
09 003E8    33F001C7 01         MTW,-1  CMDPOS          BACK UP SCAN POSITION
                         ****   AI,4    -1              DECREMENT LEN
09 003E9    680003EF            B       GETTDN          AND WE ARE DONE
09 003EA    31800009 A   GETTQC  CW,8    9               SAME QUOTE?
09 003EB    683003EF            BE      GETTDN          YES, DONE
09 003EC    20400001 A   GETTLPI AI,4    1               COUNT CHARACTER
09 003ED    214000FF A          CI,4    X'FF'           TOO MANY?
09 003EE    691003DB            BL      GETTLP          PROCESS NEXT
09 003EF    32800003 A   GETTDN  LW,8    3               SOURCE ADDRESS
09 003F0    75400008 A          STB,4   8               SET LENGTH
09 003F1    33000382 01         MTW,0   O:VERBOSE
09 003F2    68300403            BEZ     GETTOK          DONE
09 003F3    32200003 A          LW,2    3
09 003F4    32300004 A          LW,3    4
        09 003F5                PB:PRINT,NF (TEXTC,'GETTOKEN: "')
09 003FB    6A100000 X          BAL,1   PB:TEXT
        09 003FC                PB:PRINT,NF (TEXTC,'"')
09 00402    6A100000 X          BAL,1   PB:FLUSH
09 00403    02200050 A   GETTOK  LCI     5
09 00404    8A100000 A          PLM,1   *0              RESTORE R1-5
09 00405    68020001 A          B       1,1             DONE, RETURN TOKEN STRING
09 00406    02200050 A   GETTFAIL LCI    5
09 00407    8A100000 A          PLM,1   *0
09 00408    68020000 A          B       0,1             DONE, TAKE ERROR RETURN
```

```
                              **
                              ** GET A KEYWORD
                              **
                              ** R1-3, R8 VOLATILE
                              ** RETURNS SKIPPING R7=MATCHED ENTRY
                              ** ANY MATCH OF 3 CHARACTERS OR GREATER IS ALLOWED AS LONG AS
                              ** THE OPTION DOES NOT END IN !, IN WHICH CASE A COMPLETE MATCH
                              ** IS REQUIRED
                              **
09 00409    89100000 A   GETKEYWORD PSW,1 *0             SAVE LINK
09 0040A    6A1003C5             BAL,1   GETTOKEN
09 0040B    680003AF             B       CMDR0           NOTHING, RETURN IT
09 0040C    72100008 A           LB,1    8               TOKEN LENGTH
09 0040D    21100003 A           CI,1    3               AT LEAST 3 BYTES REQUIRED
09 0040E    691003AF             BL      CMDR0           INSUFFICIENT LENGTH
09 0040F    22700000 06          LI,7    OPTIONS
09 00410    321E0000 A   GETKCHK LW,1    0,7             ENTRY TYPE
09 00411    683003B0             BEZ     CMDRX           END OF TABLE, NOTHING FOUND
09 00412    72100008 A           LB,1    8               GET LENGTH OF TOKEN
09 00413    32300007 A           LW,3    7               WA(OPTION ENTRY)
09 00414    20300003 A           AI,3    3               WA(OPTION NAME)
09 00415    25300002 A           SLS,3   2               MAKE BA
09 00416    72260000 A           LB,2    0,3             GET ENTRY LENGTH
09 00417    30200003 A           AW,2    3               POINT AT LAST CHARACTER
09 00418    72240000 A           LB,2    0,2             GET LAST CHARACTER
09 00419    2120005A A           CI,2    '!'             ABBREVIATIONS OK?
09 0041A    6930041D             BNE     GETKAB          B/YES
09 0041B    71160000 A           CB,1    0,3             SAME LENGTH?
09 0041C    69300422             BNE     GETKNXT         NO, NO MATCH
09 0041D    20300001 A   GETKAB  AI,3    1               FIRST CHAR
09 0041E    32200008 A           LW,2    8               BA(TOKEN)
09 0041F    75100003 A           STB,1   3               SET LENGTH
09 00420    60200000 A           CBS,2   0               CHECK IT
09 00421    6830042C             BE      GOTKW           RETURN THIS ENTRY
09 00422    20700003 A   GETKNXT AI,7    3
                                 TXINC,1 7
                                 TXINC,1 7               NEXT ENTRY
09 0042B    68000410             B       GETKCHK
09 0042C    33000382 01  GOTKW   MTW,0   O:VERBOSE
```

```
797   09 0042D    683003C3            BEZ     CMDR1           NO DEBUGGING MESSAGES
798         09 0042E                  PB:PRINT,NF (TEXTC,'GETKEYWORD: "')
799   09 00434    6A100000 X          BAL,1   PB:TEXT8
800         09 00435                  PB:PRINT,NF (TEXTC,'"')
801   09 0043B    6A100000 X          BAL,1   PB:FLUSH
802   09 0043C    680003C3            B       CMDR1
```

```
                          **
                          ** GET A PATTERN
                          ** R3 = MAX LEN
                          ** R4 = DESTINATION
                          ** 11 = LINK
                          ** R1-2 VOLATILE
                          ** RETURNS CC4: NO MORE DATA IN CCBUF, R8 STOP CHAR
                          **
09 0043D    22200000 A    GETPATTERN LI,2 0
09 0043E    F5200004 A            STB,2   *4              SET DESTINATION COUNT
09 0043F    31200003 A    GPATCF  CW,2    3               LENGTH OK?
09 00440    E810000B A            BGE     *11             FULL, EXIT (MORE DATA AVAILABLE)
09 00441    6A1003A5              BAL,1   CMDNX           GET NEXT PATTERN CHARACTER
09 00442    6830044A              BEZ     GPATS4          NO MORE, RETURN SETTING CC4
09 00443    218000B1 A            CI,8    '\'             ESCAPE?
09 00444    6930044C              BNE     GPATCD          NO, CHECK FOR DELIMITERS
09 00445    20200001 A            AI,2    +1
09 00446    F5840004 A            STB,8   *4,2            SAVE ESCAPE
09 00447    F5200004 A            STB,2   *4              UPDATE LENGTH
09 00448    6A1003A5              BAL,1   CMDNX           GET ESCAPED CHARACTER
09 00449    68000456              B       GPATADD         PROCESS AS NORMAL CHAR
                          *
09 0044A    02200010 A    GPATS4  LCI     1               SET CC4
09 0044B    E800000B A            B       *11             RETURN
                          *
09 0044C    33F001C7 01   GPATCD  MTW,-1  CMDPOS          BACKUP PTR
09 0044D    2180000D A            CI,8    X'0D'
09 0044E    6830044A              BE      GPATS4          DONE PARSING, SET CC4
09 0044F    2180004B A            CI,8    '.'             CHECK OTHER DELIMITERS
09 00450    E830000B A            BE      *11             END OF PATTERN, CC4 IS RESET
09 00451    2180006B A            CI,8    ','
09 00452    E830000B A            BE      *11             END OF PATTERN, CC4 IS RESET
09 00453    21800040 A            CI,8    ' '
09 00454    E830000B A            BE      *11             END OF PATTERN, CC4 IS RESET
09 00455    331001C7 01           MTW,+1  CMDPOS          EAT THIS CHARACTER
09 00456    20200001 A    GPATADD AI,2    +1
09 00457    F5840004 A            STB,8   *4,2            SAVE FILENAME BYTE
09 00458    F5200004 A            STB,2   *4              SAVE COUNT
09 00459    6800043F              B       GPATCF          GO GET NEXT
```

```
        09 0045A          GETOPER EQU     $               FIRST OPERATOR CHARCTER IN 8 -> OPNUM IN 12
   09 0045A    89100000 A          PSW,1   *0
   09 0045B    89200000 A          PSW,2   *0
   09 0045C    89800000 A          PSW,8   *0
   09 0045D    6A1003A3            BAL,1   CMDCH           PEEK AT NEXT CHAR
   09 0045E    6800046A            B       GETOPCH         NOTHING THERE
   09 0045F    21800080 A          CI,8    X'80'           PUNCTUATION?
   09 00460    6810046A            BGE     GETOPCH         NO, SINGLE CHAR
   09 00461    88100000 A          PLW,1   *0              ORIGINAL CHAR
   09 00462    25100008 A          SLS,1   8
   09 00463    49800001 A          OR,8    1               POSSIBLE TWO CHAR OPERATOR
   09 00464    5220004B            LH,2    OPERTAB         # OPERATORS
   09 00465    5184004B     GETOPC2 CH,8   OPERTAB,2
   09 00466    68300472            BE      GETOPF2         FOUND IT
   09 00467    64200465            BDR,2   GETOPC2
   09 00468    25800078 A          SLS,8   -8              TRY ORIGINAL SINGLE CHAR
   09 00469    89800000 A          PSW,8   *0
   09 0046A    88800000 A   GETOPCH PLW,8  *0
   09 0046B    5220004B            LH,2    OPERTAB         #OPERATORS
   09 0046C    5184004B     GETOPC1 CH,8   OPERTAB,2
   09 0046D    68300473            BE      GETOPF          FOUND IT
   09 0046E    6420046C            BDR,2   GETOPC1
   09 0046F    88200000 A          PLW,2   *0
   09 00470    88100000 A          PLW,1   *0
   09 00471    68020000 A          B       0,1             NOT FOUND
   09 00472    331001C7 01  GETOPF2 MTW,+1 CMDPOS          EAT SECOND CHAR
   09 00473    32C00002 A   GETOPF  LW,12  2
   09 00474    88200000 A          PLW,2   *0
   09 00475    33000382 01         MTW,0   O:VERBOSE
   09 00476    683003C3            BEZ     CMDR1
        09 00477                   PB:PRINT (TEXTC,'GETOPER: #'),(WORD,12,PBF(HEX))
   09 00481    680003C3            B       CMDR1
```

```
                          **
                          **
                          ** FIND SUBSTRING UP TO THE FIRST WILDCARD CHRACTER
                          ** 6=LEN/SOURCE, 7=DEST, RETURNS SKIPPING NEXT INSTRUCTION IF NO WILDCARDS
                          **
         09 00482         GETPREFIX EQU   $
   09 00482    22400001 A          LI,4    1
   09 00483    22500000 A          LI,5    0
   09 00484    F1400006 A   GETPRE1 CB,4    *6
   09 00485    6920049C            BG      GETPREX         DONE, NO WILDCARD
   09 00486    F2880006 A          LB,8    *6,4
   09 00487    20400001 A          AI,4    +1
   09 00488    218000B1 A          CI,8    '\'
   09 00489    6930048D            BNE     GETPRE2
   09 0048A    F2880006 A          LB,8    *6,4
   09 0048B    20400001 A          AI,4    +1
   09 0048C    68000497            B       GETPRE3
   09 0048D    2180005C A   GETPRE2 CI,8    '*'
   09 0048E    6830049A            BE      GETPREW
   09 0048F    2180004E A          CI,8    '+'
   09 00490    6830049A            BE      GETPREW
   09 00491    2180006F A          CI,8    '?'
   09 00492    6830049A            BE      GETPREW
   09 00493    21400002 A          CI,4    2               ! IS WILD ONLY IF FIRST CHARACTER
   09 00494    69300497            BNE     GETPRE3
   09 00495    2180005A A          CI,8    '!'
   09 00496    6830049A            BE      GETPREW
   09 00497    20500001 A   GETPRE3 AI,5    +1
   09 00498    F58A0007 A          STB,8   *7,5
   09 00499    68000484            B       GETPRE1
   09 0049A    F5500007 A   GETPREW STB,5   *7
   09 0049B    68020000 A          B       0,1
   09 0049C    F5500007 A   GETPREX STB,5   *7
   09 0049D    68020001 A          B       1,1
```

```
                          **
                          ** COUNT THIS FILE, AND PRINT ACCOUNT HEADING IF NEEEDED
                          **
09 0049E    33000144 01  COUNTF  MTW,0   ACFILES         FIRST?
09 0049F    693004AE             BNE     COUNTNH         NO, SKIP HEADING
09 004A0    6A100000 X           BAL,1   PI:PAGE
09 004A1    02200020 A           LCI     2
09 004A2    8B200000 A           PSM,2   *0              SAVE 2-3
09 004A3    223007B1 01          LI,3    BA(ACTEXT)+9
09 004A4    22200008 A           LI,2    8
09 004A5    75200003 A           STB,2   3
09 004A6    22200818 01          LI,2    BA(CURRENTA)
09 004A7    61200000 A           MBS,2   0
09 004A8    222001EA 01          LI,2    ACTEXT
09 004A9    6A100000 X           BAL,1   PI:TEXTC
09 004AA    22200008             LI,2    ACHEAD
09 004AB    6A100000 X           BAL,1   PI:TEXTC
09 004AC    02200020 A           LCI     2
09 004AD    88200000 A           PLW,2   *0              RESTORE 2-3
                          **
                          **      PB:PRINT (TEXTC,ACTEXT),(TEXT,BA(CURRENTA),8)
                          **      PB:PRINT (TEXTC,ACHEAD)
09 004AE    32100146 01  COUNTNH LW,1    FILGRANS        GRANULES FOR THIS FILE
09 004AF    66100145 01          AWM,1   ACTGRANS
09 004B0    33100144 01          MTW,+1  ACFILES         COUNT FILES THIS ACCOUNT AND TOTAL.
09 004B1    33100140 01          MTW,+1  TOTFILES
09 004B2    E800000B A           B       *11


                          **
                          ** PRINT SUMMARY AT END
                          **
09 004B3    33000382 01  SUMMARY MTW,0   O:VERBOSE       PRINT MORE STUFF?
09 004B4    693004C6             BNEZ    SUMGO           YES DO IT
09 004B5    32200143 01          LW,2    TOTACS
09 004B6    21200001 A           CI,2    1               MORE THAN 1?
09 004B7    692004C6             BG      SUMGO           YES, DO IT
09 004B8    33000142 01          MTW,0   DELFILES        ANY DELETES?
09 004B9    693004C6             BNEZ    SUMGO           B/YES
```

```
09 004BA    33000141 01          MTW,0   ERRFILES        ANY ERRORS?
09 004BB    693004C6             BNEZ    SUMGO           B/YES
09 004BC    33000140 01          MTW,0   TOTFILES        ANY FILES AT ALL?
09 004BD    E920000B A           BGZ     *11             NO SUMMARY NEEDED
      09 004BE                   PB:PRINT (TEXTC,'NO MATCHING FILES FOUND')
09 004C5    680004F1             B       SUMDONE
      09 004C6           SUMGO   EQU     $
                                 PB:PRINT,NF (TEXTC,DOTS),;
                                     (WORD,TOTFILES,PBF(DECIMAL)),(TEXTC,' TOTAL FILES IN '),;
      09 004C6                       (WORD,TOTACS,PBF(DECIMAL)),(TEXTC,' ACCOUNTS')
09 004D6    33000142 01          MTW,0   DELFILES
09 004D7    683004E4             BEZ     SUMERR
                                 PB:PRINT (TEXTC,STARS),(WORD,DELFILES,PBF(DECIMAL)),;
      09 004D8                           (TEXTC,' WERE DELETED ** ')
09 004E4    33000141 01  SUMERR  MTW,0   ERRFILES
09 004E5    683004F1             BEZ     SUMDONE
                                 PB:PRINT,NF (TEXTC,COMMA),;
      09 004E6                       (WORD,ERRFILES,PBF(DECIMAL)),(TEXTC,' INACCESSIBLE')
09 004F1    6A100000 X   SUMDONE BAL,1   PB:FLUSH
09 004F2    E800000B A           B       *11
```

```
                          **
                          ** COMMAND HAS BEEN PARSED, PREPARE TO LIST MATCHING FILES
                          **
        09 004F3          GOLIST  EQU     $
09 004F3    33000000 X            MTW,0   J:JIT
09 004F4    681004F7              BGEZ    GOCMD
09 004F5    33000382 01           MTW,0   O:VERBOSE
09 004F6    6830050C              BEZ     GOLIST1
                          ** NOT ONLINE, OR VERBOSE IS SET, PRINT COMMAND DETAILS.
09 004F7    22300001 A    GOCMD   LI,3    1
09 004F8    6A100000 X            BAL,1   PI:SPACE
        09 004F9                  PB:PRINT (TEXT,CMDBUF,*CMDLEN)
        09 00502                  PB:PRINT (TEXTC,'RUN AT '),(TEXTC,RUNTEXT)
09 0050B    6A100000 X            BAL,1   PI:SPACE
                          **
09 0050C    72800200 01   GOLIST1 LB,8    APATTERN        ANY ACCOUNT SPECIFIED?
09 0050D    6930051A              BNE     GOLISTA         YES
09 0050E    02200020 A            LCI     2               COPY USER ACCOUNT
09 0050F    2A400000 X            LM,4    J:ACCN
09 00510    25400378 A            SCD,4   -8              SAVE 1-7 WITH A BYTE FOR COUNT
09 00511    15400200 01           STD,4   APATTERN
09 00512    72800004 A            LB,8    4               BYTE #8
09 00513    75800202 01           STB,8   APATTERN+2
09 00514    22700008 A                    LI,7    8
09 00515    22800040 A            LI,8    X'40'
09 00516    718E0200 01   ACCOUNT CB,8    APATTERN,7
09 00517    69300519              BNE     $+2
09 00518    64700516              BDR,7   ACCOUNT         DELETE TRAILING BLANKS
09 00519    75700200 01           STB,7   APATTERN
        09 0051A          GOLISTA EQU     $
                          ** DETERMINE NON-WILDCARD PREFIXES OF ACCOUNT AND FILENAME
09 0051A    226001F0 01           LI,6    FPATTERN
09 0051B    227001F8 01           LI,7    FPREFIX
09 0051C    6A100482              BAL,1   GETPREFIX
09 0051D    02000000 A            NOP
09 0051E    22600200 01           LI,6    APATTERN
09 0051F    22700203 01           LI,7    APREFIX
09 00520    6A100482              BAL,1   GETPREFIX
09 00521    6800052C              B       AWILD           ACCOUNT CONTAINS WILDCARDS
```

```
                          **
09 00522    32200007      SETACC  LW,2    WBLANK
09 00523    35200206 01           STW,2   CURRENTA
09 00524    35200207 01           STW,2   CURRENTA+1
09 00525    22300818 01           LI,3    BA(CURRENTA)    DESTINATION
09 00526    72200203 01           LB,2    APREFIX         LENGTH
09 00527    75200003 A            STB,2   3
09 00528    2220080D 01           LI,2    BA(APREFIX)+1   SOURCE
09 00529    61200000 A            MBS,2   0               MOVE ACCOUNT
09 0052A    6A10056C              BAL,1   LISTFILES       LIST FILES IN ACCOUNT
09 0052B    6800056A              B       ALLDONE
                          **
                          ** LOOP THROUGH ALL ACCOUNTS
                          **
        09 0052C          AWILD   EQU     $
        09 0052C          NXTA    EQU     $
        09 0052C          ,NXAFPT Q:OPEN  F:LA,NXTA,TEST,(ABN,AA)
09 0052D    22400080 07           LI,4    BA(F:LA:AC)
                          ** CHECK IF ACCOUNT > LAST POSSIBLE (I.E. > PREFIX)
09 0052E    2250080D 01           LI,5    BA(APREFIX)+1
09 0052F    72800203 01           LB,8    APREFIX
09 00530    68300534              BEZ     ACMATCH         NO PREFIX, MUST MATCH
09 00531    75800005 A            STB,8   5
09 00532    60400000 A            CBS,4   0
09 00533    6920056A              BG      ALLDONE
                          **
                          ** DOES IT MATCH?
                          **
09 00534    22700007 A    ACMATCH LI,7    7
09 00535    722E0020 07   ACMATT  LB,2    F:LA:AC,7
09 00536    21200040 A            CI,2    X'40'
09 00537    69300539              BNE     ACMATL          HAVE LAST NON BLANK CHARACTER
09 00538    64700535              BDR,7   ACMATT
09 00539    20700001 A    ACMATL  AI,7    +1
09 0053A    22200080 07           LI,2    BA(F:LA:AC)     ACCOUNT BYTE ADDRESS
09 0053B    75700002 A            STB,7   2               SET LENGTH
09 0053C    22300200 01           LI,3    APATTERN
09 0053D    6A100000 X            BAL,1   MATCH
09 0053E    6930054C              BCS,3   ACNOMAT         NO MATCH, GET NEXT ACCOUNT
```

```
09 0053F    33000382 01          MTW,0   O:VERBOSE
09 00540    68300547             BEZ     ACMATGO
09 00541    222001EA 01          LI,2    ACTEXT
09 00542    6A100000 X           BAL,1   PB:TEXTC
09 00543    22200080 07          LI,2    BA(F:LA:AC)
09 00544    22300008 A           LI,3    8
09 00545    6A100000 X           BAL,1   PB:TEXT
09 00546    6A100000 X           BAL,1   PB:FLUSH
09 00547    02200020 A   ACMATGO LCI     2
09 00548    2A400020 07          LM,4    F:LA:AC         GET ACCOUNT
09 00549    15400206 01          STD,4   CURRENTA        AND SAVE
09 0054A    6A10056C             BAL,1   LISTFILES       LIST FILES IN THIS ACCOUNT
09 0054B    6800052C             B       NXTA            GO GET NEXT
09 0054C    33000382 01  ACNOMAT MTW,0   O:VERBOSE
09 0054D    6830052C             BEZ     NXTA            GO GET NXT
09 0054E    6A100000 X           BAL,1   PB:CLEAR
09 0054F    222001EA 01          LI,2    ACTEXT
09 00550    6A100000 X           BAL,1   PB:TEXTC
09 00551    22200080 07          LI,2    BA(F:LA:AC)
09 00552    22300008 A           LI,3    8
09 00553    6A100000 X           BAL,1   PB:TEXT
09 00554    2220002C             LI,2    TXNOMATCH
09 00555    6A100000 X           BAL,1   PB:TEXTC
09 00556    6A100000 X           BAL,1   PB:FLUSH
09 00557    6800052C             B       NXTA
                         **
                         ** NEXT ACCOUNT ERROR
                         **
09 00558    0C40D5E7 A   AEMESS  TEXTC   ' NXTA FAILED'
09 00559    E3C140C6 A
09 0055A    C1C9D3C5 A
09 0055B    C4404040 A
                         *
      09 0055C           AA      EQU     $
09 0055C    7220000A A           LB,2    SR3
09 0055D    21200002 A           CI,2    2               END OF ACCOUNTS?
09 0055E    6830056A             BE      ALLDONE
09 0055F    22304002 A           LI,3    PBF(HEX,2)
09 00560    6A100000 X           BAL,1   PB:WORD
```

```
09 00561    2220001D            LI,2    DASH
09 00562    6A100000 X          BAL,1   PB:TEXTC
09 00563    3220000A A          LW,2    SR3
09 00564    2520006F A          SLS,2   -17
09 00565    4B200004            AND,2   M7F
09 00566    6A100000 X          BAL,1   PB:WORD
09 00567    22200558            LI,2    AEMESS
09 00568    6A100000 X          BAL,1   PB:TEXTC
09 00569    6A100000 X          BAL,1   PB:FLUSH
      09 0056A           ALLDONE EQU     $
09 0056A    6AB004B3            BAL,11  SUMMARY
09 0056B    04900001 A          CAL1,9  1
                           **
                           ** LIST MATCHING FILES IN A SPECIFIC ACCOUNT
                           **
      09 0056C           LISTFILES EQU   $
09 0056C    3510030A 01         STW,1   LFRETURN
                           **
                           ** SET UP ACCOUNT TOTALS
                           **
09 0056D    33100143 01         MTW,+1  TOTACS          COUNT THIS ACCOUNT
09 0056E    22100000 A          LI,1    0
09 0056F    35100144 01         STW,1   ACFILES
09 00570    35100145 01         STW,1   ACTGRANS
                           **
                           ** GET FIRST FILE IN ACCOUNT -- ** FIX THIS TO SUPPLY FILENAME PREFIX
                           **
09 00571    32200006            LW,2    W02020202
09 00572    3520030E 01         STW,2   ADJVLP
09 00573    12200206 01         LD,2    CURRENTA
09 00574    3520030F 01         STW,2   ADJVLP+1
09 00575    35300310 01         STW,3   ADJVLP+2
09 00576    0410030B 01         CAL1,1  ADJUST          PUT ACCOUNT INTO DCB
                           *
09 00577    12200000    LSTFRST LD,2    FIRSTFILE       **SHOULD BE PREFIX, SEE BELOW **
09 00578    02200020 A          LCI     2
09 00579    2B200010 02         STM,2   NXFFPT+6
09 0057A    22900008 A          LI,9    X'8'
09 0057B    4790000B 02         STS,9   NXFFPT+1        SET VLP INDICATOR
```

```
        09 0057C          NXTFIL  EQU     $
09 0057C    22100000 A          LI,1    0
09 0057D    351001C2 01         STW,1   ERRCODE         CLEAR ERROR
09 0057E    35100146 01         STW,1   FILGRANS        CLEAR GRANS THIS FILE
                          ,NXFFPT Q:OPEN  F:LF,NXTF,(ABN,OPNABN),(ERR,OPNABN),(FPARAM,FPARAM),;
        09 0057F                          (FILE,'$',' $$$$$$$'),(IN,EXCL)
09 00580    22800000 A          LI,8    0
09 00581    2290000F A          LI,9    X'F'            RESET FLAGS TO INDICATE
09 00582    4780000B 02         STS,8   NXFFPT+1          NO VLPS PRESENT
                          ** DOES IT MATCH?
                          ** CHECK IF > PREFIX
09 00583    2240020B 01  CHKPFX LI,4    FPARAM+1        WA(TEXTC(FILENAME))
09 00584    722001F8 01         LB,2    FPREFIX         LENGTH
09 00585    6830058C            BEZ     NXTFIL$         NO PREFIX, MUST MATCH
                          **
                          ** THE FOLLOWING OPTIMIZATION ONLY WORKS IF THE PREFIX IS USED TO SET
                          ** THE FIRST FILENAME AT LSTFRST ABOVE
                          **      CB,2    *4
                          **      BG      DONEAC          PREFIX IS LONGER, MUST BE DONE THIS ACCOUNT
09 00586    225001F8 01         LI,5    FPREFIX
09 00587    25400102 A          SLD,4   2               MAKE BYTE ADDRESSES
09 00588    75200005 A          STB,2   5               SET UP COMPARE LENGTH
09 00589    20500001 A          AI,5    +1
09 0058A    60400001 A          CBS,4   1
09 0058B    69200742            BG      DONEAC          DONE THIS ACCOUNT
                          **
09 0058C    2220020B 01  NXTFIL$ LI,2   FPARAM+1        TEXTC(FILENAME)
09 0058D    F2300002 A          LB,3    *2
09 0058E    25200002 A          SLS,2   2
09 0058F    20200001 A          AI,2    +1
09 00590    75300002 A          STB,3   2               SET LENGTH
09 00591    223001F0 01         LI,3    FPATTERN
09 00592    6A100000 X          BAL,1   MATCH
09 00593    6930078D            BCS,3   NXTOPN          NO MATCH, GET NEXT FILE
                          **
                          ** CHECK ATTRIBUTES AND BUILD A BUFFER TO LIST THIS FILE
                          **
09 00594    22100000 A          LI,1    0               INITIALIZE VARIOUS ATTRIBUTES
09 00595    3510013E 01         STW,1   SAVEDATE
```

```
09 00596    3510013F 01          STW,1   SAVEDATE+1
09 00597    35100148 01          STW,1   ORGTYPE
09 00598    3510014A 01          STW,1   RDACCTS
09 00599    35100172 01          STW,1   WRACCTS
09 0059A    3510019A 01          STW,1   EXACCTS
09 0059B    6A100000 X           BAL,1   PB:CLEAR        CLEAR BUFFER
09 0059C    2240020A 01          LI,4    FPARAM
09 0059D    F2800004 A   VLPLP   LB,8    *4              GET CODE
09 0059E    21800015 A           CI,8    X'15'
09 0059F    692005A7             BG      VLPLP1          DON'T WANT IT
09 005A0    22100002 A           LI,1    2
09 005A1    F3020004 A           MTB,0   *4,1
09 005A2    683005A7             BEZ     VLPLP1          PARAMETER NOT PRESENT
09 005A3    89400000 A           PSW,4   *0
09 005A4    32100008 A           LW,1    8
09 005A5    670205B0             EXU     VLPTAB-1,1      GET PARAMETER OR NOP
09 005A6    88400000 A           PLW,4   *0
      09 005A7           VLPLP1  EQU     $
09 005A7    20F00000 A           AI,15   0               SUPPRESS THIS FILE?
09 005A8    6820078D             BLEZ    NXTOPN          B/YES
09 005A9    22100001 A           LI,1    1
09 005AA    F3020004 A           MTB,0   *4,1            SEE IF LAST ENTRY
09 005AB    6930067F             BNEZ    LISTIT          YES, LIST THE FILE
09 005AC    22100003 A           LI,1    3
09 005AD    F2820004 A           LB,8    *4,1            GET LENGTH
09 005AE    30400008 A           AW,4    8
09 005AF    20400001 A           AI,4    1
09 005B0    6800059D             B       VLPLP           NEXT PARAMETER
                         *
      09 005B1           VLPTAB  EQU     $
09 005B1    6AF005C6             BAL,15  FNAME
09 005B2    02000000 A           NOP
09 005B3    02000000 A           NOP
09 005B4    6AF00639             BAL,15  EDATE
09 005B5    6AF005CC             BAL,15  RACCT           READ ACCT
09 005B6    6AF005CE             BAL,15  WACCT           WRITE ACCT
09 005B7    02000000 A           NOP                     INSN
09 005B8    02000000 A           NOP                     OUTSN
09 005B9    6AF005EF             BAL,15  ORG
```

```
09 005BA    6AF0062D            BAL,15  MDATE           MODIFICATION (UPDATE) DATE
09 005BB    02000000 A          NOP                     SYNON
09 005BC    02000000 A          NOP                     ?
09 005BD    6AF005E2            BAL,15  GRANS
09 005BE    6AF00631            BAL,15  CDATE           CREATION DATE
09 005BF    6AF00635            BAL,15  ADATE           ACCESS DATE
09 005C0    6AF00659            BAL,15  SDATE           SAVED DATE
09 005C1    02000000 A          NOP
09 005C2    02000000 A          NOP
09 005C3    02000000 A          NOP
09 005C4    6AF005D0            BAL,15  EXACCT          EXECUTE ACCOUNTS
09 005C5    02000000 A          NOP                     UNDER
                          *
09 005C6    2230003A A   FNAME   LI,3    58
09 005C7    6A100000 X          BAL,1   PB:SETPOS
09 005C8    32200004 A          LW,2    4               PARAM WORD
09 005C9    20200001 A          AI,2    +1              TEXTC FILENAME
09 005CA    6A100000 X          BAL,1   PB:TEXTC
09 005CB    E800000F A          B       *15
                          ** COPY ACCOUNT LISTS FOR LATER
09 005CC    2230014A 01  RACCT   LI,3    RDACCTS
09 005CD    680005D2            B       CPYACC          COPY READ ACCOUNTS
09 005CE    22300172 01  WACCT   LI,3    WRACCTS
09 005CF    680005D2            B       CPYACC          COPY WRITE ACCOUNTS
09 005D0    2230019A 01  EXACCT  LI,3    EXACCTS
09 005D1    680005D2            B       CPYACC          COPY EXECUTE ACCOUNTS
09 005D2    22100002 A   CPYACC  LI,1    2
09 005D3    F2120004 A          LB,1    *4,1            NUMBER OF ENTRIES
09 005D4    21100013 A          CI,1    19              CHECK MAX
09 005D5    682005D7            BLE     $+2
09 005D6    22100013 A          LI,1    19              TRUNCATE
09 005D7    20400001 A          AI,4    +1              START OF ACCOUNTS
09 005D8    32280000 A   CPYALP  LW,2    0,4
09 005D9    35260000 A          STW,2   0,3
09 005DA    32280001 A          LW,2    1,4
09 005DB    35260001 A          STW,2   1,3
09 005DC    20300002 A          AI,3    +2
09 005DD    20400002 A          AI,4    +2
09 005DE    641005D8            BDR,1   CPYALP
```

```
09 005DF    35160000 A           STW,1   0,3             APPEND A ZERO ENTRY
09 005E0    35160001 A           STW,1   1,3
09 005E1    E800000F A           B       *15
                           *
09 005E2    22100001 A   GRANS   LI,1    1
09 005E3    B2220004 A           LW,2    *4,1
09 005E4    35200146 01          STW,2   FILGRANS        SAVE GRANULES THIS FILE
09 005E5    22300004 A           LI,3    4
09 005E6    6A100000 X           BAL,1   PB:SETPOS
09 005E7    22303005 A           LI,3    PBF(DECIMAL,5)
09 005E8    6A100000 X           BAL,1   PB:WORD
      09 005E9           GRANCHK EQU     $
09 005E9    32300348 01          LW,3    O:GRAN
09 005EA    E820000F A           BLEZ    *15
09 005EB    22400349 01          LI,4    O:GRAN+1
09 005EC    22600000 A           LI,6    0
09 005ED    32700146 01          LW,7    FILGRANS        GET GRANULE COUNT AGAIN
09 005EE    68000716             B       COMPINT         GO CHECK AGAINST FILTER
                           **
09 005EF    22300000 A   ORG     LI,3    0
09 005F0    35300148 01          STW,3   ORGTYPE
09 005F1    6A100000 X           BAL,1   PB:SETPOS       START OF LINE
09 005F2    20400001 A           AI,4    +1
09 005F3    F2200004 A           LB,2    *4
09 005F4    72240018             LB,2    CCKR,2
09 005F5    75200148 01          STB,2   ORGTYPE         IN HIGH BYTE
09 005F6    6A100000 X           BAL,1   PB:CHAR
09 005F7    212000D2 A           CI,2    'K'             KEYED?
09 005F8    693005FD             BNE     KEYCHK          NO, MAKE SURE NO KEYLEN FILTER
09 005F9    22200001 A           LI,2    1
09 005FA    F2240004 A           LB,2    *4,2            KEYM
09 005FB    22303002 A           LI,3    PBF(DECIMAL,2)
09 005FC    6A100000 X           BAL,1   PB:WORD         PRINT KEYLEN
09 005FD    32300352 01  KEYCHK  LW,3    O:KEYLEN        IS THERE A FILTER FOR KEYLEN?
09 005FE    6830060A             BE      ORGCHK          B/NO
09 005FF    221000D2 A           LI,1    'K'
09 00600    71100148 01          CB,1    ORGTYPE
09 00601    6930073F             BNE     COMPFAIL        NOT KEYED, KEYLEN FAILS
09 00602    22600000 A           LI,6    0
```

```
09 00603    32700002 A          LW,7    2               KEYLENGTH
09 00604    22400353 01         LI,4    O:KEYLEN+1      SPECIFICATIONS
09 00605    32E0000F A          LW,14   15
09 00606    6AF00716            BAL,15  COMPINT         CHECK THAT
09 00607    20F00000 A          AI,15   0               DID IT FAIL?
09 00608    E820000E A          BLEZ    *14             YES, DON'T LIST THIS FILE
09 00609    32F0000E A          LW,15   14              NO,RESTORE
09 0060A    32300364 01 ORGCHK  LW,3    O:ORG           ANY ORG SPECS?
09 0060B    68200614            BLEZ    CALCRECS        NO, CHECK RECORD COUNT
09 0060C    22400365 01         LI,4    O:ORG+1         START OF SPECS
                        ** CONVERT THIS INTO AN UNSIGNED 32-BIT INTEGER COMPARISON
09 0060D    22600000 A          LI,6    0               CLEAR
09 0060E    32700148 01         LW,7    ORGTYPE         ('C', 'K', OR 'R')
09 0060F    32E0000F A          LW,14   15              COPY RETURN
09 00610    6AF00716            BAL,15  COMPINT         CHECK AS 4 BYTE INTEGER
09 00611    20F00000 A          AI,15   0               DID IT FAIL?
09 00612    E820000E A          BLEZ    *14             YES, SKIP
09 00613    32F0000E A          LW,15   14              NO, RESTORE
                        **
      09 00614          CALCRECS EQU    $
09 00614    330001C2 01         MTW,0   ERRCODE         WAS THERE AN ERROR?
09 00615    E930000F A          BNEZ    *15             YES/ DONT COUNT RECORDS
09 00616    72300148 01         LB,3    ORGTYPE         ORG LETTER
09 00617    213000D9 A          CI,3    'R'             IS ORG RANDOM?
09 00618    E830000F A          BE      *15             YES - NO RECORD COUNT
09 00619    22200064 A          LI,2    100             INITIALIZE PREC COUNT
09 0061A    04100326 01 CALCRLP CAL1,1  PRECFPT         DO PRECORD TO GET NUM OF RECS
09 0061B    330001C4 01         MTW,0   BREAK
09 0061C    6930061F            BNEZ    CALCBRK
09 0061D    20200064 A          AI,2    100
09 0061E    6800061A            B       CALCRLP
                        **
      09 0061F          CALCBRK EQU     $
09 0061F    32100004 08 PRECABN LW,1    F:LF+4
09 00620    2510006F A          SLS,1   -17             GET ARS FROM DCB
09 00621    38200001 A          SW,2    1               COMPUTE NUM OF RECS IN FILE
09 00622    35200147 01         STW,2   FILRECS         SAVE THAT
09 00623    22300009 A          LI,3    9
09 00624    6A100000 X          BAL,1   PB:SETPOS
```

```
09 00625    2230300A A          LI,3    PBF(DECIMAL,10)
09 00626    6A100000 X          BAL,1   PB:WORD
09 00627    3230036B 01         LW,3    O:RECORDS
09 00628    E820000F A          BLEZ    *15
09 00629    2240036C 01         LI,4    O:RECORDS+1
09 0062A    22600000 A          LI,6    0
09 0062B    32700147 01         LW,7    FILRECS         GET RECORD COUNT AGAIN
09 0062C    68000716            B       COMPINT         GO CHECK AGAINST FILTER
                          **
09 0062D    2230001E A   MDATE  LI,3    30
09 0062E    73300003 A          MTB,+3  3               LONG FORMAT
09 0062F    2270035B 01         LI,7    O:MODIFIED
09 00630    6800066D            B       PDATE
                          *
09 00631    22300015 A   CDATE  LI,3    21
09 00632    73200003 A          MTB,+2  3               SHORT
09 00633    22700333 01         LI,7    O:CREATED
09 00634    6800066D            B       PDATE
                          *
09 00635    2230002F A   ADATE  LI,3    47
09 00636    73200003 A          MTB,+2  3               SHORT
09 00637    2270032A 01         LI,7    O:ACCESSED
09 00638    6800066D            B       PDATE
                          *
09 00639    22100000 A   EDATE  LI,1    0
09 0063A    3510013C 01         STW,1   EXPYDATE        CLEAR EXPIRY
09 0063B    3510013D 01         STW,1   EXPYDATE+1
09 0063C    35100149 01         STW,1   ISEXPIRED
09 0063D    32580002 A          LW,5    2,4             2ND WORD OF DATE
09 0063E    32480001 A          LW,4    1,4             1ST WORD OF DATE
09 0063F    31400793            CW,4    ='NEVE'         IS IT NEVER?
09 00640    69300644            BNE     EXCHK           B/NO
09 00641    3300033E 01         MTW,0   O:EXPIRED       ONLY LIST EXPIRED FILES?
09 00642    6930073F            BNEZ    COMPFAIL        YES, DON'T LIST THIS ONE
09 00643    E800000F A          B       *15
                          ** CHECK IF EXPIRED
09 00644    1540013C 01  EXCHK  STD,4   EXPYDATE
09 00645    6A100000 X          BAL,1   JD:FROMCPV      R5=JULIAN DAY
09 00646    22400000 A          LI,4    0
```

```
09 00647    1540013A 01          STD,4   EXPYJD          SAVE FOR POSSIBLE COMPARISON
09 00648    37400000 X           MW,4    MSPERDAY        TIMESTAMP
09 00649    1140012E 01          CD,4    RUNTIME         TIMESTAMP FROM START OF RUN
09 0064A    6910064F             BL      EISX            B/EXPIRED
09 0064B    3300033E 01          MTW,0   O:EXPIRED       ONLY LIST EXPIRED FILES?
09 0064C    6930073F             BNEZ    COMPFAIL        B/YES, NOT EXPIRED, SO SKIP
09 0064D    2220004B A           LI,2    '.'             HAS EXPIRY, BUT NOT EXPIRED
09 0064E    68000651             B       EXSET
09 0064F    33100149 01  EISX    MTW,+1  ISEXPIRED       REMEMBER
09 00650    222000E7 A           LI,2    'X'             IS EXPIRED
09 00651    22300037 A   EXSET   LI,3    55
09 00652    6A100000 X           BAL,1   PB:SETPOS
09 00653    6A100000 X           BAL,1   PB:CHAR
09 00654    3230033F 01          LW,3    O:EXPIRES
09 00655    E830000F A           BEZ     *15             NO SELECTION ON EXPIRY DATE
09 00656    22400340 01          LI,4    O:EXPIRES+1
09 00657    1260013A 01          LD,6    EXPYJD          VALUE
09 00658    68000716             B       COMPINT         FILTER THAT
                             *
09 00659    32580002 A   SDATE   LW,5    2,4             2ND WORD OF DATE
09 0065A    32480001 A           LW,4    1,4             1ST WORD OF DATE
09 0065B    1540013E 01          STD,4   SAVEDATE        SAVE IN CASE OF DETAIL
09 0065C    33000381 01          MTW,0   O:UNSAVED       LIST ONLY SAVED FILES?
09 0065D    6930073F             BNEZ    COMPFAIL        B/YES, SAVED, SO SKIP
09 0065E    22300038 A           LI,3    56
09 0065F    6A100000 X           BAL,1   PB:SETPOS
09 00660    2220005C A           LI,2    '*'
09 00661    6A100000 X           BAL,1   PB:CHAR
09 00662    32300378 01          LW,3    O:SAVED         FILTERING SAVED DATES?
09 00663    E820000F A           BLEZ    *15             NO, DONE
09 00664    6A100000 X           BAL,1   JD:FROMCPV      GET JULIAN DAY
09 00665    32700005 A           LW,7    5               COPY IT
09 00666    22600000 A           LI,6    0
09 00667    22400379 01          LI,4    O:SAVED+1       FILTER TABLE
09 00668    68000716             B       COMPINT         FILTER SAVED DATE
                             *
09 00669    00002009 A   PDFMT   DATA    PBF(DATE,9)
09 0066A    00002005 A           DATA    PBF(DATE,5)
09 0066B    00002007 A           DATA    PBF(DATE,7)
```

```
09 0066C    0001200E A          DATA    PBF(DATE,14,LONG)
                         *
09 0066D    72600003 A   PDATE   LB,6    3               0=REGULAR, 1=REALLY SHORT, 2=SHORT, 3=LONG
09 0066E    6A100000 X          BAL,1   PB:SETPOS
09 0066F    20400001 A          AI,4    +1              POINT AT DATE STRING
09 00670    32280002 A          LW,2    2,4             PICK UP TIME IN CASE
09 00671    32580001 A          LW,5    1,4
09 00672    32480000 A          LW,4    0,4
09 00673    323C0669            LW,3    PDFMT,6         GET FORMAT
09 00674    6A100000 X          BAL,1   PB:DWORD        PRINT IT
                         ** NOW CONVERT TO JULIAN DAY AND CHECK AGAINST FILTERS
09 00675    330E0000 A          MTW,+0  0,7             ANYTHING TO FILTER?
09 00676    E820000F A          BLEZ    *15             B/NO
09 00677    32300007 A          LW,3    7               OP, VALUE SPECS
09 00678    6A100000 X          BAL,1   JD:FROMCPV      R5=JULIAN DAY
09 00679    32400003 A          LW,4    3
09 0067A    32380000 A          LW,3    0,4             #FILTER VALUES
09 0067B    20400001 A          AI,4    +1
09 0067C    22600000 A          LI,6    0
09 0067D    32700005 A          LW,7    5               GET GRANULE COUNT AGAIN
09 0067E    68000716            B       COMPINT         GO CHECK AGAINST FILTER
                         *
        09 0067F         LISTIT  EQU     $
09 0067F    6AB0049E            BAL,11  COUNTF          DO HEADING IFF REQUIRED
09 00680    6A100000 X          BAL,1   PB:FLUSH
09 00681    3300033D 01         MTW,0   O:DETAIL        MORE?
09 00682    683006EC            BEZ     DELCHK          NO, CHECK FOR DELETE
                         **
                         ** PRINT DETAIL LINES:
                         **   EXPIRES ..., SAVED ON ...
                         **   READ: ...
                         **   WRITE: ...
                         **   EXECUTE: ...
                         **
        09 00683         DETAILS PB:PRINT,NF (TEXTC,'  ')
09 00689    1220013C 01         LD,2    EXPYDATE
09 0068A    69300692            BNEZ    DETEXP
        09 0068B                PB:PRINT,NF (TEXTC,'NEVER EXPIRES, ')
09 00691    680006AE            B       DETEXN
```

```
09 00692    33000149 01  DETEXP  MTW,0   ISEXPIRED       EXPIRED?
09 00693    6830069B             BEZ     DETNEX          NO
      09 00694                   PB:PRINT,NF (TEXTC,'EXPIRED ON ')
09 0069A    680006A1             B       DETEXD
      09 0069B           DETNEX  PB:PRINT,NF (TEXTC,'WILL EXPIRE ')
      09 006A1           DETEXD  PB:PRINT,NF (DWORD,EXPYDATE,PBF(DATE,9))
      09 006A8                   PB:PRINT,NF (TEXTC,', ')
09 006AE    1220013E 01  DETEXN  LD,2    SAVEDATE
09 006AF    693006B7             BNEZ    DETSAV
      09 006B0                   PB:PRINT,NF (TEXTC,'NEVER SAVED')
09 006B6    680006C0             B       DETFL
      09 006B7           DETSAV  PB:PRINT,NF (TEXTC,'SAVED ON '),(DWORD,SAVEDATE,PBF(DATE,9))
09 006C0    6A100000 X   DETFL   BAL,1   PB:FLUSH        DONE WITH LINE
```

```
                          **
                          ** READ, WRITE, EXECUTE, ...
                          **
09 006C1    2260002F             LI,6    RDTEXTC
09 006C2    22700528 01          LI,7    BA(RDACCTS)
09 006C3    6A1006CB             BAL,1   LISTACCTS       LIST READ ACCOUNTS
09 006C4    22600032             LI,6    WRTEXTC
09 006C5    227005C8 01          LI,7    BA(WRACCTS)
09 006C6    6A1006CB             BAL,1   LISTACCTS       LIST WRITE ACCOUNTS
09 006C7    22600035             LI,6    EXTEXTC
09 006C8    22700668 01          LI,7    BA(EXACCTS)
09 006C9    6A1006CB             BAL,1   LISTACCTS       LIST EXECUTE ACCOUNTS
                          **
09 006CA    680006EC             B       DELCHK          GO SEE IF DELETE WAS SPECIFIED
                          **
        09 006CB          LISTACCTS EQU   $
09 006CB    730E0000 A           MTB,0   0,7             ANY?
09 006CC    68320000 A           BEZ     0,1             NO, RETURN
09 006CD    89100000 A           PSW,1   *0
        09 006CE                 PB:PRINT,NF (TEXTC,*6)
09 006D4    22600000 A           LI,6    0               NUMBER OF ACCOUNTS SO FAR
        09 006D5          LISTACLP PB:PRINT,NF (TEXT,*7,8)
09 006DC    20700008 A           AI,7    +8
09 006DD    20600001 A           AI,6    +1
09 006DE    21600013 A           CI,6    19              MAX ACCOUNTS
09 006DF    681006E9             BGE     LISTACDN        DONE
09 006E0    721E0000 A           LB,1    0,7             ANY MORE ACCOUNTS?
09 006E1    21100040 A           CI,1    X'40'           VALID ACCOUNT NAME?
09 006E2    682006E9             BLE     LISTACDN        B/NO
09 006E3    21600007 A           CI,6    X'7'
09 006E4    694006D5             BANZ    LISTACLP
09 006E5    6A100000 X           BAL,1   PB:FLUSH        NEED ANOTHER LINE
09 006E6    22300008 A           LI,3    8
09 006E7    6A100000 X           BAL,1   PB:SETPOS
09 006E8    680006D5             B       LISTACLP
09 006E9    6A100000 X        LISTACDN BAL,1  PB:FLUSH
09 006EA    88100000 A           PLW,1   *0
09 006EB    E8000001 A           B       *1              DONE
```

```
09 006EC    3300033C 01  DELCHK  MTW,0   O:DELETE        DELETE MATCHED FILES?
09 006ED    6830078D             BEZ     NXTOPN
                         ** DELETE THIS FILE IF POSSIBLE
09 006EE    52100000 08          LH,1    F:LF            FLAG BITS
09 006EF    21100020 A           CI,1    X'20'           IS IT OPEN?
09 006F0    6840078D             BAZ     NXTOPN          NO
      09 006F1                   Q:CLOSE  F:LF,(ABN,ERRDEL),(ERR,ERRDEL)
      09 006F2                   Q:OPEN  F:LF,(INOUT,EXCL)
      09 006F3                   Q:CLOSE F:LF,(REL)
      09 006F4                   PB:PRINT (TEXTC,' ** DELETED **')
09 006FB    33100142 01          MTW,+1  DELFILES
09 006FC    6800078D             B       NXTOPN
                         **
09 006FD    25A00168 A   ERRDEL  SLD,SR3 -24             MAJOR CODE -> LOW BYTE
09 006FE    35A001C2 01          STW,SR3 ERRCODE         SAVE IT
09 006FF    25A00107 A           SLD,SR3 +7              ISOLATE MINOR CODE
09 00700    4BA00004             AND,SR3 M7F             EXTRACT MINOR CODE
09 00701    35A001C3 01          STW,SR3 SUBCODE
                                 PB:PRINT (TEXTC,STARS),(WORD,ERRCODE,PBF(HEX,2)),;
                                         (TEXTC,DASH),(WORD,SUBCODE,PBF(HEX,2)),;
      09 00702                           (TEXTC,STARS),(TEXTC,' DELETE FAILED')
09 00715    6800078D             B       NXTOPN
```

```
                          ** COMPARE INTEGER ATTRIBUTES
                          ** THIS IS USED FOR SIMPLE ITEMS LIKE GRANULE AND RECORD COUNT
                          ** IT IS ALSO USED TO COMPARE DATES.   THE DATES ARE CONVERTED TO
                          ** JULIAN DAY NUMBERS.
                          **
                          ** BRANCH DIRECTLY TO COMPINT FROM THE ATTRIBUTE FORMAT
                          ** CODE IN ORDER TO CHECK THE VALUE AND RETURN TO THE VLP
                          ** PROCESSING LOOP, OR BAL HERE USING R15 TO RETURN SOMEWHERE
                          ** ELSE.
                          **
                          **      R6/7=VALUE TO COMPARE.
                          **      R3=NUMBER OF COMPARITOR ENTRIES
                          **      R4=TABLE ENTRY
                          **
         09 00716         COMPINT EQU     $
   09 00716    33000382 01         MTW,0   O:VERBOSE
   09 00717    6830071B            BEZ     COMPIGO
   09 00718    32100004 A          LW,1    4
   09 00719    20100003 A          AI,1    3
         09 0071A                  Q:SNAP  'COMPINT',(*4,*1)
   09 0071B    32180000 A   COMPIGO LW,1    0,4             OPERATOR AND LENGTH
   09 0071C    52200001 A          LH,2    1               OPERATOR
   09 0071D    4B200003            AND,2   M0F             TRUNCATE
   09 0071E    5120004B            CH,2    OPERTAB         CHECK VALIDITY
   09 0071F    6920073F            BG      COMPFAIL        TABLE IS DAMAGED.
   09 00720    4B100004            AND,1   M7F             BYTE LENGTH
   09 00721    21100008 A          CI,1    8               64 BITS?
   09 00722    69300727            BNE     COMPI32         NO, ASSUME 32
   09 00723    32880001 A          LW,8    1,4             HIGH WORD
   09 00724    32980002 A          LW,9    2,4             LOW
   09 00725    20400003 A          AI,4    3               POINT AT NEXT
   09 00726    6800072C            B       COMPI64
   09 00727    22800000 A   COMPI32 LI,8    0
   09 00728    32980001 A          LW,9    1,4             GET 32 BIT VALUE
   09 00729    6810072B            BGEZ    $+2
   09 0072A    32800006 A          LW,8    6               COPY SIGN
   09 0072B    20400002 A          AI,4    2               POINT AT NEXT ENTRY
   09 0072C    11600008 A   COMPI64 CD,6    8               COMPARE
   09 0072D    6704072E            EXU     $+1,2           DISPATCH ON OPERATOR
```

```
09 0072E    6800073B            B       COMPNO
                          **
09 0072F    69100735     COMPILT BL      COMPOK
09 00730    68200735     COMPILE BLE     COMPOK
09 00731    68300735     COMPIEQ BE      COMPOK
09 00732    69300735     COMPINE BNE     COMPOK
09 00733    68100735     COMPIGE BGE     COMPOK
09 00734    69200735     COMPIGT BG      COMPOK
                          **
        09 00735          COMPOK  EQU     $
                          ** CHECK IF THERE IS ANOTHER 'AND' ENTRY
09 00735    203FFFFF A          AI,3    -1              ANY?
09 00736    E820000F A          BLEZ    *15             NO/PASSED FILTER
09 00737    33080000 A          MTW,0   0,4             SIGN SET IF START OF NEW (AND) BLOCK
09 00738    69100716            BLZ     COMPINT         GO CHECK OTHER SIDE OF AND
09 00739    6A100104            BAL,1   NXENT           POINT AT NEXT
09 0073A    68000735            B       COMPOK          GO CHECK IT
                          **
        09 0073B          COMPNO  EQU     $
                          ** CHECK IF THERE IS ANOTHER 'OR' ENTRY
09 0073B    203FFFFF A          AI,3    -1              ANY MORE TERMS?
09 0073C    6820073F            BLEZ    COMPFAIL        NO/FAILED
09 0073D    33080000 A          MTW,0   0,4             SIGN REUSET IF 'OR'
09 0073E    68100716            BGEZ    COMPINT         CHECK THAT
                          **
                          ** OR IS HIGHER PRECEDENCE THAN AND.
                          ** THIS BLOCK HAS FAILED, SO THE WHOLE FILTER FAILED.
                          **
        09 0073F          COMPFAIL EQU $
                          ** FAILED FILTER. RETURN WITH LINK REGISTER ZEROED
09 0073F    3210000F A          LW,1    15
09 00740    22F00000 A          LI,15   0
09 00741    E8000001 A          B       *1
```

```
                          **
                          **
09 00742    6A100000 X   DONEAC  BAL,1   PB:CLEAR
09 00743    33000382 01          MTW,0   O:VERBOSE
09 00744    6830074F             BEZ     DONEACT         NO MESSAGE
09 00745    222001EA 01          LI,2    ACTEXT
09 00746    6A100000 X           BAL,1   PB:TEXTC
09 00747    22200080 07          LI,2    BA(F:LA:AC)
09 00748    22300008 A           LI,3    8
09 00749    6A100000 X           BAL,1   PB:TEXT
09 0074A    22200028             LI,2    TXDONEAC
09 0074B    6A100000 X           BAL,1   PB:TEXTC
09 0074C    22200017 08          LI,2    F:LF+23
09 0074D    6A100000 X           BAL,1   PB:TEXTC
09 0074E    6A100000 X           BAL,1   PB:FLUSH
09 0074F    52100000 08  DONEACT LH,1    F:LF            FLAG BITS
09 00750    21100020 A           CI,1    X'20'           IS IT OPEN?
09 00751    68400754             BAZ     DONEACC         NO
      09 00752                   Q:SETDCB F:LF,(ABN,DONEACC),(ERR,DONEACC)
      09 00753                   Q:CLOSE F:LF
      09 00754           DONEACC EQU     $
09 00754    33000144 01          MTW,0   ACFILES         ANYTHING FOR THIS ACCOUNT?
09 00755    E820030A 01          BLEZ    *LFRETURN       NO, NO TOTALS THEN
09 00756    2220001E             LI,2    DOTS
09 00757    6A100000 X           BAL,1   PB:TEXTC
09 00758    32200144 01          LW,2    ACFILES
09 00759    22303000 A           LI,3    PBF(DECIMAL)
09 0075A    6A100000 X           BAL,1   PB:WORD
09 0075B    22200021             LI,2    TXFILES
09 0075C    6A100000 X           BAL,1   PB:TEXTC
09 0075D    32200145 01          LW,2    ACTGRANS
09 0075E    22383000 A           LI,3    PBF(DECIMAL,0,UNSIGNED)
09 0075F    6A100000 X           BAL,1   PB:WORD
09 00760    22200024             LI,2    TXGRANS
09 00761    6A100000 X           BAL,1   PB:TEXTC
09 00762    6A100000 X           BAL,1   PB:FLUSH
09 00763    6A100000 X           BAL,1   PI:SPACE
09 00764    E800030A 01          B       *LFRETURN
```

```
                            **
                            **
        09 00765            OPNABN  EQU     $
09 00765    35A001C2 01             STW,SR3 ERRCODE         SAVE THIS
09 00766    22800000 A              LI,8    0
09 00767    2290000F A              LI,9    X'F'            RESET FLAG TO INDICATE
09 00768    4780000B 02             STS,8   NXFFPT+1          NO VLPS PRESENT
09 00769    7280000A A              LB,8    SR3
09 0076A    21800002 A              CI,8    X'02'           SEE IF END OF ACCOUNT
09 0076B    68300742                BE      DONEAC
09 0076C    33000382 01             MTW,0   O:VERBOSE       ALWAYS PRINT?
09 0076D    69300777                BNEZ    PRTERR          YES
09 0076E    33000351 01             MTW,0   O:IGNORE        NEVER PRINT?
09 0076F    6930078D                BNEZ    NXTOPN          YES
                            **
                            ** DOES IT MATCH?
                            **
09 00770    22200017 08             LI,2    F:LF+23
09 00771    F2300002 A              LB,3    *2
09 00772    25200002 A              SLS,2   2
09 00773    75300002 A              STB,3   2               SET LENGTH
09 00774    223001F0 01             LI,3    FPATTERN
09 00775    6A100000 X              BAL,1   MATCH
09 00776    6930078D                BCS,3   NXTOPN          NOPE
```

```
                          **
                          ** PROCESS ERROR
                          *
09 00777    6A100000 X    PRTERR  BAL,1   PB:CLEAR        REMOVE ANY JUNK IN BUFFER
09 00778    6AB0049E              BAL,11  COUNTF          PRINT HEADING IF REQUIRED
09 00779    33100141 01           MTW,1   ERRFILES        COUNT FILES THAT CAN'T BE OPENED
09 0077A    2220001F              LI,2    STARS
09 0077B    6A100000 X            BAL,1   PB:TEXTC
09 0077C    722001C2 01           LB,2    ERRCODE         MAJOR ERROR CODE
09 0077D    22304002 A            LI,3    PBF(HEX,2)
09 0077E    6A100000 X            BAL,1   PB:WORD
09 0077F    2220001D              LI,2    DASH
09 00780    6A100000 X            BAL,1   PB:TEXTC
09 00781    322001C2 01           LW,2    ERRCODE
09 00782    2520006F A            SLS,2   -17
09 00783    4B200004              AND,2   M7F             EXTRACT SUB CODE
09 00784    22304002 A            LI,3    PBF(HEX,2)
09 00785    6A100000 X            BAL,1   PB:WORD         FORMAT
09 00786    2220001F              LI,2    STARS
09 00787    6A100000 X            BAL,1   PB:TEXTC
09 00788    2230003A A            LI,3    58
09 00789    6A100000 X            BAL,1   PB:SETPOS
09 0078A    22200017 08           LI,2    F:LF+23         FILENAME
09 0078B    6A100000 X            BAL,1   PB:TEXTC
09 0078C    6A100000 X            BAL,1   PB:FLUSH
                          **
        09 0078D          NXTOPN  Q:SETDCB F:LF,(ABN,NXTFIL),(ERR,NXTFIL)
09 0078E    52100000 08           LH,1    F:LF            FLAG BITS
09 0078F    21100020 A            CI,1    X'20'           IS IT OPEN?
09 00790    6840057C              BAZ     NXTFIL          NO
        09 00791                  Q:CLOSE  F:LF
09 00792    6800057C              B       NXTFIL
```

```
   1682                          **
   1683                          **
   1684         09 00090                  END      START
          09 00793    D5C5E5C5 A

CONTROL SECTION SUMMARY: 01 00384   PT 0     02 00032   PT 0     03 00000   PT 1     04 0005C   PT 0
                         05 00000   PT 1     06 000CD   PT 1     07 00036   PT 0     08 00036   PT 0
                         09 00794   PT 1
```

```
*    SYMBOL VALUES
        AA*09 0055C             ACCOUNT*09 00516        ACFILES*01 00144        ACHEAD*09 00008
        ACMATCH*09 00534        ACMATGO*09 00547        ACMATL*09 00539         ACMATT*09 00535
        ACNOMAT*09 0054C        ACTEXT*01 001EA         ACTGRANS*01 00145       ADATE*09 00635
        ADJUST*01 0030B         ADJVLP*01 0030E         AEMESS*09 00558         ALLDONE*09 0056A
        AOP*00000005            APATTERN*01 00200       APREFIX*01 00203        AWILD*09 0052C
        BLANK8*09 00019         BREAK*01 001C4          CALCBRK*09 0061F        CALCRECS*09 00614
        CALCRLP*09 0061A        CCKR*09 00018           CDATE*09 00631          CHARTEXT*01 001E9
        CHARTYPE*09 00050       CHKPFX*09 00583         CMDANLZ*09 003AB        CMDBUF*01 001C8
        CMDCH*09 003A3          CMDCKN*09 003B4         CMDCR*09 000A7          CMDLEN*01 001C6
        CMDNX*09 003A5          CMDNXS*09 003BF         CMDPOS*01 001C7         CMDRC*09 003BB
        CMDRX*09 003B0          CMDR0*09 003AF          CMDR1*09 003C3          CMDSKB*09 003B2
        CMDS1*09 003AE          COMMA*09 0001C          COMPFAIL*09 0073F       COMPIEQ*09 00731
        COMPIGE*09 00733        COMPIGO*09 0071B        COMPIGT*09 00734        COMPILE*09 00730
        COMPILT*09 0072F        COMPINE*09 00732        COMPINT*09 00716        COMPI32*09 00727
        COMPI64*09 0072C        COMPNO*09 0073B         COMPOK*09 00735         COUNTF*09 0049E
        COUNTNH*09 004AE        CPYACC*09 005D2         CPYALP*09 005D8         CTL*00000008
        CURACC*09 00195         CURRENTA*01 00206       DASH*09 0001D           DATBUF*01 00208
        DELCHK*09 006EC         DELFILES*01 00142       DETAILS*09 00683        DETEXD*09 006A1
        DETEXN*09 006AE         DETEXP*09 00692         DETFL*09 006C0          DETNEX*09 0069B
        DETSAV*09 006B7         DIG*00000004            DONEAC*09 00742         DONEACC*09 00754
        DONEACT*09 0074F        DOTS*09 0001E           EDATE*09 00639          EISX*09 0064F
        ENDOPTS*06 000C8        ERRCODE*01 001C2        ERRDEL*09 006FD         ERRFILES*01 00141
        ERRFULL*09 00169        ERRPARSE*09 00038       ERRTEXT*09 0015F        EXACCT*09 005D0
        EXACCTS*01 0019A        EXCHK*09 00644          EXPYDATE*01 0013C       EXPYJD*01 0013A
        EXSET*09 00651          EXTEXTC*09 00035        F*00004000              F:LA-07 00000
        F:LA:AC*07 00020        F:LF-08 00000           FILEMUST*09 00046       FILGRANS*01 00146
        FILRECS*01 00147        FINPARSE*09 00042       FIRSTFILE*09 00000      FNAME*09 005C6
        FORMATWORD*00004002     FPARAM*01 0020A         FPATTERN*01 001F0       FPREFIX*01 001F8
        GETKAB*09 0041D         GETKCHK*09 00410        GETKEYWORD*09 00409     GETKNXT*09 00422
        GETOPCH*09 0046A        GETOPC1*09 0046C        GETOPC2*09 00465        GETOPER*09 0045A
        GETOPF*09 00473         GETOPF2*09 00472        GETPATTERN*09 0043D     GETPREFIX*09 00482
        GETPREW*09 0049A        GETPREX*09 0049C        GETPRE1*09 00484        GETPRE2*09 0048D
        GETPRE3*09 00497        GETTDN*09 003EF         GETTFAIL*09 00406       GETTLP*09 003DB
        GETTLPI*09 003EC        GETTOK*09 00403         GETTOKEN*09 003C5       GETTQ*09 003D3
        GETTQC*09 003EA         GETTQS*09 003E6         GOCMD*09 004F7          GOLIST*09 004F3
        GOLISTA*09 0051A        GOLIST1*09 0050C        GOTKW*09 0042C          GPATADD*09 00456
        GPATCD*09 0044C         GPATCF*09 0043F         GPATS4*09 0044A         GRANCHK*09 005E9
        GRANS*09 005E2          HELPTEXT*09 00198       HELPX*09 003A2          I*00000000
```

```
      INTO_EURO*00001000       INTO_LEADZEROS*00002000  INTO_PRTSIGN*00004000
      INTO_SEPARATE*00000800   INTO_STARS*00000200      INTO_TRUNCLEFT*00000400
      INTO_UNSIGNED*00008000   ISEXPIRED*01 00149       KEYCHK*09 005FD          L*00000002
      LCA*00000002             LFRETURN*01 0030A        LISTACCTS*09 006CB       LISTACDN*09 006E9
      LISTACLP*09 006D5        LISTFILES*09 0056C       LISTIT*09 0067F          LISTOBC*01 001C5
      LSTFRST*09 00577         M*00000003               MDATE*09 0062D           MFF*09 00005
      MOREHELP*09 00371        M0F*09 00003             M08000000*09 00002       M7F*09 00004
      NAT*00000003             NOTCR*09 000A8           NXAFPT*02 00007          NXENT*09 00104
      NXFFPT*02 0000A          NXTA*09 0052C            NXTFIL*09 0057C          NXTFIL$*09 0058C
      NXTOPN*09 0078D          O*00000000               O:ACCESSED*01 0032A      O:CREATED*01 00333
      O:DELETE*01 0033C        O:DETAIL*01 0033D        O:EXPIRED*01 0033E       O:EXPIRES*01 0033F
      O:GRAN*01 00348          O:IGNORE*01 00351        O:KEYLEN*01 00352        O:MODIFIED*01 0035B
      O:ORG*01 00364           O:RECORDS*01 0036B       O:SAVED*01 00378         O:UNSAVED*01 00381
      O:VERBOSE*01 00382       OPCK4*09 00137           OPERTAB*09 0004B         OPNABN*09 00765
      OPTBOOL*09 000F4         OPTCKB*09 0015B          OPTCKC*09 00120          OPTDESC*09 00395
      OPTDISP*09 000EF         OPTEQ*09 00126           OPTGO*09 00115           OPTHELP*09 00379
      OPTICL*09 00151          OPTIND*09 00113          OPTIONS*06 00000         OPTI8*09 0013C
      OPTNEXT*09 00154         OPTNXI*09 00116          OPTST4*09 00139          OPTTXT*09 00140
      OPTVAL*09 0010C          OPTVINT*09 00133         ORG*09 005EF             ORGCHK*09 0060A
      ORGTYPE*01 00148         PBF/FUNC                 PDATE*09 0066D           PDFMT*09 00669
      PRECABN*09 0061F         PRECFPT*01 00326         PRINTHELP*09 00370       PRTERR*09 00777
      PUN*00000007             RACCT*09 005CC           RDACCTS*01 0014A         RDTEXTC*09 0002F
      ROP*00000006             RUNTEXT*01 00130         RUNTIME*01 0012E         SAVEDATE*01 0013E
      SCAN*09 000AD            SCANCMA*09 000C1         SCANDONE*09 00192        SCANFNP*09 000B2
      SCANKER*09 00180         SCANOE*09 00172          SCANOKW*09 000E1         SCANOPT*09 000BF
      SCANPE*09 00175          SCANPOP*09 000DA         SDATE*09 00659           SETACC*09 00522
      SONLINE*09 0009E         SPC*00000000             SPD*01 0012C             SR3*0000000A
      SSIZE*0000012C           STACK*01 00000           STARS*09 0001F           START*09 00090
      SUBCODE*01 001C3         SUMDONE*09 004F1         SUMERR*09 004E4          SUMGO*09 004C6
      SUMMARY*09 004B3         TOTACS*01 00143          TOTFILES*01 00140        TXDONEAC*09 00028
      TXFILES*09 00021         TXGRANS*09 00024         TXNOMATCH*09 0002C       UCA*00000001
      VLPLP*09 0059D           VLPLP1*09 005A7          VLPTAB*09 005B1          W*00000000
      WACCT*09 005CE           WBLANK*09 00007          WRACCTS*01 00172         WRTEXTC*09 00032
      W02020202*09 00006
*    EXTERNAL SYMBOLS
      F:LA-DEF       F:LF-DEF       INT:FROMTEXT-REF              J:ACCN-REF     J:CCBUF-REF
      J:JIT-REF      JB:CCARS-REF   JD:FROMCPV-REF                JD:FROMTEXT-REF
      M:DO-REF       MATCH-REF      MSPERDAY-REF   PB:CHAR-REF    PB:CLEAR-REF   PB:DONE-REF
      PB:DWORD-REF   PB:FLUSH-REF   PB:INIT-REF    PB:SETPOS-REF  PB:TABPOS-REF  PB:TEXT-REF
```

```
        PB:TEXTC-REF   PB:TEXT8-REF   PB:WORD-REF    PB:WRAP-REF    PD:DCB-REF     PD:LINEWIDTH-REF
        PI:PAGE-REF    PI:SPACE-REF   PI:TEXT-REF    PI:TEXTC-REF   TS:NOW-REF     TS:TOTEXT-REF
* NO UNDEFINED SYMBOLS
*    ERROR SEVERITY LEVEL: 0
* NO ERROR LINES
```

LYNX R:LA, R:PRINT, R:MATCH, R:UTILS OVER LA(J0)

* * ALLOCATION SUMMARY * *

PROTECTION	LOCATION	PAGES
DATA (00)	A000	3
PROCEDURE (01)	A800	9
DCB (10)	A600	1

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08:50 AUG 24,'26 ID=005A
ELAPSED JOB TIME              00:01:00
PARTITION NUMBER                     8
TOTAL CPU TIME                  .5772
  PROCESSOR EXECUTION TIME      .0190
  PROCESSOR SERVICE TIME        .0087
  USER EXECUTION TIME           .4770
  USER SERVICE TIME             .0725
CARDS:  CARDS READ                 16
PAGES:  PROCESSOR PAGES             7
        USER PAGES                151
CORE:   PEAK CORE(PAGES)           41
        PAGE.MINUTES          25.4072
I/O:    OPERATIONS               1770
        CALS                    12703
FILE SPACE
  AVLBL RAD PERMANENT              64
  PEAK DISK TEMPORARY              77
  AVLBL DISK PERMANENT          32485
RESOURCES ALLOCATED
  CO=128(PAGES)
```